

Automated support for operators executing
industrial procedures through visual analysis
of data from head mounted displays

Tim J. Schoonbeek

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PROEFSCHRIFT

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Cover photo: the design illustrates the evolution from the early industrial era to Industry 4.0. The left side is inspired by Philips' early 1900s factories in Eindhoven, The Netherlands. The right side reflects the modern, high-tech cleanroom environment of ASML, a Philips spinoff. At the center, a single worker connects these eras, symbolizing the shift from manual labor to AI-enabled work supported by augmented reality.

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Summary

Automated support for operators executing industrial procedures through visual analysis of data from head mounted displays

Augmented reality (AR) and mixed reality (MR) devices have recently seen accelerated adoption in the industry, even in highly IP-sensitive environments such as semiconductor factories, partially caused by pandemic-related travel restrictions. These AR devices capture a vast amount of data through high-resolution cameras and depth sensors which, when combined with modern artificial intelligence (AI) techniques, open new avenues for automated support to human operators. By offloading administrative tasks from on-site workers (*e.g.* automated logbooks of procedure executions) and providing warnings about potential errors or safety hazards, such systems hold the potential to increase quality and simultaneously allow operators to concentrate on the actual tasks at hand. Therefore, this thesis explores computer vision (CV) and AR techniques to advance such technology specifically to automatically provide assistance to operators during procedure executions.

The groundwork for this thesis is laid in Chapter 2 by identifying a lack of available techniques for understanding procedures and their execution at the intersection of action recognition, object state detection, and instructional AR. This lack particularly impacts industrial use cases, where the mere execution of (part of) a procedural step does not constitute the correct completion of that step. A preliminary investigation in this chapter demonstrates how combining real-time scene perception with procedural knowledge can flag potentially missing or incorrectly performed steps. The chapter concludes with requirements for vision-based industrial assistive systems and highlights how current datasets and problem definitions fail to capture the complexity of realistic industrial procedures in terms of flexibility of their execution, prevalence of errors, and availability of CAD models.

These uncovered research gaps are addressed in Chapter 3 by introducing procedure step recognition (PSR) and the IndustReal dataset. The PSR task aims to recognize correctly completed procedure steps and is evaluated with three metrics. The first measures how accurately the system recognizes the procedure execution order (procedure order similarity POS). The second quantifies the balance of recall and precision of the system (F_1 score). Finally, the third evaluates how quickly after the actual step completion, a system is able to correctly recognize that step completion (τ). Along with the task, the chapter introduces the new IndustReal dataset for benchmarking the performance of PSR systems. This dataset represents realistic industrial use cases by (i)

linking data from a head-mounted AR device to detailed CAD models, (ii) containing maintenance as well as assembly tasks, and (iii) capturing various unique procedure execution orders, including a significant amount of execution errors.

In Chapter 4, the focus shifts to detailed deployment challenges. Specifically, the research addresses the challenge of continuously evolving components and part configurations in industrial settings. To this end, a supervised representation learning approach is presented, along with a modified loss function. This approach compares procedure states, rather than memorizing pre-defined states, and leverages unlabeled intermediate steps to better recognize new or erroneous configurations. Experiments show substantially improved recognition and clustering performance compared to the classification-based baselines. On erroneous states, the system obtains an AP of 0.997 for missing parts and 0.952 for incorrectly oriented parts, respectively.

Chapter 5 proposes an error localization method, comparing reference images of CAD models with real-world video frames to pinpoint mismatches at the pixel level using a framework for change detection. To this end, the proposed system combines a synthetic data generation pipeline with comparative learning to train a model that determines meaningful differences between a rendered image of the intended assembly state and the corresponding real-world video frames of a procedure. These meaningful differences are calculated on the pixel level and constitute potential errors between the intended and performed steps. The CV algorithm can recognize errors on unseen assembly configurations even without training explicitly on errors, thereby enabling the scalability required for industrial applications.

The work in Chapter 6 extends beyond assembly checks based purely on spatial features, by leveraging temporal features to handle partially occluded states. Such states are a bottleneck to the recognition of completed procedural steps rapidly after their actual execution. PSR systems that are based exclusively on assembly state detection rely on complete, unobstructed views of the object in the procedure. This is a constraint, since such views are rare in ego-centric videos of procedures, and thus introduces significant delays between the correct completion of a procedure step and the corresponding recognition by the system. The system proposed in this chapter recognizes correct step completions, even if irrelevant parts of the assembly are out-of-view or invisible due to occlusion, which reduces delays between the actual completion and its corresponding recognition by 26% whilst maintaining the POS and F_1 scores. Additionally, PSR annotations and a benchmark performance are provided on a second ego-centric procedure dataset, MECCANO, to stimulate further research.

The aforementioned chapters focus predominantly on the technical aspects of AI for AR-based assistive systems. However, a highly relevant aspect of such systems has not yet been addressed in these chapters: user acceptance and experience. Even the most sophisticated system will not succeed if end users find it intrusive, prone to unhelpful warnings, or invasive. Therefore, Chapter 7 focuses on the human aspects by performing a user study based on an task-oriented implementation of the algorithm

from Chapter 5. This study is performed to evaluate how different groups – novices, manufacturing experts, and specialists in immersive technologies – respond to the system. The impact of false positive versus false negative predictions in error detection is investigated, because cognitive biases may lead some users to distrust a AI-based assistive system that over-warns, while others may lose confidence if mistakes go unflagged. By systematically quantifying both the real-world performance and the subjective acceptance of the system, this chapter investigates the important trade-off that can enable and/or hamper the adoption of AR-based procedure assistance. The results show significant quantitative and qualitative differences between user groups. For instance, both AR and manufacturing experts tend to ignore the system warnings at higher rates, while novices follow the instructions more closely, yielding less errors at completion of the procedure. These findings highlight the need of specifically tailoring assistive systems to the end-users. Furthermore, the real-world performance of the error localization system is quantified in this chapter, obtaining an ROC-AUC of 0.93 and an AP of 0.88.

To conclude, the efforts described in this dissertation span the creation of a dataset, the definition of the PSR task, novel AI techniques for PSR, temporal modeling, and error localization, as well as detailed user studies. This research lays the groundwork for AI-driven, automated procedural support in industrial settings. By reducing repetitive administrative burdens and providing automated feedback, the proposed solutions allow human operators to focus on high-level tasks. As these systems continue to evolve to better handle unforeseen states and respect specific user-defined factors, they hold promise of a future in which AI and humans fluidly collaborate to produce high-quality procedural outcomes and simultaneously improve operator job satisfaction.

Samenvatting

Geautomatiseerde ondersteuning voor operators tijdens de uitvoering van industriële procedures via beeld- en data-analyse van head-mounted displays

Augmented reality (AR) en mixed reality (MR) apparaten worden de laatste tijd, mede door de covidperiode, steeds vaker toegepast in de industrie, zelfs in sterk IP-beschermde omgevingen zoals halfgeleiderfabrieken. Deze AR apparaten genereren enorme hoeveelheden data met hogeresolutiecamera's en dieptesensoren die, in combinatie met moderne technieken voor kunstmatige intelligentie (AI), nieuwe mogelijkheden bieden voor geautomatiseerde ondersteuning van menselijke *operators*. Door administratieve taken bij de operators weg te nemen (bijv. door automatisch logboeken van procedures te genereren) en door waarschuwingen te geven voor potentiële fouten en/of veiligheidsrisico's, kunnen dergelijke systemen de kwaliteit verhogen en tegelijkertijd operators in staat stellen zich te concentreren op hun fundamentele taken. Daarom beschrijft dit proefschrift technieken voor beeldanalyse met als doel om operators automatisch te assisteren tijdens de uitvoering van procedures.

De basis voor dit proefschrift wordt gelegd in hoofdstuk 2 door tekortkomingen te identificeren bij de huidige beschikbare beeldanalysetechnieken voor de herkenning en het volgen van procedure-uitvoeringen. Deze technieken liggen op de grenzen van actieherkenning, detectie van de objectstatus en instructieve AR systemen. Het gebrek aan technische mogelijkheden treedt vooral op bij de industriële toepassingen, waar het uitvoeren van (een deel van) een procedurestap niet direct overeenkomt met de correcte en complete voltooiing van die stap. Een verkennend experiment in dit hoofdstuk laat zien hoe het combineren van perceptie met procedurekennis, ontbrekende of verkeerd uitgevoerde stappen kan herkennen. Het hoofdstuk sluit af met systeemeisen voor op beeldanalyse gebaseerde industriële assistentie. Eveneens wordt uitgelegd hoe de huidige datasets en probleemdefinities de complexiteit van industriële procedures onvoldoende representeren wat betreft de uitvoeringsflexibiliteit, de foutfrequentie en de beschikbaarheid van CAD-modellen.

De bovengenoemde onderzoekshiaten worden in hoofdstuk 3 geadresseerd door *procedure step recognition* (PSR) te introduceren en de IndustReal dataset te presenteren. De PSR-taak beoogt correct voltooide procedurestappen te herkennen en wordt met drie metrieke geëvalueerd. De eerste metriek stelt vast hoe nauwkeurig het systeem de juiste uitvoeringsvolgorde van de procedure herkent (*procedure order similarity*, afgekort POS). De tweede kwantificeert de F_1 score van het systeem, *i.e.* de balans tussen *recall* en *precision* (detectiescore en precisie). Tenslotte beoordeelt de derde

metriek hoe snel na de feitelijke voltooiing van elke stap, het systeem deze voltooiing correct heeft herkend (τ). Naast de PSR-taak introduceert het hoofdstuk de nieuwe IndustReal dataset als benchmark voor het evalueren van prestaties van PSR systemen. Deze dataset representeert realistische industriële toepassingen met de volgende vier aspecten, door: (i) data van een AR-apparaat te koppelen aan gedetailleerde CAD-modellen, (ii) zowel onderhouds- als assemblagetaken te gebruiken, (iii) diverse unieke uitvoeringsvolgordes van procedures te bevatten, en (iv) een aanzienlijke hoeveelheid uitvoeringsfouten op te nemen.

In hoofdstuk 4 verschuift de focus naar specifieke uitdagingen in de implementatie van industriële computervisie algoritmes. Dit hoofdstuk gaat concreet in op de uitdagingen van continu evoluerende componenten en variërende configuraties van onderdelen. Daartoe wordt een methodologie voor zelflerende representatie-algoritmes gepresenteerd, gecombineerd met een aangepaste kostenfunctie. Deze aanpak vergelijkt objectstatussen in procedures in plaats van deze, vooraf gedefinieerd, aan te leren. Daarnaast benut het algoritme niet-geannoteerde tussenstappen om nieuwe en/of foutieve configuraties beter te herkennen. Experimentele resultaten geven aanzienlijk verbeterde herkennings- en clusterperformance vergeleken met classificatiegebaseerde kostenfuncties. Voor foutieve objectstatussen behaalt het systeem een *average precision* (AP) van 0.997 voor ontbrekende onderdelen en 0.952 voor onjuist georiënteerde onderdelen.

Hoofdstuk 5 presenteert een methode voor foutlokalisatie, waarin referentieafbeeldingen van CAD-modellen worden vergeleken met echte videobeelden om met behulp van veranderingsdetectie de verschillen te identificeren en lokaliseren. Hiertoe combineert het voorgestelde systeem een procesketen voor synthetische datageneratie met vergelijkend leren om een algoritme te trainen dat zinvolle verschillen vaststelt tussen een gegenereerde afbeelding van de beoogde procedurestatus en het werkelijke videobeeld van een procedure. Deze verschillen worden op pixelniveau berekend en vormen potentiële fouten tussen de beoogde en de uitgevoerde stappen. Het beeldanalyse-algoritme kan fouten herkennen in onbekende assemblageconfiguraties, zelfs zonder expliciet te trainen op fouten, en faciliteert daarmee de schaalbaarheid die wenselijk en zelfs regelmatig vereist is voor industriële toepassingen.

Het werk in hoofdstuk 6 gaat verder dan herkenning louter gebaseerd op ruimtelijke kenmerken door temporele kenmerken aan te leren om procedurestappen te herkennen, zelfs als het betrokken object niet volledig zichtbaar is. Dergelijke oclusies vormen een bottleneck voor tijdige herkenning van voltooide procedurestappen vrijwel direct na hun feitelijke uitvoering. PSR-systemen die uitsluitend op detectie zijn gebaseerd, zijn compleet afhankelijk van onbelemmerde zichtbaarheid van het object. Deze beperking treedt op omdat dergelijke beelden zeldzaam zijn in procedurevideos met head-mounted camera's. Dit aspect introduceert een aanzienlijke vertraging tussen de correcte voltooiing van een procedurestap en de overeenkomstige herkenning van die stap. Het voorgestelde systeem herkent correct voltooide stappen zelfs wanneer (irrel-

evante) delen van de assemblage buiten beeld of door occlusie onzichtbaar zijn. Dit reduceert de vertraging tussen de feitelijke voltooiing en de overeenkomstige herkenning met 26%, terwijl de POS- en F_1 -scores vrijwel gelijk blijven. Daarnaast worden PSR-annotaties en een benchmarkprestatie gegeven op een tweede proceduredataset, MECCANO, om verder onderzoek te stimuleren.

De bovengenoemde hoofdstukken hebben zich overwegend gericht op de technische aspecten van AI voor AR-gebaseerde assistentiesystemen. Echter, een belangrijk aspect van dergelijke systemen is nog niet behandeld: gebruikersacceptatie en -ervaring. Zelfs het meest geavanceerde systeem zal niet worden gebruikt als eindgebruikers het opdringerig, onduidelijk of het gebruik van een (bepaald type) headset te invasief vinden. Daarom richt hoofdstuk 7 zich op de menselijke aspecten door een gebruikersstudie uit te voeren op basis van een taakgerichte implementatie van het algoritme uit hoofdstuk 5. Deze studie wordt uitgevoerd om te onderzoeken hoe verschillende groepen – beginners, experts uit de industrie, en specialisten in immersieve technologieën – op het systeem reageren. De impact van foutpositieve versus foutnegatieve voorspellingen bij foutdetectie wordt onderzocht, omdat cognitieve vooroordelen ertoe kunnen leiden dat sommige gebruikers een AI-gebaseerd assistentiesysteem wantrouwen dat te veel waarschuwt, terwijl anderen het vertrouwen verliezen als fouten niet altijd worden aangegeven. Door zowel de reële prestaties als de subjectieve acceptatie van het systeem systematisch te kwantificeren, levert dit hoofdstuk inzage in een belangrijke afweging: de acceptatie van AR-gebaseerde procedureassistentie. De resultaten tonen significante kwantitatieve en kwalitatieve verschillen tussen de gebruikersgroepen. Zo neigen zowel AR- als industrie-experts er sterker toe om systeemwaarschuwingen te negeren, terwijl beginners de instructies nauwgezet volgen. Dit leidt tot minder fouten bij de voltooiing van procedures door beginners. Deze bevindingen onderstrepen de noodzaak om assistentiesystemen specifiek op de eindgebruikers af te stemmen. Daarnaast worden de reële prestaties van het foutlokalisatiesysteem gekwantificeerd, met een ROC-AUC van 0.93 en een AP van 0.88.

Samenvattend beschrijft deze dissertatie de presentatie van een belangrijke dataset, de definitie van de nieuwe PSR taak, innovatieve AI-technieken voor PSR gebruikmakend van tijdsmodellering en foutlokalisatie, evenals een gedetailleerde gebruikersstudie. Dit onderzoek legt de basis voor AI-gedreven geautomatiseerde procedurele ondersteuning in de industriële context. Door administratief werk te verlichten en automatisch feedback te geven, stellen de besproken oplossingen de menselijke operators in staat zich te richten op taken van hoger niveau. Naarmate deze systemen zich verder ontwikkelen om onvoorziene toestanden beter te hanteren en gebruikersspecifieke factoren te adresseren, wordt de mogelijkheid gecreëerd waarin AI en mensen vloeiend samenwerken om enerzijds goede procedurele uitkomsten te garanderen en anderzijds hoogwaardig werk voor operators te realiseren.

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List of Abbreviations

ACR	Action Recognition
ANOVA	Analysis of Variance
AP	Average Precision
AR	Augmented Reality
ASR	Assembly State Recognition
ASD	Assembly State Detection
BIT	Bi-temporal Image Transformer
CAD	Computer-Aided Design
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CT	Computed Tomography
CV	Computer Vision
DamLev	Damerau-Levenshtein Distance
DL	Deep Learning
DUV	Deep Ultra-Violet
ECU	Electronic Control Unit
EUV	Extreme Ultra-Violet
ERB	Ethical Review Board
FN	False Negative
FNC	False Negative Condition
FoV	Field of View
FP	False Positive
FPC	False Positive Condition
fps	Frames per Second
FSS	Flow Short Scale
GCA	Global Cross-Attention
GLMM	Generalized Linear Mixed Model
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HHD	Hand-Held Device
HL2	HoloLens 2
HMD	Head-Mounted Display
HPLV	High Precision, Low Volume
HPU	Holographic Processing Unit
IMU	Inertial Measurement Unit

IoU	Intersection over Union
ISIL	Intermediate-State Informed Loss
KCAS	Key-Clip Aware Sampling
KFS	Key-Frame Sampling
KNN	k-Nearest Neighbors
LCA	Local Cross-Attention
Lev	Levenshtein Distance
LMM	Linear Mixed Model
LTM	Long-Term Memory
mAP	Mean Average Precision
mIoU	Mean Intersection over Union
MLP	Multi-Layer Perceptron
MR	Mixed Reality
MRI	Magnetic Resonance Imaging
MSA	Multi-headed Self-Attention
nQD	Norm of Quaternion Difference
PSR	Procedure Step Recognition
POS	Procedure Order Similarity
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
ROI	Region of Interest
RTLX	Raw Task Load Index
SLR	Systematic Literature Review
SEM	Scanning Electron Microscopy
SNN	Siamese Neural Network
STORM-PSR	Spatio-Temporal Occlusion-Resilient Modeling for Procedure Step Recognition
SupCon	Supervised Contrastive (Loss)
SUS	System Usability Scale
TAM	Technology Acceptance Model
TCP	Transmission Control Protocol
TN	True Negative
TP	True Positive
UMAP	Uniform Manifold Approximation and Projection
VL	Visible Light
ViT	Vision Transformer
VR	Virtual Reality
WM	Working Memory
XR	eXtended Reality

1.1 Background

The complexity of engineered systems has increased substantially in recent decades. This trend is evident across domains, from mass-produced consumer electronics and vehicles to one-off aerospace and semiconductor equipment. Modern systems increasingly integrate dense assemblies of specialized hardware, programmable components, embedded electronics, and high-precision sensors and actuators. Examples of such increased complexity, both in hardware and software, are numerous. In 2011, only premium cars contained 100 electronic control units (ECUs), whereas in 2021, high-end cars contain 150 or more, and even low-end vehicles approximate 100 ECUs [1]. Similarly, the number of lines of code on the most recent 787 Dreamliner airplane (Boeing, Seattle, USA) is three times as high as the line count for its predecessor, the 777 [2]. Finally, the number of transistors on the top-of-the-line iPad (Apple, Cupertino, USA) chips has increased from 3 billion in 2015 [3] to 28 billion in 2024 [4].

While high-volume markets can absorb this rising complexity through scale, the high-precision, low-volume (HPLV) manufacturing sectors, such as semiconductor lithography, aerospace, and medical imaging, face a fundamentally different challenge. These systems are not only more complex but also highly customized and often subject to extensive regulation. Each unit is effectively unique, produced either as a one-off or in small batches, and often pushed to the limits of current engineering capabilities. As a result, domain knowledge frequently does not directly transfer between systems, making manufacturing and maintenance of these machines significantly more burdensome than in mass-production environments. Moreover, these machines involve highly specialized processes executed at extremely tight tolerances, and the limited production volume makes the cost of training specialized engineers disproportionately high.

A prominent example within high-precision manufacturing is the semiconductor lithography equipment developed by ASML (Veldhoven, the Netherlands). Lithography represents a critical bottleneck in semiconductor fabrication, since it directly influences the ability to produce increasingly compact, powerful, and energy-efficient



Figure 1.1. EUV lithography machine. Source: ASML, 2020.

integrated circuits. This manufacturer is the global leader in this field and produces lithography systems such as the extreme ultraviolet (EUV) machine shown in Fig. 1.1. In semiconductor fabrication, lithography involves precisely patterning integrated circuits onto silicon wafers. Advanced modern chips contain billions of transistors, structured across dozens of layers, each requiring precise alignment and uniformity. The continuous advancement in lithographic technology has been pivotal in sustaining Moore's Law, which predicts the doubling of transistor density approximately every two years [5]. To meet these requirements, ASML has progressively transitioned from deep ultraviolet (DUV) to EUV lithography, facilitating smaller feature sizes at the cost of higher system complexity.

This complexity is not unique to the manufacturing and maintenance of lithography machines. Other high-technology sectors face similar engineering challenges. For example, Philips (Eindhoven, the Netherlands) produces advanced computed tomography (CT) and magnetic resonance imaging (MRI) machines, each embodying considerable complexity due to stringent performance requirements in healthcare and precision imaging capabilities. Similarly, the aerospace industry, exemplified by maintenance on the Rolls-Royce (London, United Kingdom) aircraft engines shown in Fig. 1.2, demonstrates significant levels of complexity. Aircraft engines require meticulous engineering precision and rigorous quality assurance procedures to meet safety, efficiency, and reliability standards. Additional examples of industries characterized



Figure 1.2. Aircraft engine maintenance. Source: Rolls-Royce, 2025.

by such technological complexity include space exploration and nuclear power plants.

Given the complexity and high degree of interdependence among components in HPLV systems, even minor errors during manufacturing or maintenance actions can propagate and lead to significant system downtime and potentially much worse. Ensuring consistent quality across all processes is therefore critical, not only to meet performance requirements but also to maintain safety, uptime, and regulatory compliance. The following section examines some of the manufacturing and servicing aspects of these machines.

1.2 Manufacturing and servicing in the HPLV industry

These high-precision, low-volume (HPLV) systems are typically composed of well-defined, tightly integrated subsystems, allowing for partial outsourcing of both development and production. This enables specialized suppliers to manufacture critical components with high precision. For example, ASML develops and integrates the lithography machines, while key optical subsystems are provided by Zeiss (Oberkochen, Germany). In deep ultraviolet (DUV) systems, Zeiss supplies high-precision lenses, whereas extreme ultraviolet (EUV) systems rely on sets of mirrors instead of lenses to

guide and shape the light beams¹. These relationships illustrate how HPLV machines are designed from highly specified, interdependent subsystems, each engineered to tight tolerances and subject to rigorous design constraints.

Since the overall system structure is explicitly defined and subsystem interfaces are well described, procedures are typically developed with a high degree of specificity. However, executing these procedures reliably in low-volume, high-complexity environments remains nontrivial, due to the lack of scalable automated quality-assurance validation and the increasing cognitive load placed on service engineers.

Quality-control processes in high-volume manufacturing environments tend to be relatively straightforward due to *e.g.* the availability of extensive datasets that can readily be collected. Automating quality-assurance validation in sectors producing large volumes, is economically feasible despite significant upfront investments in data collection and model development for each new product model. For example, automobiles are typically produced by the hundreds of thousands per model, allowing manufacturers to justify substantial initial expenses in automation technologies. In contrast, low-volume manufacturing sectors face critical scalability challenges. Rolls-Royce, for instance, sold 529 jet engines in 2024 across multiple engine variants [7]. Similarly, ASML sold 42 NXE EUV machines and just 2 EXE EUV machines (with an estimated price of 350 million euros per machine [8]) in 2024, each machine being customized with unique configurations [9]. In such low-volume contexts, collecting sufficient real-world data for automated quality-assurance validation becomes infeasible, except for specific scenarios characterized by high-frequency failures. Consequently, many quality-control techniques implemented in mass-production settings cannot be directly transferred to these specialized manufacturing processes.

The inherent complexity of these systems further amplifies this issue, rendering traditional paper-based manuals insufficient for effective quality control and maintenance. To address such complexity, alternative technologies have been explored. For example, Boeing has employed augmented reality (AR) systems since 2018 for guiding the installation of wiring looms in aircraft. Each aircraft requires approximately 210 kilometers of wiring, making conventional two-dimensional paper diagrams excessively difficult to interpret. This AR implementation demonstrated a 90% increase in first-time quality and simultaneously reduced procedure time by 30% [10]. This case clearly illustrates the value of advanced assistive systems for tasks with significant complexity, yet such systems remain limited to specific scenarios with high failures rates.

For ASML, implementing similarly advanced quality control and guidance systems holds significant potential. The economic consequences of unexpected downtime

¹Contrary to light at DUV wavelengths, light at EUV wavelengths gets absorbed by air and glass. Therefore, high-precision mirrors have to be used within a vacuum environment, as opposed to lenses at atmospheric pressure. These mirrors are manufactured within such tight tolerances, that if one were to scale them up to the size of Germany, the largest unevenness would only be one tenth of a millimeter [6].

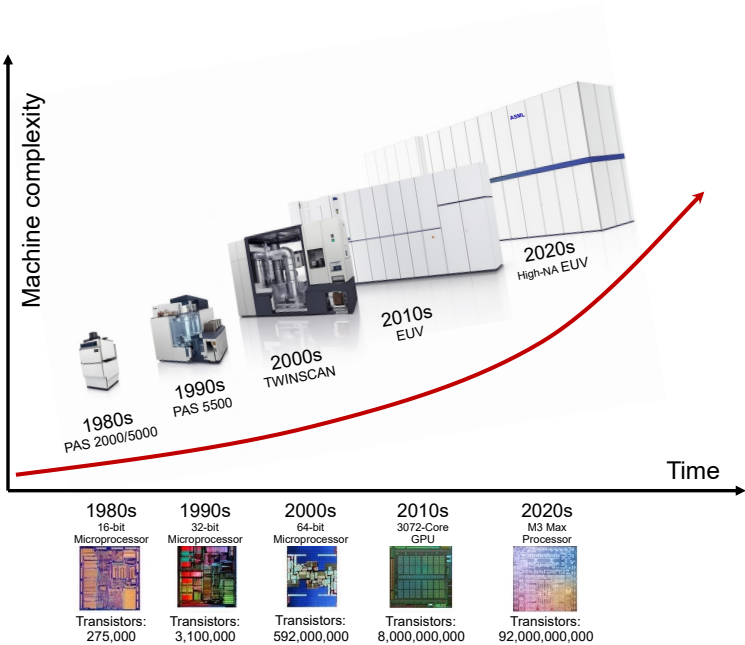


Figure 1.3. Visualization of the increasing complexity of high-precision, low-volume manufacturing systems over time, highlighting the growing need for automated AI-based support methods to ensure quality and operational efficiency.

in semiconductor manufacturing underscore this urgency, since ASML machines incur costs approximated at 100000 euros per hour of unplanned downtime². With escalating complexity, as illustrated in Fig. 1.3, performing maintenance and servicing procedures becomes increasingly challenging, as minor errors during these tasks can cause substantial operational disruptions.

Currently employed quality-assurance methods include extensive multi-year training programs for service engineers, the “four-eyes” principle involving cross-checking of work by a second engineer, and textual logging of maintenance procedures. However, these conventional methods are increasingly insufficient given the pace at which machine complexity is growing. Because of the increase in compute and memory availability in modern computing platforms, combined with recent advancements in artificial intelligence (AI) techniques, this thesis concentrates on the growing potential for providing engineers with automated AI-based support to facilitate and validate the execution of procedures.

²Information based on [11] and adjusted for inflation by the author.

1.3 Potential of AI-based procedure support

In HPLV settings, omitting or incorrectly executing even a single step during assembly and maintenance procedures can incur significant costs, ranging from halting production for hours or days (in the semiconductor industry) to loss of human lives (in the aerospace industry). Maintaining high quality and avoiding unexpected downtime for increasingly complex machines, requires advancing support mechanisms for engineers, exemplified by the augmented reality (AR) implementation example for wiring loom installations from the previous section. Currently, support includes extensive engineer training, pre-procedure briefings highlighting potential issues, and adherence to the four-eyes principle, where one engineer verifies the work of another. This thesis proposes an *additional* support method that allows real-time tracking and monitoring of procedure execution, enabled by head-worn sensors equipped with video cameras and processing of the related output signals with advanced deep learning (DL) algorithms.

Deploying camera-based sensors in IP-sensitive manufacturing environments, such as semiconductor factories, poses unique challenges. Notably, Samsung (Seoul, South Korea) implemented restrictions that ban camera-equipped mobile phones within its facilities as early as 2003 [12]. Nevertheless, constraints arising from the COVID-19 pandemic, which limited experts from traveling, significantly accelerated the adoption of AR and mixed reality (MR) technologies for remote expert support in the semiconductor industry [13]. As a result, previously stringent intellectual property (IP) protection policies were adjusted to accommodate AR devices, including Microsoft (Redmond, USA) HoloLens 2 (HL2) [14], even within cleanrooms. Therefore, this breakthrough in the use of such devices in critical industrial environments has recognized the initial potential for using such devices. This thesis argues that the opportunities for these devices is not limited to remote support, but enable a significant further advancement to automatically provide enhanced support to on-site engineers.

Automated generation of searchable logs. Recent substantial progress in deep learning methodologies further amplifies these aforementioned opportunities. By precisely recognizing and documenting the actions performed by engineers during maintenance and assembly procedures, AI systems can potentially automatically generate precise, objective, and searchable procedure logs. Such automated documentation enhances traceability and facilitates root-cause analysis by clearly indicating actions taken and components involved. This is particularly valuable during the troubleshooting that follows after unexpected downtime. For example, when machines experience sudden failures, part replacements often occur under pressure. In such cases, manually writing service logs is generally not prioritized, making it difficult to gain any clarity on which component finally resolves the issue. An algorithm that executes on a computing platform can be capable of automatically generating service logs can address this problem, providing clarity and traceability that is currently unattainable.

Just-in-time warnings. Moreover, automated real-time tracking enables targeted procedural alerts precisely when they become relevant, rather than relying on pre-procedure briefings alone. Engineers currently receive warnings regarding safety or quality hours earlier than the corresponding procedural steps, incurring the risk that these messages are forgotten. AI-based assistive systems can deliver context-specific reminders, improving quality assurance, for instance by emphasizing the installation of easily overlooked components, and enhancing safety by alerting engineers to hazards precisely at the desired moment. Therefore, such a procedural AI-assistant cannot only improve the traceability of procedures through searchable logs, but also directly increase quality and safety.

Best practice and error recognition. Expanding these capabilities, AI-based assistive systems can identify and encourage best-practice procedures, provide immediate feedback, and proactively warn engineers of potential errors when they occur. Additionally, automated part recognition supports configuration management by systematically tracking and documenting machine components and installation states throughout their life cycle. Consequently, such systems enhance component traceability, while simultaneously increasing engineers' productivity during procedural tasks through immediate feedback on best-practice executions and potential mistakes.

Reduction of training time. By automatically tracking procedures, providing real-time guidance, and detecting procedural errors, AI-based assistive systems also hold potential to reduce extensive training requirements. This reduction addresses ongoing challenges related to skill shortages and employee turnover within high-tech industries, enabling engineers to focus more directly on their primary tasks of operating and maintaining advanced machinery rather than extended training sessions. This shift not only streamlines workforce integration, but also supports sustained productivity in environments constrained by limited skilled labor.

The opportunities outlined above offer clear potential for improving procedural support through automation, but implementing them is technically challenging. Accurate interpretation of complex tasks, under strict industrial constraints, remains a major challenge. The next section examines these difficulties in the context of assistive systems for the HPLV industry.

1.4 Challenges for assistive systems in the HPLV industry

Developing assistive systems for use in high-precision, low-volume industry involves a range of technical and practical constraints that differ from those found in academic settings. This section outlines the key challenges that shape the design and deployment of such systems in real industrial environments.

The nature of available datasets. Industrial computer vision applications in the HPLV

setting poses significant challenges that clearly differ from those typically addressed by academic research. In academia, most CV datasets have a large scale, are carefully curated, and are primarily intended to evaluate algorithmic improvements. Common datasets include Cityscapes [15], ImageNet [16], KITTI [17], and Kinetics [18], [19]. These datasets typically employ clearly defined, measurable performance metrics that focus solely on algorithmic evaluation, rather than real-world effectiveness. For example, mean intersection-over-union (mIoU) is a common metric used to evaluate algorithms on semantic segmentation at the pixel level. However, it is widely recognized that metrics for such datasets do not necessarily correlate well with practical value, which is exemplified by the mIoU metric for medical applications³ [20].

Availability of industrial data modalities. Furthermore, these academic datasets generally operate within open-world conditions. In contrast, HPLV industrial applications present more constrained and structured scenarios that typically involve limited sets of possible components. Although constrained, these scenarios remain highly complex, since the actions executed within these settings are usually highly domain specific. On one hand, collecting extensive real-world data across all potential industrial scenarios is generally infeasible, particularly due to this variability across different tasks. On the other hand, the detailed computer-aided design (CAD) data and explicitly defined procedural instructions are widely available within industrial settings, and can be leveraged to address the scarcity of real-world data. However, existing datasets, even when intended for industrial scenarios, such as Assembly101 [21] and MECCANO [22], [23], do not include these essential modalities.

Recognition of correct and complete actions. Another industrial-specific requirement, typically absent in general research on video understanding, is the necessity to recognize the completion and correctness of actions. For instance, recognizing general actions such as “running” or “cooking” in video clips that contain only one action, does not require an understanding of completion or correctness of that specific action. In industrial scenarios, such measures are crucial and can define the difference between a successful procedure execution or extended downtime.

Scalability of algorithms. In HPLV applications, systems are often deployed for unique configurations that may only be assembled once. As a result, algorithms should function reliably from the first execution, without relying on task-specific real-world training data. In practice, components are regularly updated and procedures frequently undergo minor revisions. Current models typically require full data collection and retraining or fine-tuning to accommodate such changes, which imposes high costs and delays. This lack of adaptability presents a major barrier to deployment. To be viable in HPLV settings, algorithms should therefore generalize across (minor)

³For the medical application of cancer segmentation, the exact pixel-level metrics that indicate the quality of the predicted cancerous regions are not nearly as relevant as the localization of the regions. For example, if one large and one small cancer region are visible, precisely outlining the large tumor at the cost of completely missing the small tumor. In that case, the resulting mIoU score may increase, but the algorithm has significantly reduced clinical value.

procedural variations and hardware updates without requiring retraining, enabling immediate usability and long-term scalability.

Difficulty of objects for CV algorithms. Objects in HPLV settings present particularly unfavorable conditions for computer vision. Components are often made of plastic or aluminum, resulting in low-texture, reflective surfaces that hinder feature extraction. Additionally, many parts are highly symmetrical, reducing the number of unique geometric cues available for reliable detection and pose estimation. This combination of objects that are reflective, have a high degree of symmetry and a low degree of textures is especially challenging for current CV techniques. While this thesis focuses on procedural understanding rather than object recognition, the inherent visual ambiguity of such components remains an obstacle for deploying robust vision-based systems in HPLV environments.

IP-sensitive environments. Another critical challenge is the stringent intellectual property (IP) protection within industry. Even when cameras are not prohibited, postprocessing of stored video recordings frequently is. Consequently, solutions should operate in real time. Within the context of this thesis, real-time operation is not strictly defined by throughput rate, *e.g.* 30 frames per second (fps), but rather by avoiding the persistent storage of video frames and providing feedback to the operator in seconds, rather than hours or days. For example, an AI-based assistive system capable of identifying missed steps can substantially reduce delays, even if the processing time is in the order of seconds, especially when compared to the current scenario, where omissions are only identified hours later.

Users with domain expertise. An additional, distinct consideration is regarding user experience. Many augmented reality (AR) assistive systems are primarily developed for linear, step-by-step procedures, optimized for novices and training procedures because the systems lack context-aware understanding. Conversely, if the algorithms proposed in this thesis prove effective, AR systems could extend their utility beyond training to proactively support experienced users. However, such expert users likely have different requirements than novices. Therefore, determining how AR-based systems should adaptively address expert users' distinct needs and preferences remains an essential consideration.

The challenges discussed above, including constrained datasets, scalability, strict data policies, and diverse user requirements, highlight the complexity of developing effective assistive systems in HPLV applications. These issues result in the formulation of the problem statement and the research questions presented in the following section.

1.5 Problem statement and research questions

This section formulates a problem statement grounded in the preceding discussion and identifies specific research questions that emerge from it.

1.5.1 Problem statement

This thesis concentrates on automated understanding of procedural executions in HPLV settings and details this in four connected problems. The first is the development of a benchmark and dataset that represent the real-world constraints and available data types in industrial procedure executions. The second issue involves the scalability of assembly state recognition and error localization algorithms that operate effectively in procedures. The third problem concerns the robust temporal modeling of procedures to enable detection of correctly completed actions even when objects are partially occluded. Finally, the fourth problem concerns the integration of CV algorithms into user-friendly AR systems to guide assembly processes, with a focus on user expertise.

1.5.2 Research questions

The overarching research question guides this thesis is *how can AI systems on head-mounted devices reliably recognize the execution of procedural steps, from first assembly, under data constraints, and in ways expert users accept?* Within a unified framework, we address this question along four complementary dimensions: (i) what to recognize, (ii) how to scale, (iii) how to be timely, and (iv) how to be accepted by various user groups. Specific research questions are formulated for each dimension in this section, and addressed in the corresponding chapters throughout this thesis.

A Industrial benchmarks for procedure understanding

Advancements in CV techniques and the adoption of camera-equipped devices in the HPLV industry open the door for advanced, AI-based support to service engineers. At the same time, current academic benchmark tasks and data neither adequately address the industry needs, nor is it obvious what these industry needs exactly are. Furthermore, it is unclear how state-of-the-art CV models perform on tasks that do meet the HPLV industry requirements. These observations lead to the following research questions.

- RQ2.1 *Value proposition.* In what way can AR devices be used to provide support to engineers during the execution of procedures, such as maintenance and assembly?
- RQ2.2 *Gap in literature.* What is missing in current computer vision research to provide automated support to engineers during procedure executions?
- RQ2.3 *Requirements for automated support systems.* What specific requirements can be induced from the high-precision, low-volume manufacturing industry for automated support systems?
- RQ3.1 *Procedure understanding task definition.* How can a task be defined, such that it combines visual perception and explicit procedural knowledge

to reliably recognize (unique sequences of) correctly completed steps in procedural videos?

- RQ3.2 *Dataset*. How can a dataset be created to effectively capture the variability in procedure executions, error types, and data modalities typical to industrial procedures?
- RQ3.3 *Benchmark*. To what extent can existing state-of-the-art methods from related tasks generalize to this newly proposed procedure understanding task?

B Scalability of object recognition algorithms

A large body of research on CV focuses on techniques, relying on large-scale datasets of real-world data that minimize the domain shift between training data and data from novel, open-world settings. While such real-world data are scarce in HPLV industrial settings, the settings are more constrained due to well-defined procedures, and other data modalities (such as CAD data) being widely available. Furthermore, procedures are often executed with slight variations in assembly configuration and parts are frequently updated. Algorithms that seamlessly handle this fine-grained variation between objects, without the requirement to retrain the models for all potential assembly configurations (including errors), have not been established. This leads to the following research questions.

- RQ4.1 *Scalability to procedure updates*. How can an ASR algorithm be designed, such that it does not require re-training when object states or procedures are modified?
- RQ4.2 *Pre-defined error states*. How can erroneous assembly states be identified, without requiring the pre-definition of all potential procedure errors and what error recognition performance can be obtained by such an ASR algorithm, when it is not trained on any error assembly states?
- RQ5.1 *Error localization*. How can an algorithm localize the exact regions containing a potential error, rather than merely recognizing that a mistake is present somewhere in the assembly?
- RQ5.2 *Synthetic training data*. To what extent can an algorithm be trained solely on synthetic data, while still accurately recognizing errors in real assembly states, and without being trained on potential errors occurring in the real world?

C Temporal modeling of procedural steps

Object-centric recognition algorithms typically require the object of interest to be entirely within the camera FoV, from an informative viewing angle, and without any

occlusions. Such video frames are rare, particularly in ego-centric videos, due to frequent head movement and hand occlusions. Therefore, the robust performance that such algorithms provide on “clean” images comes at the cost of a large delay between a procedural step’s completion and its subsequent recognition. Temporal modeling techniques can directly optimize for step completions, without such stringent requirements on object observability, yet remain under-investigated in the HPLV setting. This leads to the formulation of the following three research questions.

- RQ6.1 *Addition of temporal features.* How can a spatio-temporal neural network be trained directly for the PSR task?
- RQ6.2 *Partial visibility of objects.* As the algorithm is directly optimized for PSR, can it recognize step completions with a reduced delay, since it intrinsically does not have to wait for images with the assembly object unoccluded and entirely visible within the image?
- RQ6.3 *Generalization.* How does a spatio-temporal neural network for PSR perform on another dataset, and how does this compare to the approach outlined in Chapter 3?

D User experience with industrial AI assistants

The experience of the end user is crucial to the adoption of AI-based assistants within the HPLV setting. While this thesis proposes that most automated assistance will be provided to experienced users, existing research focuses disproportionately on AR-guided procedures with novice users, which likely does not generalize to users with domain expertise. Therefore, the following research questions are formulated.

- RQ7.1 *System performance in a real setting.* Can the algorithms presented in this thesis be integrated into a real-time AR system to provide advanced support during a procedure and what is the real-world system performance?
- RQ7.2 *System feedback trade-off.* Should the system feedback be tuned to over-warn/under-warn users about potential mistakes, and how does this relate to the different user groups?
- RQ7.3 *User experience.* How do users, having different background expertise, experience such an AR assistive system and are they receptive to using it in the workplace?

1.6 Contributions

This section provides an overview of the scientific contributions presented in this thesis. These contributions are linked to four categories, elaborated below.

1.6.1 Industrial benchmarks for procedure understanding

The first major contribution of the work in this thesis is the definition of a new task in video understanding, named procedure step recognition (PSR). The objective of PSR is to recognize *correctly completed* procedural steps. This task fills the gap that was previously existing between action recognition (ACR), assembly state detection (ASD), and instructional AR systems. Alongside the PSR task, three metrics are defined, namely the procedure order similarity (POS), F_1 score, and average delay τ . The POS compares the actual order in which procedural steps are executed to the recognized execution order, the F_1 score evaluates the prediction quality, and τ measures the average delay (in seconds) between the *actual* and *recognized* correct step completion. These metrics are based on industrial demands and should be interpreted jointly, since they are intricately connected.

Providing new annotations for existing datasets is not sufficient to overcome the shortcomings of the currently available, industrial-like datasets. Therefore, together with the PSR task, this thesis presents the new IndustReal dataset. This dataset is novel in three important aspects. First, it contains a wide variety of execution and omission errors. The dataset comprises 38 unique errors (in 84 procedure executions), of which a significant part is exclusively kept to the test set. Second, the participants for IndustReal are presented with subgoal-oriented procedures. This results in a variety of different execution strategies, with 48 unique execution orders, of which again some are kept exclusively for the test set. The third aspect is that 3D geometries are open sourced. This type of data is frequently available within industry, yet rarely combined with computer vision datasets. Additionally, the dataset is 3D printed, which means it can be reproduced and extended anywhere and at any time as long as a printer is available.

The two aforementioned contributions culminate to the final contribution of this thesis to the benchmarking of industrial procedure understanding, namely establishing a baseline performance on IndustReal. Experiments with different baseline algorithms towards PSR provide valuable insights. Firstly, the experiments show that the best-performing baseline obtains a POS of 0.797, F_1 score of 0.883, and τ of 22.4 seconds. This performance notably drops for videos that contain at least one procedure error. The experiments also demonstrate that training a model exclusively on synthetic data reduces its performance, but the gap between real-world and synthetic training is significantly reduced when explicit procedure information is injected into the algorithm. Besides PSR labels, annotations for ASD and ACR are also provided. Therefore, state-of-the-art algorithms are trained and tested on those tasks, demonstrating that IndustReal is challenging and relevant not just for PSR, but for a broader range of tasks. The broader implications and future outlook of the PSR task are examined in Chapter 8.

1.6.2 Scalability of object recognition algorithms

Scalability is an important requirement that the HPLV industry places on CV algorithms for the recognition of objects and potential errors within such objects. Specifically, algorithms should neither have to be retrained for all potential minor variations between systems, nor for minor updates to procedures and parts. Additionally, given the complexity, it is infeasible to define all potential execution errors beforehand. To this end, the work presented in this thesis provides four contributions.

First, a representation learning approach for scalable assembly state recognition (ASR) is presented. Specifically, a function is learned that projects images into a meaningful embedding space, enabling a model to distinguish between assembly states based on visual similarity rather than explicitly memorizing each possible state. Because the model is *comparative*, rather than relying on memorization, the model is able to recognize entirely unseen assembly states without requiring re-training or fine-tuning of the model weights. Additionally, it outperforms classification-based models even on assembly configurations observed during training.

Secondly, a modification to contrastive loss functions is proposed, named the intermediate-state informed loss (ISIL). The ISIL leverages intermediate states, which occur during the transition between two well-defined assembly states. These intermediate states remain unlabeled, as no appropriate labels can be assigned to them⁴. The proposed ISIL leverages these intermediate states by using them exclusively as negative samples within the contrastive loss function. This does not significantly affect classification performance, but increases clustering performance regardless of backbone type (ResNet vs. Transformer) or contrastive loss function (Batch Hard vs. Supervised Contrastive) by 5-22% *MAP@R*.

Third, an extensive evaluation of the aforementioned algorithm is performed on assembly images with assembly errors. The models have not been exposed to any erroneous states during training, nor have potential error types been a-priori specified, but are capable of distinguishing between erroneous and correct assembly states for various error types. The method demonstrates especially high performance on missing and incorrectly oriented parts, with average precision (AP) scores of 0.997 and 0.952, respectively.

Finally, an error localization network based on change detection is presented. This model compares a rendered image, based on the CAD model of an assembly configuration, to real-world assembly images. Because the rendered image does not contain any assembly errors, any meaningful changes detected between the rendered and real-world image constitute errors. The algorithm is trained exclusively on synthetic data and is therefore scalable to new systems and can be used already for the first real-world assembly. Furthermore, the synthetic data-generation pipeline allows

⁴It is impractical to label every possible transitional state, such as “bolt is 10% tightened”, “bolt is 11% tightened”, etc. as separate classes given the amount of possibilities. Therefore, these are labeled as a transition between two well-defined states, or remain unlabeled.

for full control over the expected (normal) variability between the image pairs.

1.6.3 Temporal modeling of procedural steps

Object-oriented algorithms provide robust recognition on images that do not contain (partial) occlusions or object parts that fall outside of the field of view (FoV). However, data from ego-centric video cameras frequently contain occlusions from hands and tools. Furthermore, the camera follows the operator's head movement, resulting in a significant amount of frames in which the objects are only partially visible within FoV. Therefore, this thesis also provides contributions to temporal modeling of procedure step completions.

A spatio-temporal model is presented, the first algorithm to be directly optimized for the task of PSR. By combining spatial and temporal features, the model is able to recognize the correct completion of procedural steps based on parts relevant to the actual completion of a step. Furthermore, due to an understanding of temporal consistency, the model is able to handle temporary loss of observability of object parts, *i.e.* due to occlusions or non-favorable viewpoints. Because the model does not rely on pristine, unobstructed video frames, it is able to improve the average prediction delay between the actual completion and recognition of a procedural step (τ) by 26% while maintaining state-of-the-art POS and F_1 scores.

Additionally, key-frame sampling (KFS) and key-clip aware sampling (KCAS) are proposed to effectively train the spatio-temporal model. KFS is a weakly supervised pretraining technique for the spatial encoder that maximizes similarity between identical step completions from different videos, while minimizing that similarity for different steps. KCAS is a new sampling strategy for training the temporal encoder, based on a bi-modal distribution that over-samples hard negatives (clips that do not contain step completions) right before step completions, under-samples the ambiguous frames around the exact moment of a step completion, and over-samples positives (clips that contain step completions). Without both KFS and KCAS, the spatio-temporal model does not converge to state-of-the-art performance.

Finally, the MECCANO [22], [23] dataset is extended with PSR labels. Benchmark performances are reported from both the object-oriented PSR baseline, as well as the spatio-temporal model. Both methods perform significantly worse on the MECCANO dataset, in large part due to the absence of CAD models and sparsity of step completions in the dataset. The best model achieves a POS of 0.377, F_1 of 0.497, and τ of 88.6 seconds.

1.6.4 User experience with industrial AI assistants

AI-based industrial assistive systems can theoretically provide significant business value, but only if the target users are willing to use these systems. Since many AR-guidance systems are tested on novice users, with the intention of training rather

than providing real-time support, many aspects of user experiences for such assistants remain unclear. The final contributions of this work directly address this research gap.

The object recognition algorithms proposed in this thesis are implemented into a real-world system for error recognition on a quadcopter assembly. The algorithm is evaluated throughout the user study, yielding the quantification of real-world performance, both in terms of algorithmic performance and overall latency of the system (that includes computation time and streaming latency). The presented system enables an area under the receiver-operator curve (ROC-AUC) of 0.93 and an AP of 0.88, without training on any real-world data, explicit domain adaptation techniques, or predefining erroneous states. The system is able to reliably detect various (but not all) error types. A computation time of 1.8 seconds per evaluation is measured, which can be reduced without an impact on the performance by another 57%.

Using this system, an extensive user study is performed to evaluate the cognitive biases across three different population groups, namely domain experts (*i.e.* manufacturing), AR/VR experts, and novices. The AI-based assistive systems in HPLV settings are projected for usage with a high degree of domain expertise, rather than novices. However, there is no study that has investigated the expert-specific cognitive biases across different types of expertise, or how the experts' performances and attitudes differ and overlap. To this end, the effect of incorrect warnings (false positives) and missed errors (false negatives) on assembly outcome, user perception, and user expertise is systematically assessed in the user study. From this study, the following notable findings are presented.

The study demonstrates that over-warning results in statistically significant more accurate assemblies compared to under-warning. Whereas over-warning significantly elevates cognitive load, it does not lead to a statistically significant increase in task completion time. The results demonstrate that this effect is moderated by user expertise. Novices and domain experts benefit from over-warning and experience improved flow, whereas AR/VR experts exhibit similar patterns under under-warning conditions. These findings indicate that effective system design should consider the user's background expertise. Additionally, cognitive biases tend to shape users' tendencies to follow instructions and their preferences for warning intensity. Both domain and AR/VR experts frequently disregard (correct) warnings provided by the system, drawing on prior experience with systems providing inaccurate outputs.

1.7 Thesis outline and scientific background

This section provides an overview of the thesis chapters, including the manner in which the chapters are linked. It also briefly highlights the contributions of each chapter, including the associated scientific publications that form the basis of the individual chapters. The layout of the thesis is illustrated in Fig. 1.4. After the generic introduction, the problem space of this thesis is formulated and the research gap is

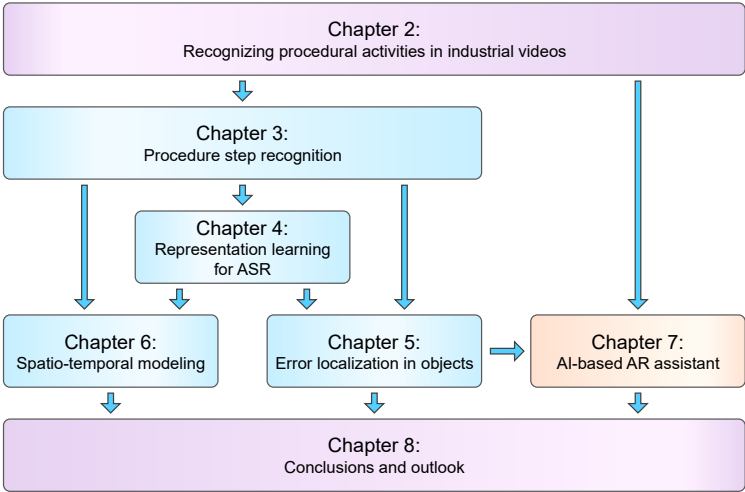


Figure 1.4. Schematic layout of the thesis content.

investigated (Chapter 2). The insights obtained from this analysis are used to define the new PSR task and IndustReal dataset, specifically relevant to the HPLV industry (Chapter 3). This dataset is used in the three subsequent chapters to investigate the scalability of CV algorithms (Chapter 4), generalization to unseen error types (Chapter 5), and spatio-temporal modeling of procedure executions (Chapter 6). The insights obtained from those chapters culminate in a user study, wherein three groups, with a varying degree and type of expertise, execute procedural tasks with automatically generated AR guidance (Chapter 7). The last chapter (Chapter 8) concludes the research and provides an outlook.

Chapter 2 defines the problem space of the thesis by narrowing down to the task of automated procedure logging. Generating such automated logs creates objective, searchable and interpretable logs, rather than relying on handwritten or free-form text. A research gap between action recognition (ACR), assembly state detection (ASD), and instructional AR systems is identified. Furthermore, a proof-of-concept framework provides insights into the system requirements for automated logging systems.

The proof-of-concept framework for automated logging of procedure actions was published in the Proc. of the London Imaging Meeting (LIM) 2022 [24] and the research gap identification in the IEEE Conf. on Virtual Reality and 3D User Interfaces Abstracts (VR) in 2023 [25].

Chapter 3 is dedicated to the definition of the procedure step recognition (PSR) task and its three accompanying metrics. The task and the metrics are defined based on the insights from Chapter 2. Crucially, PSR focuses on recognizing steps that are *completed correctly*. Together with PSR, the IndustReal dataset is released, containing many procedural errors and various execution orders and CAD data for all components, a

data modality that is widely available within industry but rarely provided in academic datasets. Benchmark performances are provided for various PSR methods.

The IndustReal dataset, PSR task and evaluation metrics, and benchmark performances were published in the Proc. of the IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV) in 2024 [26].

Chapter 4 introduces a framework for ASR based on supervised representation learning. By generating meaningful embeddings from images instead of memorizing a fixed set of configurations, the model can recognize assembly states in IndustReal that are unseen during training. It also introduces the ISIL modification, that leverages unlabeled intermediate states as negative samples in the contrastive loss, improving clustering across architectures and loss functions. The models also distinguish correct assembly configurations from assemblies with missing or incorrectly oriented parts.

The work on scalable ASR was published in the IEEE Robotics and Automation Letters (RA-L) journal in 2024 [27] and presented at the 2025 IEEE Int. Conf. on Robotics Automation (ICRA) [28]. The underlying ideas of this work were applied to the inspection of patterns on silicone wafers, and resulted in a European and world Patent application [29].

Chapter 5 focuses on the exact localization of assembly errors by introducing the first assembly error localization algorithm based on change detection. Whereas the algorithm from the previous chapter recognizes that an error is present, it cannot exactly localize the error. Therefore, this chapter demonstrates that this error localization algorithm is able to find errors between synthetic images of CAD models and real-world assembly images, while being trained exclusively on synthetic data and on correct assembly configurations only.

The research on the localization of assembly errors was published at the European Conf. on Computer Vision (ECCV) in 2024 at the Workshop on Vision-based Industrial Inspection (VISION) [30]. The fundamentals of these techniques, adapted to defect inspection on silicone wafers, resulted in an European Patent application [31].

Chapter 6 addresses the main limitations from the two previous chapters, assuming full-object visibility within the FoV and minimal occlusion, causing delays between step completion and recognition. Therefore, this chapter introduces a spatio-temporal network to model step completions even under partial occlusions. The network is evaluated on IndustReal and MECCANO [22]. The model reduces the average delay τ while obtaining state-of-the-art performance on the other metrics.

This research has been accepted for publication in the Elsevier Journal on Computer Vision and Image Understanding (CVIU) [32].

Chapter 7 shifts the focus to user interaction, which is a critical factor in system adoption. This chapter studies the trade-off between over-warning and under-warning operators for potential mistakes by implementing the error localization algorithm

from Chapter 5 into a real-time system and evaluating three user groups with varying expertise. The system detects assembly errors without training on real-world data or erroneous assemblies and the user study underscores the need for evaluation across user backgrounds.

The system implementation and user study has been submitted to the IEEE Trans. on Visualization and Comp. Graphics (TVCG) and is currently under review.

Chapter 8 summarizes the main findings of the thesis by reviewing the results presented in each chapter in relation to the corresponding research questions. It concludes with an outlook on future work and identifies future research challenges.

Recognizing procedural activities for automatic procedure logging

2.1 Introduction

The preceding chapter has underscored the complexity inherent in the high-precision, low-volume manufacturing industry and emphasized the potential of automated understanding of procedure executions to enhance quality control, root-cause analysis, and machine uptime. It has also discussed how the integration of augmented reality devices and the availability of large-scale computational resources create new opportunities for automation through combining computer vision techniques and vision-based deep learning methods. The objective of the current chapter is to establish the technical foundations underlying the core research areas of this thesis and to identify relevant gaps in the current literature that motivate the proposed contributions.

Modern AR devices, such as the HoloLens 2 (HL2), integrate RGB, depth, eye-tracking and inertial measurements in a single, intrinsically calibrated package. We pose that these multi-modal signals can be used to provide real-time, contextual support to engineers during the execution of complex procedures. A fundamental starting point for this support translates into the automated logging of procedural steps, where the system infers progress and outcomes without human intervention. However, even when simplifying the procedure, constraining it to a small set of well-defined steps, existing computer vision methods struggle to deliver robust, generalized, and scalable solutions. These methods typically require large-scale, domain-specific and real-world training data which are infeasible to collect in many industrial settings due to time, cost, and proprietary constraints. Additionally, and particularly for the high-precision, low-volume manufacturing industry, it is crucial that the automated support can be provided from the first assembly, as every machine can be unique. Therefore, it is simply not possible to first collect real-world data.

Despite the clear benefits of automated action logging, there is no consolidated overview of the research landscape that integrates insights from action recognition,

assembly state detection, and AR-based instructional systems. Furthermore, a clear set of requirements and demands, relevant to the high-tech industry, remains absent. Such requirements should account for the unique challenges of these industrial settings, including non-repetitive workflows with stringent quality objectives. The above discussions reveal three core points for this chapter, leading to the formulation of the following research questions.

- RQ2.1 *Value proposition.* In what way can AR devices be used to provide support to engineers during the execution of procedures, such as maintenance and assembly?
- RQ2.2 *Gap in literature.* What is missing in current computer vision research to provide automated support to engineers during procedure executions?
- RQ2.3 *Requirements for automated support systems.* What specific requirements can be induced from the high-precision, low-volume manufacturing industry for automated support systems?

To address these research questions, this chapter first argues that automated procedure understanding (and thereby logging) is the fundamental task that facilitates further automated procedure support systems. Subsequently, the capabilities and shortcomings of state-of-the-art systems in providing that type of support during maintenance and procedure actions are investigated. This investigation demonstrates that existing research addresses only the individual facets of the problem. Action recognition (ACR) methods classify short video clips, assembly-state detection (ASD) identifies the physical outcome of a step, and instructional-AR systems overlay information in the user's field of view (FoV). However, none of these approaches, implemented alone, can effectively leverage explicit procedure knowledge, recognize *correctly and completed* steps, handle flexible execution orders, or remain robust even under partial occlusions. Bridging this gap is therefore essential for industrial application.

Accordingly, this chapter explores whether AR devices can facilitate automated logging during maintenance and assembly tasks through a proof-of-concept system on a simplified procedure, as depicted in Fig. 2.1. The implementation of this proof of concept helps to identify (i) which shortcomings in the current literature hinder industrial implementation, and (ii) what explicit requirements are imposed on automated support systems by the high-precision, low-volume manufacturing industry. Experimental findings demonstrate that meaningful logs can be produced without enforcing a rigid step sequence and without storing any videos, provided that strong assumptions are imposed (*i.e.* fiducial markers on all components). These constraints are gradually relaxed throughout the subsequent chapters to move toward a fully deployable solution with sufficient flexibility.

In summary, this chapter contains the following contributions.

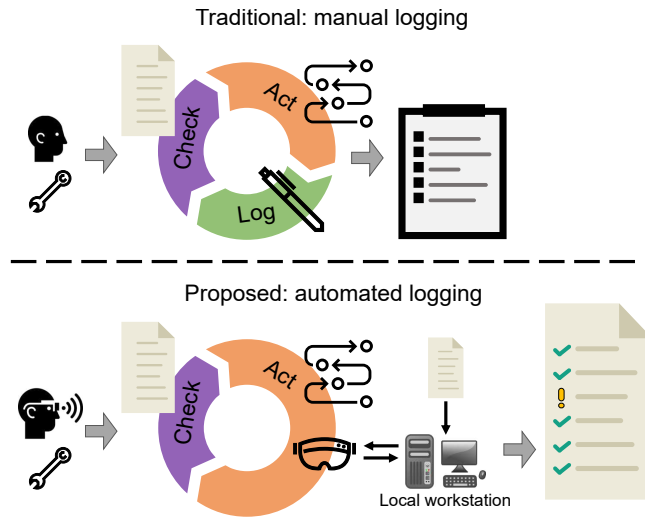


Figure 2.1. Conventionally, an engineer checks the tasks to be performed, then executes those tasks and manually logs the completion of each step during the procedure. This chapter proposes to automatically generate a searchable and interpretable text log of these procedural steps by perceiving the progress and comparing this to the expected procedure, without enforcing a restrictive step-by-step sequence order.

- The task of recognizing industrial actions through visual perception and structured procedural knowledge is framed, with the objective of generating interpretable and searchable logs of procedure executions.
- A comprehensive overview of related work in automated procedure understanding is provided, identifying a gap between complementary research domains.
- A set of design requirements is identified, grounded in the literature review and a proof-of-concept study, for AR-based support systems in high-precision, low-volume manufacturing contexts.

The chapter proceeds as follows. Section 2.2 delineates the problem space of industrial procedure understanding by specifying where automated support holds significant potential, followed by an overview of the related work in the field of procedure understanding in Section 2.3. The proof-of-concept system is described in Section 2.4, followed by the empirical evaluation in Section 2.5. The system requirements are presented in Section 2.6, after which conclusions are drawn in Section 2.7.

2.2 Automated logging of procedure executions

Various types of additional support can be provided through AR devices such as the HL2. For instance, work instructions can be displayed directly within the field of view (FoV) of the engineer [33]–[35]. Additionally, it allows for remote collaboration

between on-site generalists and remote experts [36]–[38]. However, this type of support requires significant human input, either from the executing engineer (selecting the relevant instructions) or the remote expert. This thesis focuses on providing automated support, specifically by perceiving the engineer's workflow during procedure executions. With an understanding of the status of all steps within a procedure, including whether they have been correctly executed up to a certain point in the procedure, the engineer can automatically be warned for potentially forgotten steps. Additionally, messages about updated steps of a procedure, as well as moments of frequent failures, can be automatically provided directly when they are most relevant¹. Furthermore, with an understanding of the visual scene and procedural knowledge, objective and searchable procedure logs can be automatically created.

Procedure logs are used to perform root-cause analysis in case of failure and for configuration management. In its present state, logs are created either by writing or typing (retrospectively) the performed actions. This is time-consuming and in free-form text data, requiring complex algorithms to convert technical language into standardized, searchable and optimizable logs [39]–[41]. Moreover, the creation of logs, particularly while being under pressure to finish a procedure quickly, is frustrating to the service engineers, who denote administrative duties as their main frustration. Alternatively, logs can be automatically generated based on the timestamps of electronically opened procedure instructions. These resulting logs are often poorly interpreted by humans, and more critically, do not necessarily represent the actual execution of a procedure. Engineers know parts of the procedure by heart, hence they may not open the corresponding instruction at all². Simply storing a video as a log also has significant downsides. For example, storing videos from sensitive factories and cleanrooms in the semiconductor industry pose serious IP risks and videos are not easily searchable without spending significant time [42], [43]. Therefore, these types of logging have significant flaws, resulting in incomplete or unusable logs. For this reason, we pose that automated logging, based on both real-time visual scene perception and procedural knowledge, is the main value aspect for providing advanced support, since this enables the creation of objective, searchable, and human-interpretable logs. Therefore, the capability of a system to automatically recognize (and thus log) the execution of procedure steps is the fundamental objective for providing automated support. Furthermore, algorithms capable of real-time logging, can readily be extended, without significant additional computational overhead, to also supply real-time warnings of (potential) forgotten steps and communicate instructions to the engineer at the exact point of the procedure when it is relevant.

¹Often, procedures take hours to complete, but warnings for points of frequent failure are provided only at the briefing *before* the start of the procedure. Therefore, the actual moment for which the warning is intended, typically occurs hours after the warning is provided.

²It is not uncommon to see logs where an engineer has apparently worked on the first step of a procedure for hours longer than expected, only to “execute” the remaining steps in a few seconds by clicking through the corresponding instructions.

Existing AR frameworks based on perceptual understanding are already able to actively assist operators in various ways, *e.g.* by training novice operators for assembly tasks [44]–[47], or visualizing instructions that are hard to interpret from 2D representations [10], [48]. Whilst such approaches clearly provide benefits, most are only intended to display instructions, without confirming whether those instructions were actually executed. Systems that actually confirm whether actions are completed, verify this in a fixed and concatenated sequence, meaning that operators must perform the involved steps from first to last in a sequential order. Such rigid sequences can be beneficial for novice users, but inherently restrict the more experienced operators, since many service tasks may be correctly performed in different orders. Additionally, the sequence provided in the instruction is not necessarily optimal. We pose that a support system should allow for flexible following of the procedure, while still allowing to compare observations to the expected actions based on procedural information. Analyses of the sequence in which the actions were actually executed also enable for identifying ways for optimizing execution strategies, *e.g.* finding best practices.

2.3 Related work on CV for procedure understanding

The task of automatically providing support and logging human actions during manufacturing and maintenance tasks touches several research fields, namely (procedural) action recognition (ACR), assembly state detection (ASD), and intelligent AR applications.

The task of procedural action recognition focuses on determining human behavior in relation to procedure execution. Existing work on recognizing and tracking procedural videos takes multiple approaches, such as action recognition (ACR) [21], [22], [49], [50], procedural activity understanding [51]–[53], keystep recognition [49], [54]–[56], and, as will be further introduced in Chapter 3, procedure step recognition [57]. In procedural action recognition, the objective is to categorize actions performed by humans in short video clips. Several datasets are available for this task, *e.g.* determining interactions in kitchens [58], [59], home situations [60], and industrial settings [22], [61]. In contrast, the task of procedural *activity* recognition focuses on understanding longer videos. Within this, the task of *keystep* recognition aims to find key moments and their sequences in long videos of procedure executions.

Recognizing these interactions between objects and humans can provide valuable information when trying to understand what a human is doing, but does not provide conformation of whether a task is actually completed. For instance, if a prediction is made that an operator is “placing block X”, it remains unclear whether it was placed correctly and if the task was completed or stopped half-way through. Conventional ACR methods often stop at labeling temporally bounded actions, without assessing whether those actions led to valid task outcomes that align with the procedural constraints. This disconnect becomes particularly relevant in industrial domains,

where partial or incorrect completions can lead to downstream faults, rework, or even safety issues. As such, merely recognizing that an action occurred is insufficient — support systems must also determine whether procedural steps have been fully and correctly carried out. To address this gap, assembly state detection (ASD) offers an outcome-oriented perspective by evaluating the physical configuration of components, thereby providing a direct signal of task completeness rather than just human activity.

The task of ASD provides the current state of an object given all of its possible states during assembly. Several possible approaches to ASD have been successfully implemented, using markers on each object part [46], applying region-of-interest crops based on a single marker [47], and deploying convolutional neural networks (CNNs) [62]–[64]. CNN-based methods operate without any markers and therefore do not require alterations to object (parts) and are robust to marker occlusions, which frequently occur when humans are interacting with such objects. However, marker-based detections do not require image datasets for training and provide a higher accuracy with significantly less computation power. Another approach to ASD is presented by Jones *et al.* [65], who apply a partially-observable Markov decision process to understand assembly states of Duplo blocks during assembly. While the aforementioned research all provides the state of an object during its assembly or disassembly, three fundamental limitations remain.

First, ASD algorithms typically require all components within the assembly to be (i) within the image boundaries and (ii) unoccluded. Such images are rare in ego-centric videos, where hands and tools frequently interact with the objects in a procedure. The second limitation is regarding scalability. ASD methods rely either on fiducial markers [46], fixed RGB-D cameras [45], or neural networks trained for classification [62]–[64]. Placing fiducial markers on all relevant components is impractical for many real-world applications due to the associated time, effort, and disruption to standard manufacturing processes. While fixed-camera setups are effective for tasks involving small objects on constrained workbenches, they become impractical for large or complex assemblies where frequent occlusions by the operator and limited viewpoints result in incomplete or obstructed observations. Therefore, ASD algorithms that use deep neural networks for the classification of assembly states fit the setting of this thesis best. However, whenever even a minor change to the procedure or objects within the procedure is made, new data must be collected and the entire network needs to be re-trained. This is costly and time consuming, significantly hindering the adoption within industry, since parts and procedures are frequently updated. The final limitation of ASD is that no physical constraints and procedural information are explicitly leveraged. Since object states can look identical from several viewing angles during assembly, there is a need for a more intelligent framework, which keeps track of observed states and determines which future states are to be expected.

Gupta *et al.* [45] present such an intelligent framework, which guides a user into the correct assembly of Duplo block models with visual aids, using multiple RGB-D

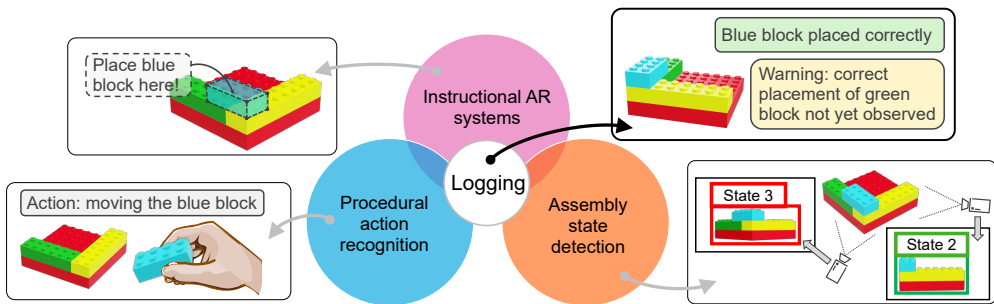


Figure 2.2. Automated logging lies at the intersection of three research fields. It aims at recognizing the completion of procedural actions and determines whether those actions, as well as their order, are in line with the expected procedure.

cameras on a fixed work-bench setup. Westerfield *et al.* [46] propose a framework using markers on each relevant component and automatically detects whether a user completed an assembly correctly. Yin *et al.* [47] propose a system which automatically determines whether an engineer's current step is completed by simultaneously recognizing hand motion and visual completeness of the object on which the engineer is working. All the previously mentioned approaches rely on a rigid, strictly sequential order, whereas we hypothesize that a worker-centric framework must allow for a flexible and/or modified sequential order.

In parallel to ACR and ASD, instructional augmented reality (AR) systems have been developed to guide users through procedures by overlaying visual cues and instructions in their field of view. These systems have shown promise in training novice operators and assisting with complex tasks, but typically focus on displaying information rather than verifying task completion [10], [66], [67]. Furthermore, most instructional AR implementations assume a fixed procedure flow, offering limited support for flexible execution or procedural deviations that are common in real-world manufacturing and maintenance contexts.

These three domains (ACR, ASD, and instructional AR) each contribute valuable capabilities, but none fully address the needs of automated procedural understanding in industrial settings. As shown in Fig. 2.2, the core challenge lies at their intersection: a system must perceive actions, assess object states, and reason over procedural logic in real time to support flexible, correct, and efficient task execution. This gap forms the foundation for the contributions presented in this thesis. The remainder of this chapter introduces a proof-of-concept prototype that explores real-time procedural support using visual perception. Building on the insights from this prototype, the subsequent chapters formalize the task of procedure step recognition and present scalable, data-driven approaches for industrial deployment.

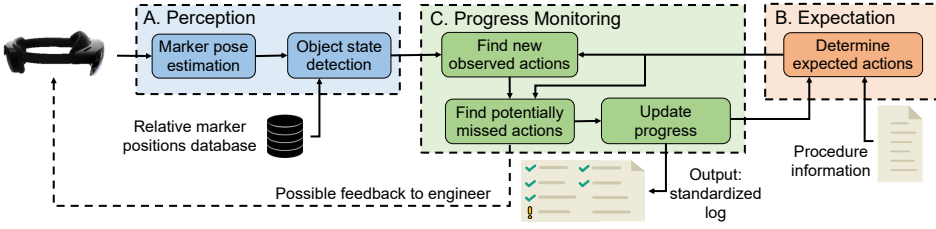


Figure 2.3. Illustration of the proposed method to automatically log manufacturing and maintenance tasks. The approach consists of three main modules, namely perception, expectation, and progress monitoring. The latter automatically generates standardized, easily interpretable logs.

2.4 System prototype

To gain insights into industrial assistive systems within the scope of this thesis, we present a proof-of-concept for an automatic logging system based on visual perception which allows for flexible execution of procedural steps. Fig. 2.3 outlines the proposed method and shows the proposed perception, progress monitoring, and expectation modules. The remainder of this section describes each module in detail.

2.4.1 A. Perception: assembly state detection

For the perception module, we opt for ASD using ArUco markers [68], an approach similar to that proposed by Westerfield *et al.* [46]. The ASD exploits relative poses between markers attached to connecting parts. Consequently, this approach requires markers on all components and suffers when markers are (partially) occluded. However, it is computationally efficient, does not require the collection of a large image dataset or scanning of 3D objects, and does not impose a lengthy retraining of a CNN when minor changes are made to the manufacturing procedure. Our approach only requires a Database D_{rp} of relative poses between markers on correctly assembled parts.

This Database D_{rp} is created by taking a set of images $I_d = \{i_d^0, i_d^1, \dots, i_d^n\}$ of the object with relevant parts properly connected, ensuring that each Marker m has at least one image where the marker is visible for each connecting part. Clearly, not all object parts have meaningful connections with all other parts, requiring a specification of meaningful connections beforehand.

Assuming a relative pose from any object a to b be $p_{a \rightarrow b}$, such that

$$p_{a \rightarrow b} = \{\mathbf{r}_{a \rightarrow b}, \mathbf{t}_{a \rightarrow b}\}, \quad (2.1)$$

where $\mathbf{r}_{a \rightarrow b} \in \mathbb{R}^{1 \times 3}$ is the rotation vector and $\mathbf{t}_{a \rightarrow b} \in \mathbb{R}^{1 \times 3}$ the translation vector from a to b . For the j^{th} image $i_d^j \in I_d$, showing correctly assembled object parts, the pose of each Marker A with respect to the camera C ($p_{A \rightarrow C}$) is calculated, using the ArUco API [68] and the camera's known intrinsic and distortion parameters. Then,

the relative pose from A to each relevant other and correctly assembled Marker B in i_d^j can be determined by solving

$$p_{A \rightarrow B}^* = p_{A \rightarrow C} + p_{B \rightarrow C}^{-1}, \quad (2.2)$$

where $p_{A \rightarrow B}^*$ is the desired relative pose from Marker A to B for correctly assembled parts, and $p_{B \rightarrow C}^{-1}$ the inverse perspective of the relative pose from Marker B to the camera. As p consists of a set of rotation and translation vectors, $p_{B \rightarrow C}$ and $p_{A \rightarrow C}^{-1}$ cannot simply be summed and, instead, two shift-and-rotation transformations are performed, which are specified by

$$\mathbf{r}_{A \rightarrow B}^* = \mathbf{R}^{-1}(\mathbf{R}(\mathbf{r}_{A \rightarrow C}) \cdot \mathbf{R}(\mathbf{r}_{B \rightarrow C}^{-1})), \quad (2.3)$$

$$\mathbf{t}_{A \rightarrow B}^* = \mathbf{R}(\mathbf{r}_{A \rightarrow C}) \cdot \mathbf{t}_{B \rightarrow C} + \mathbf{t}_{A \rightarrow C}, \quad (2.4)$$

where $\mathbf{r}_{A \rightarrow B}^*$ and $\mathbf{t}_{A \rightarrow B}^*$ are the desired rotation and translation vectors for correctly assembled parts, $\mathbf{R}$ denotes the Rodrigues rotation vector to a rotation-matrix transformation and $\mathbf{R}^{-1}$ its inverse [69], [70]. Finally, the relative poses between all markers are stored in matrices.

Having created D_{rp} , an unseen image i can be analyzed to determine the state of an object. First, the frame is converted to gray-scale and afterwards markers are detected. Then, the relative pose of each marker with respect to the camera is calculated, e.g. $p_{A \rightarrow C}$ for Marker A . Next, the relative pose of the detected marker to all relevant other markers are loaded from D_{rp} , e.g. $p_{A \rightarrow B}^*$. Using Eqs. (2.3) and (2.4), the desired pose $p_{B \rightarrow C}^*$ from all relevant markers to the camera is determined. Finally, the expected corner locations B^* of Marker B are projected using $p_{B \rightarrow C}^*$, and compared to the actual detected corners of Marker B . If the average distance per corner between B^* and B is less than τ_c pixels in the horizontal and vertical direction, the parts are deemed properly connected. If the average distance is larger than τ_d pixels, with $\tau_c > \tau_d$, they are deemed disconnected. Otherwise, the observation is considered to be ambiguous and not used. This process is repeated for all markers detected in i .

2.4.2 B. Expectation: extracting procedure information

As previously mentioned, this work aims to log procedural actions performed in a flexible sequence, rather than in a step-by-step pre-defined order. However, perceived object states should still be compared to an expectation of the procedure, based on current progress and procedural information. Therefore, pre-conditions are extracted from work instructions, determining which actions should be completed prior to performing each of those actions. These conditions can be based on physical limitations (one cannot “Place X on Y”, if Y is yet not assembled) or procedural limitations (one should not “Place X on Y” yet, if this prevents an earlier action from being completed). For each action in a work instruction, we store its pre-conditions, whether it is connecting or disconnecting an object part, and the corresponding marker IDs of relevant (object) parts.

2.4.3 C. Progress monitoring

The progress-monitoring module tracks the progress of a procedure by comparing new observations from the perception module to the set of expected actions, provided by the expectation module. When an assembly state change is observed that does not violate any pre-conditions, the progress is updated and a log entry is created. Such a log entry contains a description of the action, a confidence value indicating that the state change has been observed, together with its date and time. If an assembly state change violates a pre-condition, the perception module either has missed a prior state change or the engineer has skipped a procedural step. In order to keep the system user-friendly, the framework assumes it has missed a prior state change and creates a log entry both for the observed action, and the action(s) that should have preceded the observed action. The latter is logged with a confidence value indicating that the change was not observed but explicitly assumed, which should be reviewed at the end of the service action. Once the engineer stops streaming video data, a standardized overview with all log entries is automatically generated, including the involved confidence levels.

2.5 Exploratory experiment

To qualitatively evaluate the performance of the proposed method, an experiment is performed on the task of assembling and disassembling a Duplo block model. This section outlines the implementation details for this experiment, the procedure, and an evaluation of the proposed method.

2.5.1 Implementation details

A HoloLens 2 is used to stream the first-person-view video to a laptop, at a resolution of 720×1280 pixels ($H \times W$) and 30 frames per second, where the live video is processed. The laptop is a T480 with an i7-8650U CPU at 1.90 GHz and 16 GB of RAM, on which the framework is executed with an average computation time of 13.0 ms per frame. To each Duplo model sub-part, a 20×20 -mm ArUco marker from the 4x4_50 ArUco dictionary [68] is placed. For the perception module, thresholds $\tau_c = 10$ pixels and $\tau_d = 50$ pixels are chosen. Given the focal length of the HoloLens 2, these thresholds correspond to a marker at a distance of 1 m being deemed properly connected if its corners are within 1 cm of its desired position, and deemed improperly connected if its corners are not within 5 cm.

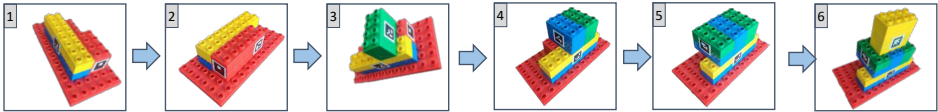


Figure 2.4. Assembly of the block model used in the experiment.

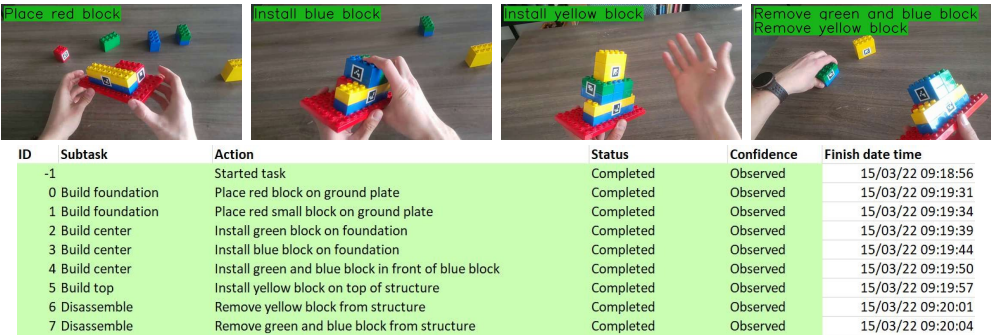


Figure 2.5. Result where all actions were correctly observed in expected order, showing examples frames of the actions (top) and the output log (bottom).

2.5.2 (Dis)assembly procedure

To test the proposed framework, its performance is evaluated on the task of assembling and disassembling a Duplo block model consisting of 15 blocks, divided into 7 subparts. One possible assembly sequence is shown in Fig. 2.4. Whilst the performance of the framework is demonstrated on the relatively simple task of a block model (dis)assembly, it generalizes easily to other objects, since the perception module is based solely on marker detection. Therefore, the framework can be used on a variety of different (dis)assembly tasks, even if they contain traditionally hard-to-detect components, such as objects with a high degree of symmetry or a low degree of texture. The only two requirements are that (1) the objects contain a flat surface of sufficient surface area to place an ArUco marker and that (2) each marker has at least one relevant other marker visible during the procedure.

The database D_{rp} , containing the relative poses of markers on correctly assembled components, is created by assembling the block model and simultaneously determining the relative poses. Therefore, it does not require images to be stored during this process. Pre-conditions for the procedure are created based on physical constraints and one procedural condition, namely to prevent a marker occlusion between Steps 4 and 5 in Fig. 2.4.

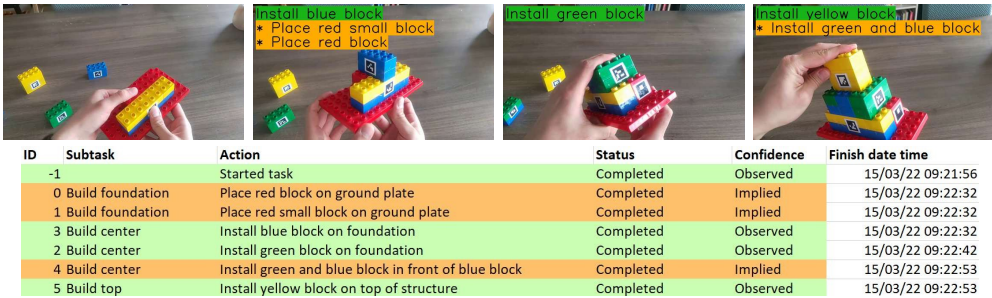


Figure 2.6. Results where action IDs 0, 1, and 4 were not observed (due to occlusions) but implied based on subsequent observed actions. This is explicitly highlighted in the log at the bottom of the figure. In the frames shown in the top of this figure, * indicates an explicitly assumed action.

2.5.3 Evaluation

The performance of the framework is evaluated by qualitatively analyzing its performance on the aforementioned (dis)assembly operation, executed by a single operator. Fig. 2.5 shows the results for one execution, where every action is detected exactly in the order of which the engineer has executed the action. Fig. 2.5 shows the detections in real-time as well as the standardized log generated from this action. Since all actions were observed without any pre-condition being violated, no specific actions are explicitly highlighted.

Fig. 2.6 outlines a scenario where an operator did not show the ArUco markers to the HoloLens on several occasions, specifically the markers corresponding to action IDs 0, 1, and 4. Without detecting the markers, the action cannot be detected. Because the markers of the succeeding actions were being detected, the system successfully assumed which actions were missed by the perception module (or potentially not performed). These missed actions are explicitly highlighted in the log. As shown with these results, the performance of the framework is limited by the visibility of the markers. If the engineer does not rotate the block model with markers of newly placed components facing the camera, these components will not be registered. In such cases, the system has to assume the completion of the corresponding procedure steps.

2.6 Requirements for vision-based assistive systems

The proof-of-concept study confirms that automatic procedure logging is feasible, yet with significant simplifications on the computer vision perception through the use of fiducial markers. Additionally, as demonstrated in the first sections of this chapter, recent works highlight a research gap between three overlapping research fields. Action recognition pipelines aim to classify short video clips, but do not verify whether a step was *completed* and performed *correctly*. Assembly state detection (ASD) approaches either require fiducial markers that are accurate yet fundamentally not scalable, or

neural classifiers that demand complete and unobstructed views, real-world datasets, and costly retraining whenever a part or sequence changes. Finally, instructional AR systems rely on the operator to confirm task completion, forcing strict step-by-step sequences and manual “next-step” triggers that disregard the objective state of the procedure. These limitations point to missing capabilities at the intersection of perception, procedural reasoning, and user-centered deployment. Therefore, the remainder of this section defines requirements for systems in this new space of automated procedure logging. The requirements are divided into three groups, namely *perception*, *procedural reasoning*, and *user acceptance*.

Prior to enumerating the individual requirements, the intended scope of the ensuing lists is first clarified. They are not intended to constitute a comprehensive enumeration of all possible requirements. Instead, they constitute a prioritized subset of issues that, in our assessment, constitute the principal obstacles to the deployment of vision-based assistive systems within high-precision, low-volume manufacturing environments. Each specified requirement is addressed by at least one subsequent chapter, while topics lying beyond this immediate scope are acknowledged in the final chapter as avenues for future investigation.

2.6.1 Perception

For vision-based systems to support procedural monitoring, perception should extend beyond object detection to progress and correctness under constraints typical of industrial settings.

- *Completeness & correctness detection.* Recognition should extend beyond verb–object labels to confirm that the physical outcome of a step aligns with the procedural intent, delivering objective feedback on the *completion* and *correctness* of every step within the procedure.
- *Scalability.* The final solution should operate without fiducial markers or fixed camera rigs, instead relying on algorithms that generalize across components. Furthermore, it should be able to operate (without laborious retraining or fine-tuning) even when procedures are updated or parts are upgraded.
- *Real-time processing.* The system should be able to provide feedback to the user within a reasonable time, *i.e.* seconds, rather than hours, after the completion of a specific step. Additionally, this avoids video storage of IP-sensitive data, since frames are processed in real time and therefore do not need to be recorded.
- *Generalization from synthetic to real data.* Because high-precision, low-volume machines are frequently one-off builds, operation on the first field execution is essential, which can be enabled by synthetic data and procedural priors. To this end, the CAD models provide particularly valuable data, as they are highly descriptive and widely available within industry.

- *Occlusion tolerance.* Inference should succeed when only the subset of discriminative parts which is required to confirm the correct completion of a procedural step is visible, regardless of hand or tool occlusions common to the ego-centric viewpoint.

The perception-oriented requirements are addressed throughout the thesis. Chapter 3 defines the task and accompanying metrics to verify the completeness and correctness of procedural steps. Chapter 4 introduces strategies that drive scalability by removing the need to re-train neural networks for each minor update in procedures or objects. Chapter 5 and Chapter 7 contain neural networks that are trained exclusively on synthetic data, and Chapter 6 demonstrates robust recognition even under partial occlusion. The remaining requirement regarding real-time processing is considered in every chapter.

2.6.2 Procedural reasoning

When automatically understanding procedures and providing advanced support to operators, the first objective is to generate machine-readable logs that are also interpretable for engineers, while remaining resilient to unseen failure modes.

- *Searchable and interpretable logs.* Detected procedure steps should be translated into a timeline that is both machine-readable and readily interpreted by engineers.
- *Exploitation of prior knowledge.* CAD models and (digital) work instructions encode strong spatial and temporal constraints. Leveraging them shrinks the search space, which can overcome the lack of real-world data for robust industrial use.
- *Flexible sequence recovery.* The system should recognize the *actual* order of executed steps, even when it differs from the prescribed procedure order. Consequently, downstream analytics can decide on correctness.
- *Implicit step inference.* Unobserved but logically required actions should be logged with explicit confidence tags, *i.e.* indicating that an action was assumed to be completed, but not perceptually confirmed.
- *Handling unseen errors.* The system should be able to handle novel errors, that are not defined prior to model training; many errors cannot be foreseen beforehand, and the errors that are unexpected are frequently the most costly.

The handling of previously unseen errors is specifically addressed in Chapter 5, where an algorithm is proposed to localize previously unseen errors. The requirements of searchable logs, exploitation of prior knowledge, flexible sequence recovery and implicit inference, are integrated into the analyses throughout the other chapters.

2.6.3 User acceptance

Regardless of technical accuracy, assistive systems will not be adopted unless they align with human workflows, cognitive constraints, and user preferences.

- *Granularity of feedback.* Warnings for potential mistakes in the procedure execution should pinpoint where the mistake occurred, either in time or in space, thereby increasing the interpretability of the system.
- *User-dependent guidance.* The level of feedback provided by the system should adapt to the expertise level of operators to prevent unnecessary increases in cognitive load.
- *Real-world system performance.* The algorithms should not only be evaluated on carefully curated datasets, but also be evaluated in a practical setting to quantify the real-world performance.

The human-centered requirements are explored in depth in Chapter 7, that analyzes how feedback granularity, expertise-adaptive guidance and real-world performance interact in a controlled user study.

The aforementioned requirements serve as a guidance for the remainder of this thesis. Subsequent chapters incrementally address them, progressively relaxing the simplifying assumptions while adding capability.

2.7 Conclusions

This chapter has outlined the fundamental capability of algorithms that aim to provide advanced automated support during procedural actions, namely the ability to automatically log procedure actions in (near) real time. This chapter has also provided an overview of the research areas related to this task (action recognition, assembly state detection, and AR systems) and highlighted the gap between these three fields that automated action logging can occupy. Finally, this chapter has explored a proof-of-concept system for automated procedure step logging, after which a set of requirements has been formulated.

Findings. This chapter showed that automated logging is a foundational step for assistive systems, identified a research gap between related fields, and derived core system requirements from a proof-of-concept implementation. The proposed perception-based framework automatically generates logs based on a head-worn AR device. In such a setting, engineers no longer have to step away from their work to perform the tedious yet important task of logging performed actions. The log highlights the steps that the framework has not visually confirmed, allowing engineers to easily review those steps after the execution is completed. The framework does not restrict engineers to rigid, step-by-step sequences. Instead, it allows engineers to

execute tasks in an order they deem suitable. As the logs are stored in standardized form, they are searchable and can therefore be used to perform root-cause analysis in case of failures, discover best practices, and optimize servicing processes.

Limitations. The proof-of-concept system presented in this chapter adopts several simplifying assumptions that limit its immediate applicability. First, it depends on ArUco fiducial markers on every component. While this expedites pose estimation, it is impractical for large-scale deployments. Second, the prototype is evaluated only on a deliberately simplified Duplo (dis)assembly task. Third, assessment solely relies on qualitative inspection of the generated logs. These constraints clearly prevent direct industrial use. However, the purpose of this presented proof-of-concept is to provide a controlled environment which identifies the fundamental capabilities that assistive systems should incorporate, to provide automated support during procedural actions. By isolating the core challenges, the chapter sets the research direction for the remainder of the thesis.

In conclusion, this chapter paves the way for research on recognizing industrial procedural actions, based on a prior knowledge of what is to be expected during procedures. The potential of recognizing such actions is not limited to automatically generating service logs, but can be extended, *e.g.* to provide warnings of forgotten or incorrectly executed steps in real time, thereby reducing the number of procedural errors, increasing the quality of the procedure outcome.

This chapter has addressed the necessity of automated logging of procedural steps and the research gap in the current body of literature. A proof-of-concept, automated logging system has exposed fundamental challenges that the remaining chapters of this thesis will discuss. The most pressing of these challenges is addressed in the following chapter, in which the new task of procedure step recognition is defined, along with a set of three evaluation metrics, a benchmark performance and a new, industrially relevant dataset.

Procedure step recognition in ego-centric videos

3.1 Introduction

The previous chapter has outlined a proof-of-concept system to automatically recognize steps in an assembly procedure, relying on fiducial markers placed on the components. Furthermore, the chapter has provided the groundwork of this thesis by identifying a critical gap between action recognition (ACR), assembly-state detection (ASD), and instructional augmented reality (AR) systems. Although these fields are individually explored, the overlap between them is particularly relevant from an industrial perspective, both in terms of available datasets and task formulation. Therefore, this chapter focuses on addressing that overlap, by proposing a new task within the space of procedural activity recognition, and presenting an accompanying dataset.

As highlighted in the previous chapter, an algorithm becomes of significant value if it can understand procedural actions by automatically recognizing and tracking steps during the execution of a procedure, *i.e.* a given set of instructions, describing the procedural actions required to correctly complete a task. A system containing such an algorithm can warn users about potential mistakes or forgotten steps [71], and summarize the execution of a procedure, thereby reducing or even potentially eliminating the need for manual logbook keeping [72]. Understanding procedures is not only highly relevant for industrial tasks, but to a broad range of procedural tasks, such as tracking the stages of surgeries to enhance post-surgical assessment and optimizing procedure workflow [73], [74]. However, as outlined in Section 2.7, such a system should be readily incorporated within the existing environment using advanced computer vision algorithms, rather than rely on *e.g.* fiducial markers.

Automated understanding of procedures with computer vision is challenging for various reasons. Firstly, there is often a limited amount of visual difference between subsequent steps. For instance, determining whether a screw is correctly mounted into a specific component requires a fine-grained visual understanding. Additionally,

there can be a high degree of symmetry and a low degree of textural variation for objects within industrial procedures [75]. Secondly, it is not feasible to collect a vast amount of data for many procedures, as they are often infrequently performed and rather specialized. Finally, procedures can be typically completed correctly via several possible execution orders. For instance, when replacing the tires of a vehicle, it is proper to remove and replace each tire one by one, but it is also appropriate to first remove all tires, and then replace them all. Therefore, it is frequently not sufficient to merely look whether a single, pre-defined step is completed correctly.

We pose that these aforementioned challenges can be effectively addressed by defining a task that combines video recordings of procedure executions with explicit procedural knowledge. Building upon this hypothesis, this chapter specifically addresses the following research questions.

- RQ3.1 *Procedure understanding task definition.* How can a task be defined, such that it combines visual perception and explicit procedural knowledge to reliably recognize (unique sequences of) correctly completed steps in procedural videos?
- RQ3.2 *Dataset.* How can a dataset be created to effectively capture the variability in procedure executions, error types, and data modalities typical to industrial procedures?
- RQ3.3 *Benchmark.* To what extent can existing state-of-the-art methods from related tasks generalize to this newly proposed procedure understanding task?

To address the aforementioned research questions, this work formally defines the novel task of *procedure step recognition* (PSR), resolving limitations within the existing body of literature, *i.e.* the reliance on large-scale datasets, the lack of a measure of completeness [76], [77], and the simplification to five or six parts used in the assembly [62], [64], [78]. Additionally, the IndustReal dataset is introduced, a publicly available dataset towards solving PSR. It is an ego-centric, multi-modal dataset where 27 participants are challenged to perform assembly and maintenance procedures on a construction-toy car, based on STEMFIE [79] and demonstrated in Fig. 3.1. The videos are annotated and benchmarked for PSR, as well as ACR, ASD. The IndustReal dataset features four novel aspects:

- *Variety of execution errors.* IndustReal features 38 errors, of which 14 are exclusive to the validation and test sets. Whilst some datasets already include procedural errors (*e.g.* omissions) [77], [80], IndustReal is the first to also include execution errors (*e.g.* wrong type of nut used).
- *Subgoal-oriented execution.* Other datasets are either open-ended assemblies [77] or contain a strict, step-by-step execution order [76]. IndustReal combines these execution types with a subgoal-oriented assembly style, where participants are given flexibility to determine the execution order between subgoals. This

approach more closely resembles industrial procedures, since it maintains a hierarchy in procedure execution whilst allowing for flexibility where possible. The IndustReal data contains 48 unique execution orders.

- *Open-source geometries.* Scalability is an important factor for many industrial tasks, where simultaneously the technical drawings are often available. Therefore, 3D models for all parts are published to stimulate use of synthetic data in procedural action understanding, *e.g.* by sim2real¹ domain adaptation or generalization.
- *3D printed parts.* To ensure reproducibility, future availability of the model, and growth via community effort, all parts are 3D printed and open source [79].

In summary, the main contributions of this chapter are (i) the definition of the PSR task, (ii) the creation and release of the IndustReal dataset to support PSR and related tasks, and (iii) a benchmarking study to assess the suitability of existing methods on this novel task.

The content of this chapter is divided as follows. In Section 3.2, the related work and comparable datasets are outlined. The formal definition of the PSR task is provided in Section 3.3, followed by the presentation of the IndustReal dataset in Section 3.4. Subsequently, Section 3.5 provides a benchmark performance on ACR, ASD, and PSR, after which the conclusions are discussed in Section 3.6.

3.2 Related work

This section reviews the existing work on procedural action understanding, assembly state detection, and datasets relevant for these tasks. Current limitations in these areas are highlighted, motivating the need for PSR and the IndustReal dataset.

3.2.1 Work on procedural action understanding

In (procedural) action recognition tasks, the objective is to classify a video clip into a set of activities expected within a certain procedure [62], [64], [78], [81]. Examples of such procedures are cooking [56], [82], [83], instruction videos [51], [84], and assembly tasks [85], [86]. Recently, focus has been directed towards recognizing procedural actions in industrial-like settings [76], [77], [80], [87], [88].

Whilst recognizing actions in procedural videos is certainly of some interest in an industrial setting, we argue that it is more valuable to know what was actually completed, rather than only performed (and therefore potentially not finished). Therefore, the PSR task is proposed to recognize the correct completion of steps and to track the order in which those steps were completed.

¹Sim2real is machine learning and robotics jargon for adapting models from simulated to real-world data.

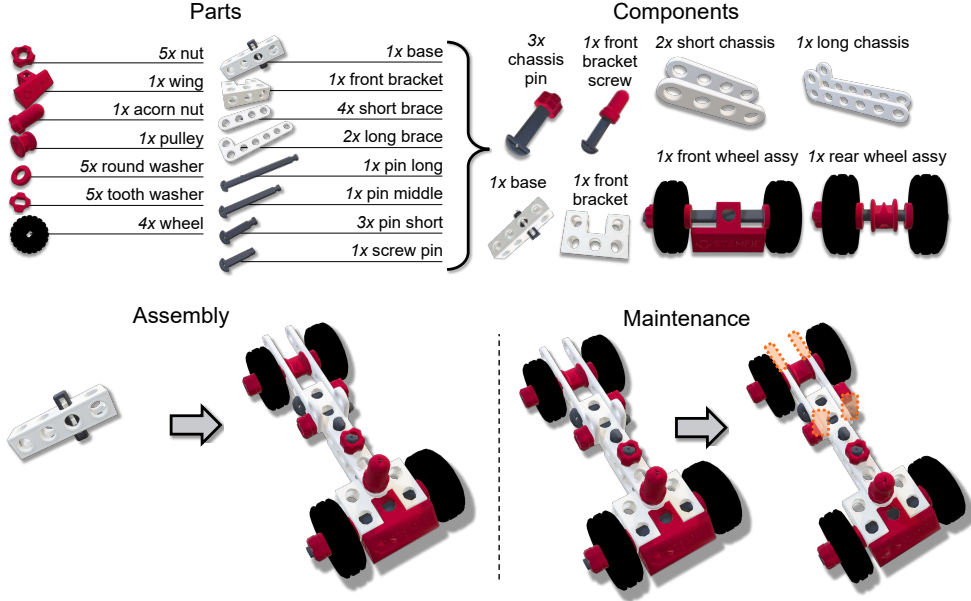


Figure 3.1. Overview of the construction-toy car's (3D printed) parts, components, and models used in IndustReal. The maintenance task represents a component upgrade and consists of replacing the long braces with short braces on the rear chassis.

Additionally, the comparable datasets outlined in Table 3.1 generally have restricted sizes, since it is challenging to record sufficient video data of people executing (nearly) the same procedural task. This is especially relevant to industrial applications, where a significant portion of the tasks is performed rather infrequently. However, procedural knowledge, in the form of work instructions or technical drawings, is generally available for those tasks. Whilst such procedural knowledge is not explicitly leveraged by the aforementioned approaches, we advocate to explicitly use this knowledge in PSR to constrain the number of possible actions expected at any given moment.

3.2.2 Literature on assembly state detection

Assembly state detection is a sub-task of object detection, where the objective is to recognize and locate the specific state of an object during an assembly procedure [47], [62]–[64], [78], [89]. ASD is significantly more challenging than object detection, since the visual variability between two states is often small. For instance, detecting the object “Car” is significantly less complicated than detecting the state “Car without window”. Su *et al.* [62] simultaneously detect the state and the pose of a coffee machine during its assembly. The authors demonstrate a good performance on the task, but the different assembly states consist of visually distinctive objects. Lui *et al.* [64]

demonstrate an attention mechanism for their convolutional neural network (CNN), detecting object states even if they are visually similar to each other. Both works demonstrate promising advances on this task. The approaches are trained predominantly on synthetic data, which can be readily generated if 3D models of each assembly state are already designed.

Nevertheless, the aforementioned approaches have two important limitations. Firstly, the algorithms must be trained on a dataset of images from each object state. This forces the models to learn a low-dimensional representation for each state. Since the procedures that the authors selected consist of only 5 parts and, at most, 6 distinctive object states, training a model in such a manner is feasible. However, it remains unclear whether this approach scales to procedures with higher complexity and more object states. Secondly, no procedural information is leveraged by either approach. Therefore, the models expect for example, the initial state in an assembly equally as much as the very last state. For tasks with sufficient visual distinction between states, this is not a critical limitation. However, for objects that appear identical from certain viewpoints for several different states, procedural information could be used to determine the most likely state. Finally, none of the above-mentioned approaches release their (test) data.

3.2.3 Published comparable datasets

The placement of IndustReal within publicly available datasets, recorded in industrial-like settings is shown in Table 3.1. The MECCANO dataset [76], [90] is most relevant to IndustReal in terms of procedure, task complexity, and dataset size. Notable differences between these two datasets are that in MECCANO users follow strict, step-by-step instructions, limiting the variety in execution order. Secondly, although MECCANO contains some procedure errors, they are nearly always corrected in the subsequent step. Therefore, later states do not have prior errors in them, whilst such cases are certainly of interest and are frequently encountered in the industrial domain. Finally, we argue that the availability of 3D models of all parts represented in the dataset is crucial for industrial applications, given the wide availability and usage of technical drawings in that domain. Unfortunately, the MECCANO dataset uses an IP-protected construction set, prohibiting the publication of such CAD models. BRIO-TA [80] and the large-scale Assembly101 [77] datasets both contain procedural mistakes, such as omissions and incorrect execution order, but do not contain (labeled) execution errors. Furthermore, none of the mentioned datasets provide labels for ASD. Lastly, to ensure future availability of the models, all parts in IndustReal are 3D printed. An added benefit of 3D printing is that it allows researchers to print in different scale, colors, or materials, to test the limits of their algorithms.

Dataset	Year	Ego	Tasks			Procedural complex.				3DM	Dataset size		
			ACR	ASD	PSR	Flex.	PEs	EEs	Parts		Seqs.	Dur.	Par.
IKEA ASM [88]	2021	✗	✓	✗	✗	✓	✗	✗	7	✗	381	35.3h	48
MECCANO [76], [90]	2021	✓	✓	✗	✗	✗	✓*	✗	49	✗	20	6.9h	20
Assembly101 [77]	2022	✓	✓	✗	✗	✓	✓	✓*	15	✗	1.01K	167.0h	53
BRIO-TA [80]	2022	✗	✓	✗	✗	✓	✓	✗	10	✗	75	2.9h	15
HA4M [91]	2022	✗	✓	✗	✗	✓	✗	✗	17	✓	217	5.9h	41
ATTACH [87]	2023	✗	✓	✗	✗	✓	✗	✗	26	✗	378	17.2h	42
IndustReal (prop.)	2024	✓	✓	✓	✓	✓	✓	✓	36	✓	84	5.8h	27

Table 3.1. Comparison of industrial-like procedural understanding datasets. ACR: action recognition, ASD: assembly state detection, PSR: procedure step recognition, flex.: more than one single execution order for the task, PEs: procedural errors (omission, execution order), EEs: execution errors (component installed incorrectly), 3DM: 3D models made publicly available, Seqs.: sequences, Dur.: duration, Par.: number of participants. *rare and not explicitly annotated.

3.3 Procedure step recognition

The previous section has identified a gap in the related work between ACR and ASD. By formalizing the task of procedure step recognition (PSR), along with an evaluation scheme, we encourage researchers to develop methods to automatically recognize the completion of steps, rather than the (partial) execution of activities. Additionally, PSR systems should explicitly leverage procedural knowledge and allow a flexible execution order for procedural tasks, when the procedure allows it.

3.3.1 Task definition

The objective of PSR is to extract an estimate of all procedure steps correctly performed by a person up to time t , based on sensory inputs $X_t = (x_t, x_{t-1}, \dots, x_{t-h})$ and a descriptive set of the procedural actions to be performed $\mathcal{P} = \{a_0, a_1, \dots, a_n\}$. Here, h is the observation horizon and $n + 1$ the total number of actions $a_i \in \mathcal{P}$ covered in the procedure. The predicted completed procedure steps $\hat{y}_t$ at time t , given some model $\mathcal{F}$, are defined such that

$$\hat{y}_t = \mathcal{F}(X_t, \mathcal{P}). \quad (3.1)$$

Here, a “predicted” (recognized) procedure step does not refer to the prediction of a future step, but rather the prediction by a model for a correctly completed step, based on the given inputs. Crucially, this definition allows for real-time operation, since contrary to existing tasks, PSR does not require a full recording of the procedure as input [51], [92].

Each unique action $a_i \in \mathcal{P}$ contains information regarding this specific action, and can be different for varying approaches to PSR. For instance, a_i can contain a description, an image of what the completed step is supposed to look like, or relative positions between components. Sensory inputs X_t may comprise of camera images, depth maps, or even user interaction with the system, during the execution of a procedure. An example of the latter would be to prompt the user to confirm whether the recognition of a step is correct. This can be especially valuable when the system makes a recognition with a low degree of confidence. Note that with the given definition, a PSR system can be an ensemble of various computer vision algorithms, *e.g.* object detection to determine the assembly state and action recognition to determine the activities.

The predicted procedure steps form an ordered list of elements, describing the completed steps and is defined as

$$\hat{y}_t = (\hat{s}_{\sigma(0)}, \hat{s}_{\sigma(1)}, \dots, \hat{s}_{\sigma(m)}), \quad (3.2)$$

where step $\hat{s}_{\sigma(i)}$ is the predicted completion of the action $a_i \in \mathcal{P}$, at prediction time $\hat{t}_{\sigma(i)}$, having a prediction confidence $c_{\sigma(i)}$, with $m + 1$ the total number of recognized procedure steps. The function σ maps a completed step $\hat{s}_j \in \hat{y}_t$ to the corresponding

action $a_i \in \mathcal{P}$. Therefore, the first predicted step $\hat{s}_{\sigma(0)}$ is not necessarily the completion of the action $a_0 \in \mathcal{P}$, but rather the first recognized step that a person completed.

The ground-truth execution order y_t at time t describes the order in which the steps are actually completed, which can differ from the prescribed order in $\mathcal{P}$, and is defined as

$$y_t = (s_{\rho(0)}, s_{\rho(1)}, \dots, s_{\rho(k)}), \quad (3.3)$$

where $s_{\rho(i)}$ is the completion of the action $a_i \in \mathcal{P}$ at time $t_{\rho(i)}$. The value $k + 1$ is the total number of actions completed and ρ a function that maps the completed steps from y_t to the steps described in $\mathcal{P}$. If the order of the steps predicted in $\hat{y}_t$ equals that of y_t , it follows that $\sigma = \rho$, signifying a perfect execution order prediction.

3.3.2 Evaluation metrics

To quantify the performance of a PSR system, three evaluation metrics are proposed. These metrics focus on predicting the procedure steps in the correct order, the number of false predictions, and the timeliness of the predictions.

Metric 1: procedure order similarity. We propose to measure the quality of a predicted sequence order for an entire recording, $\hat{y}$ (e.g., ‘ACB’), by comparing it with a similarity measure with respect to the ground truth y (e.g., ‘ABC’). This is approached as a string similarity problem, a common problem in spelling error detection [93], [94], since words consist of a sequence of characters, where order and type of character matters. In temporal action segmentation, the Levenshtein (Lev) distance, normalized over the length of the ground-truth sequence, is commonly used [92]. We propose two changes to this metric, by (1) eliminating substitution from the edit distance, preventing the metric from favoring models with many false positives, and (2) using the Damerau-Levenshtein (DamLev) [95], rather than the Lev distance because it penalizes transpositions less, since it is intuitive to penalize “ACB” less compared to “CAB”.

In contrast to Lea *et al.* [92], we propose to normalize the edit distance with respect to the length of the ground truth, rather than the length of either the ground truth or the prediction, depending on which is longer. This prevents models with many false positives from being normalized favorably. Finally, the normalized edit distance is subtracted from the unity value, resulting in a similarity metric, rather than a distance metric. Thus, the procedure order similarity (POS) between y and $\hat{y}$ is defined as

$$\text{POS} = 1 - \min\left(\frac{\text{DamLev}(y, \hat{y})}{|y|}, 1\right), \quad (3.4)$$

where $\text{DamLev}(\cdot)$ is a weighted DamLev edit distance function. Further clarification on POS may be found in the supplementary material.

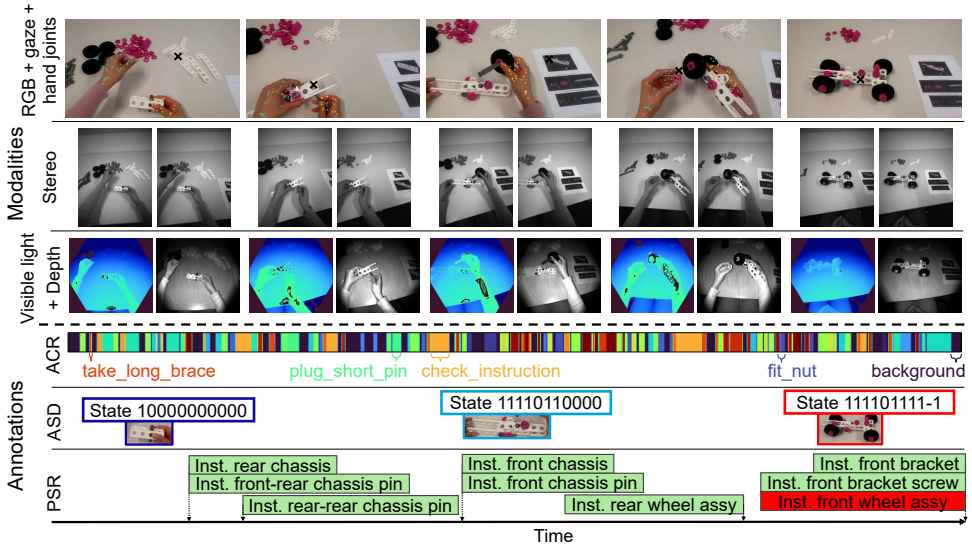


Figure 3.2. Samples from a clip in the IndustReal dataset, demonstrating the modalities and annotations for all three tasks. Gaze is indicated by the cross, detected hand joints by the dots. ACR: action recognition, ASD: assembly state detection, PSR: procedure step recognition.

Metric 2: F_1 score. A false positive (FP) is defined as a procedure step $\hat{s}_{\sigma(j)}$ that is predicted prior to the actual completion of action a_i , or if a_i is not at all completed, hence

$$(\hat{t}_{\sigma(j)} < t_{\rho(i)}) \vee (a_i \notin y). \quad (3.5)$$

A false negative (FN) is defined as a step $s_{\rho(j)}$ for a corresponding action a_i , that has indeed been completed, but is not represented in $\hat{y}$, so that

$$(a_i \in y) \wedge (a_i \notin \hat{y}). \quad (3.6)$$

Finally, a true positive (TP) is defined as the prediction of a procedure step $\hat{s}_{\sigma(j)}$, that is observed at or after the actual completion of a_i , such that

$$(\hat{t}_{\sigma(j)} \geq t_{\rho(i)}) \wedge (a_i \in y). \quad (3.7)$$

PSR systems can explicitly assume that actions have been completed based on procedural information, without relying exclusively on sensory recognition. Therefore, a distinction can be made between F_1 score at the recognition and system level. Since this definition does not contain any time restriction on true positives, procedure steps that are recognized long after the step completion are not penalized. Therefore, comparing only the procedure order similarity and F_1 score lacks a temporal component.

Metric 3: average delay. To complement the aforementioned metrics with a temporal component, the average delay τ is introduced, quantifying the time between the ground-truth completion and corresponding recognition of a step. False negatives, defined by Eq. (3.6), have an undefined delay, and similarly, false positives (defined in Eq. (3.5)) have either an undefined delay, or a negative delay. Therefore, FPs and FNs are excluded from τ , defines as

$$\tau = \frac{1}{h} \sum_{i=0}^{h-1} (\hat{t}_{\sigma(i)} - t_{\rho(i)}), \quad (3.8)$$

where n is the number of total TPs in y . Because FPs and FNs are discarded in the delay, only the combination of the three proposed metrics provides a valuable insight into the performance of a system towards solving PSR.

3.4 IndustReal dataset

This section details the IndustReal dataset, including its assembly assets, procedures, recording setup, participant protocol, and annotations

3.4.1 Construction-set car procedures

Figure 3.1 demonstrates the 36 part models used in the IndustReal dataset, based on the STEMFIE construction-toy car [79]. The model has significant complexity, consisting of multiple types of washers, pins, and braces. Some components require screwing, others need tightening, and participants frequently have to use both hands. Two procedures are defined, an assembly task, where the car has to be built from scratch, and a maintenance task, where the participants have to replace part of the rear chassis of the toy car. Printed instructions are provided in a subgoal-oriented manner, meaning that the participant builds towards subgoals rather than executing strict, step-by-step instructions, or “free-style” building towards a final assembly. Unlike related datasets, participants are allowed to create subassemblies, as commonly encountered in industrial settings.

All of the parts are 3D printed on an Ultimaker S5 at 200% scale, layer height of 0.3 mm, 15% infill, a print speed of 50 mm/s, and PLA filament and the colors white, silver metallic, magenta and black. All parts, as well as the final models, are published together with the dataset, since part geometries are commonly available in industrial settings, *e.g.* CAD models.

3.4.2 Recording and setting

The HoloLens 2 (HL2) [96] mixed reality headset is used as recording rig for the dataset. The front-facing RGB camera records at a resolution of 1280×720 pixels and

the stereo cameras provide images at 480×620 pixels. The long-throw depth and IR sensors record at a resolution of 320×288 pixels and contain normalized values. Next to the images, we also record gaze, hand, and head-pose tracking, provided by the HL2 algorithms. All sensors, visualized in Fig. 3.2, are sampled at 10 fps, except from the depth and IR sensors, as these are limited to 5 fps in hardware. The data are sent from the HL2 to a server in real time, using the HL2SS library [97].

The dataset is recorded in a setting with consistent background and lighting conditions. More specifically, participants are asked to perform the procedures on a white desk placed against a white background. Such lack of variety in background and lighting conditions is assumed to be consistent with industrial settings. Redundant washers, nuts, screws, and pins are placed amongst the required parts to reduce bias towards detecting unused parts. Prior to each recording, all parts are sorted into heaps according to color, and then placed randomly on the desk.

3.4.3 Participants and protocol

In total, 27 participants were recruited. Each participant signed a consent form and the experiment was approved by the institution's Ethical Review Board (ERB). Each participant was asked to perform the assembly procedure once, without recording, to familiarize themselves with the construction-set car. During this practice assembly, feedback was provided to the participants when required. Subsequently, the HL2 was fitted to the participant and the gaze tracking was calibrated. Then, each participant was assigned a correct assembly instruction, a correct maintenance instruction, and one or two instructions with errors, to ensure that a variety of mistakes were introduced. As anticipated, the participants exhibited errors even when they were provided with the correct instructions, and these mistakes were annotated as well. The recorded errors in the dataset vary from minor, difficult-to-observe mistakes (e.g. installing the wheels without washers), to large mistakes, such as forgetting a wheel. To minimize "learners bias" of the participants in the dataset, each participant was given the instructions in a random order. Participants were asked to remove their hands from the assembly after completing a step, to ensure a minimally occluded view on the assembly state. Finally, a horizontal line was projected in the participants' field of view, indicating the lower bound of the RGB camera's view.

3.4.4 Annotation

Since IndustReal is intended to be closely representative of industrial use cases, significant importance is given to evaluation and robustness to unseen, out-of-distribution errors. Additionally, a large real-world test set is desired when training exclusively on synthetic data. Therefore, a dataset split of 12/5/10 (No. of train/val/test participants) is chosen, which is heavily focused on the test set. Furthermore, numerous errors are exclusively present in the validation or test sets. Note that the split is made on

participants, rather than videos, to ensure sufficient variation in viewpoint, execution, and head movements between the three sets.

Although IndustReal is specifically presented to address PSR, annotations are also provided for ACR and ASD, enabling researchers to use IndustReal for various tasks, or to combine tasks for better PSR performance in future work.

A Action recognition

ACR labels consist of the frame at which the action starts, ends, and the combination of a verb and noun. Given the resemblance between MECCANO [76] and the construction-toy car used in IndustReal, we adopt the same verbs as utilized in MECCANO: *take, put, align, plug, pull, screw, unscrew, tighten, loosen, fit, check, and browse*. These verbs, the component names described in Fig. 3.1, and the additional nouns *objects, partial model, and instruction*, are used to create the annotations. Annotators were instructed to mark the action start when the participants initiate the action, rather than when participants touch the relevant object(s). Figure A.1 in Appendix A demonstrates the action class instances and long-tail distribution for action recognition annotations in the IndustReal dataset. The *check_instruction* action class is the most common, followed by *align_objects*. All actions were annotated using the ELAN tool [98].

Two annotators annotated the entire dataset, and each annotator reviewed the annotations of the other. In total, 75 fine-grained action classes with a total of 9,273 instances were annotated. The average action lasts 1.9 ± 1.4 seconds. A long-tail distribution is observed, with 80% of the data containing 29.3% of the actions, and provided together with more statistical details in the supplementary materials. Recordings have an average of 110 ± 38 actions per video, with 134 ± 32 actions per assembly and 79 ± 13 actions maintenance video. Participants frequently perform actions simultaneously, resulting in 24.2% of all instances having an overlap with at least one other action.

B Assembly state detection

To the best of our knowledge, IndustReal is the first publicly available dataset with ASD annotations. Because the participants have flexibility in execution order, all labeled states must be explicitly defined. We label the assembly states with integers, where a “1” indicates that a component at that index has been correctly installed and a “0” indicates that it has not (yet) been correctly installed. Additionally, we assign “-1” to components that have been incorrectly installed. Rather than labeling each assembly part individually with such a code, we divide the toy car into 11 components (in order): *base, front chassis, front chassis pin, rear chassis, short-rear chassis, front-rear chassis pin, rear-rear chassis pin, front bracket, front bracket screw, front wheel assy, and rear*

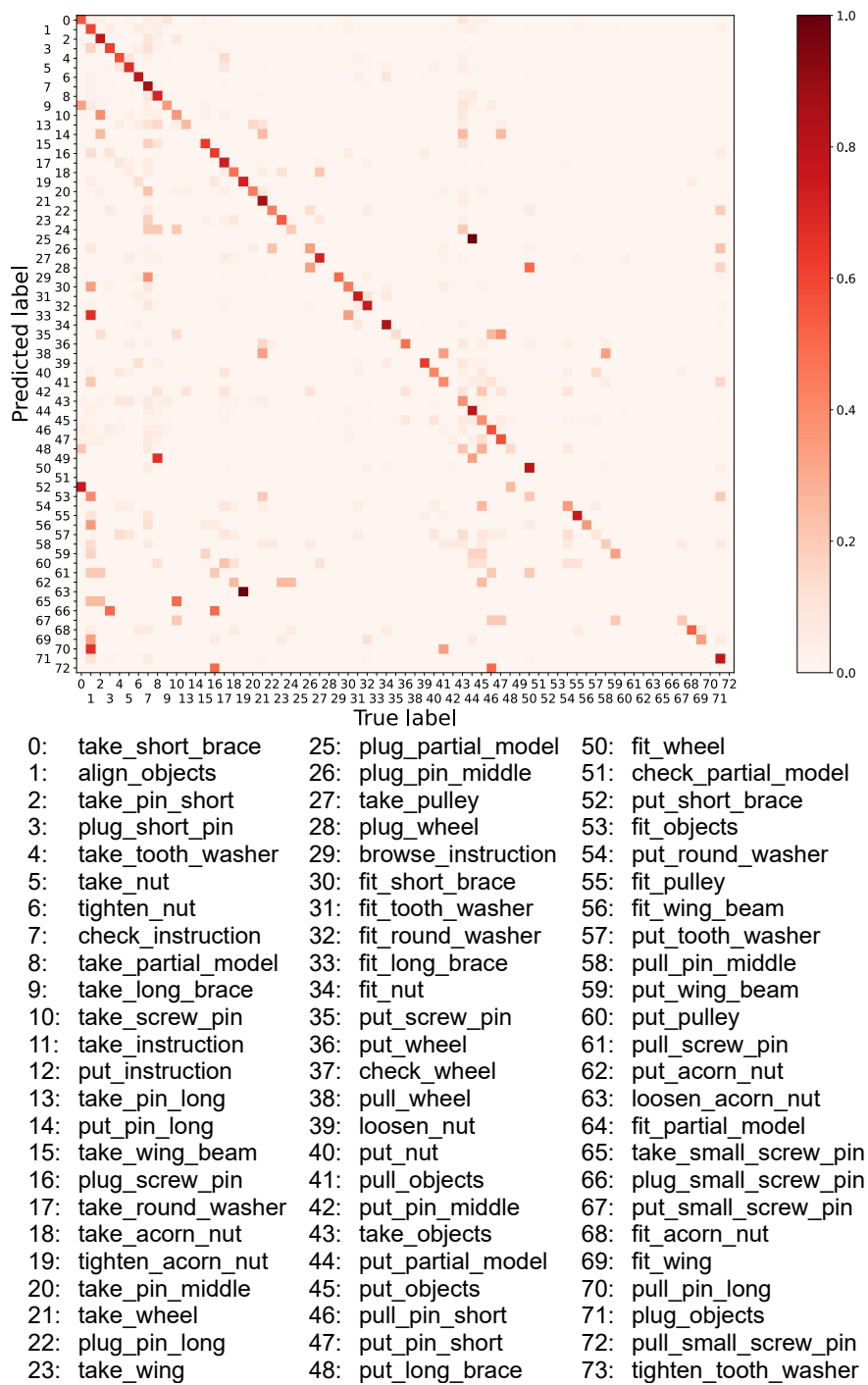


Figure 3.3. Normalized confusion matrix for action recognition on IndustReal using the MViTv2 [99] architecture and pretraining on Kinetics [18].

wheel assy (see Fig. 3.1). For example, the assembly state *11100000000* consists of a correctly installed base, front chassis and front chassis pin.

Bounding boxes and labels are provided for all 22 labeled (defined) states in IndustReal, as well as 27 different error states. Intermediate states, which occur during the assembly of components, are not labeled, as outlined in Fig. 3.2. Whilst those states could be annotated as *partial model*, such labels empirically appear to hold little value. Intermediate states differ from erroneous states in that participants in intermediate states are actively progressing towards completing a state, whereas participants in erroneous states are not trying to further complete that step. In total, 26.9K video frames (13% of total) are annotated for ASD, of which 3,569 frames show error states.

C Procedure step recognition

The PSR labels consist of the frame at which a step completion occurs and the new assembly state, as defined in the previous section. The difference between two assembly states can directly be used to determine which procedure steps are completed. A procedure step is defined as *completed* when the component relating to that step is correctly installed, which includes actions such as the tightening of a nut. Although the explicit detection of incorrect step execution is not included in the PSR task definition (Section 3.3.1), annotations of incorrect assembly states are included for qualitative analysis and future work. Therefore, two sets of PSR labels are provided, one with only correctly executed procedure steps (e.g. “Installed front chassis”), and one that also includes incorrectly completed steps, such as “Incorrectly installed front wing”. An example of a PSR-annotated sequence is outlined in Fig. 3.2. In total, IndustReal consists of 724 correct procedure step completions (8.6 ± 1.2 correct completions per recording) and 38 incorrect step completions. In total, 35 videos (42%) contain a missing or incorrectly completed procedure step. IndustReal contains 22 different correctly completed procedure execution orders, plus an additional 26 different execution orders containing error states.

3.5 Benchmark experiments

This section outlines the benchmark performance of state-of-the-art approaches to ACR and ASD on IndustReal. Furthermore, a PSR benchmark implementation is outlined and evaluated, providing a baseline performance for more sophisticated approaches towards this task.

3.5.1 Action recognition benchmark

Definition. Given a video segment $X_i = [x_{ts_i}, x_{te_i}]$ and a set of action classes $C_a = \{c_0, c_1, \dots, c_n\}$, the objective of action recognition is to classify segment X_i to the correct

Model	Modalities	Top-1 acc. [%] $\uparrow$	Top-5 acc. [%] $\uparrow$
SlowFast [100] [†]	RGB	60.39	85.21
SlowFast [100] [*]	RGB	57.83	82.87
SlowFast [100] [†]	Depth	43.20	73.98
SlowFast [100] [†]	Visible light	53.75	81.48
SlowFast [100] [†]	Stereo images	57.72	83.03
MViTv2 [99] [†]	RGB	<u>65.25</u>	<u>87.93</u>
MViTv2 [99] [*]	RGB	62.43	85.62
MViTv2 [99] [†]	Depth	49.08	76.51
MViTv2 [99] [†]	Visible light	58.59	83.50
MViTv2 [99] [†]	Stereo images	58.86	83.55
SlowFast [100] [†]	RGB, VL, stereo	62.34	85.97
MViTv2 [99] [†]	RGB, VL, stereo	66.45	88.43

Table 3.2. ACR benchmark on IndustReal for models trained on various (combinations of) sensor modalities. *MECCANO [76] pre-trained, [†]Kinetics [18] pre-trained, VL: visible light. **Bold** text indicates best performance, underlined indicates second-best.

class $c_i \in C_a$ [76]. Here, x_{ts_i} and x_{te_i} indicate the start and end frame for action $c_i \in C_a$, respectively.

Benchmark. For this task, the SlowFast [100] CNN and MViTv2-S [99] transformer are chosen to benchmark the performance. Each model is trained on IndustReal after Kinetics [18] pretraining. Additionally, since the MECCANO dataset for ACR is closely related to the IndustReal dataset, we report baselines pre-trained on MECCANO. Finally, both networks are trained on depth, visible light (VL), and stereo images, and combined to create an ensemble of models trained on various modalities.

Top-1 and top-5 accuracy are reported in Table 3.2 for the aforementioned experiments. pretraining on the MECCANO dataset does not provide a performance benefit. The MViTv2 transformer outperforms the SlowFast architecture. The best performance is observed for the MViTv2 ensemble of the modalities RGB, VL, and stereo images. Depth is excluded from this ensemble due to its under-performance compared to the other modalities. This reduced performance is due to the inability to access the short-throw depth and RGB camera simultaneously, hence the high-quality short-throw depth data are not provided. The long-throw depth data is not particularly suitable for ego-centric action recognition where the object is close to the person. Notably, for both architectures, each individual modality is outperformed by the ensemble, indicating that each modality contains some complementary form of relevant information.

Figure 3.3 shows the normalized confusion matrix for action recognition on the IndustReal test set. Specifically, it shows the performance of the MViTv2 [99] transformer, pretrained on the Kinetics dataset [18]. It is observed that although the performance is

Pre-trained	Fine-tuned	mAP (b-boxed) [-] $\uparrow$	mAP (entire videos) [-] $\uparrow$
COCO	Synthetic	0.573	0.341
COCO	IndustReal	0.753	0.553
Synthetic	IndustReal	0.779	0.575
COCO	IndustReal + synthetic	0.838	0.641

Table 3.3. ASD performance benchmark on IndustReal using YOLOv8-m [101] for various training schemes. **Bold** text indicates best performance.

generally adequate, the model confuses actions that are visually similar, such as *take* and *put* the *short_brace*, and *tighten* and *loosen* the *acorn_nut*.

3.5.2 Assembly state detection benchmark

Definition. Given a video frame X_i and a set of assembly states $Z_a = \{z_0, z_1, \dots, z_n\}$, the objective of ASD is to detect the bounding box and assembly state $z_i \in Z_a$ for the sample X_i . The states Z_a are defined in Section 3.4.4-B.

Benchmark. For this benchmark, the state-of-the-art YOLOv8-m [101] object detection network is employed. Since geometries of all parts are published with IndustReal to stimulate synthetic learning, we provide four training schemes. First, we train the model, pre-trained on COCO [102], exclusively on a synthetically generated dataset. To generate the synthetic data, Unity Perception [103] is used to generate 100K training samples, each containing one assembly state in Z_a . Secondly, we train the model (pre-trained on COCO) directly on IndustReal. Then, we pre-train on the synthetic dataset, after which we fine-tune on the IndustReal dataset. Finally, the synthetic and real-world datasets are combined for the last baseline.

The results are quantified using the mAP metric and reported on frames of the IndustReal test set containing ground-truth bounding-box annotations, as well as the entire test set. As outlined in Table 3.3, combining the synthetic and real-world images results in the highest performance. A significant performance drop of 27% is observed when evaluating entire videos, rather than only frames with a ground-truth annotation. The decreased performance is caused by false positive predictions on states with fine-grained visual differences, predominantly error states and directly prior to the completion of a procedural step. Notably, the best performing model has a false positive rate of 65% and average precision (AP) of 0.23 for assembly states containing an error. Figure 3.4 highlights the prediction of the ASD baseline on the IndustReal test set. The figure highlights correct and incorrect predictions for challenging samples. Two additional error states, and their corresponding ASD predictions, are visualized in Fig. 3.5.

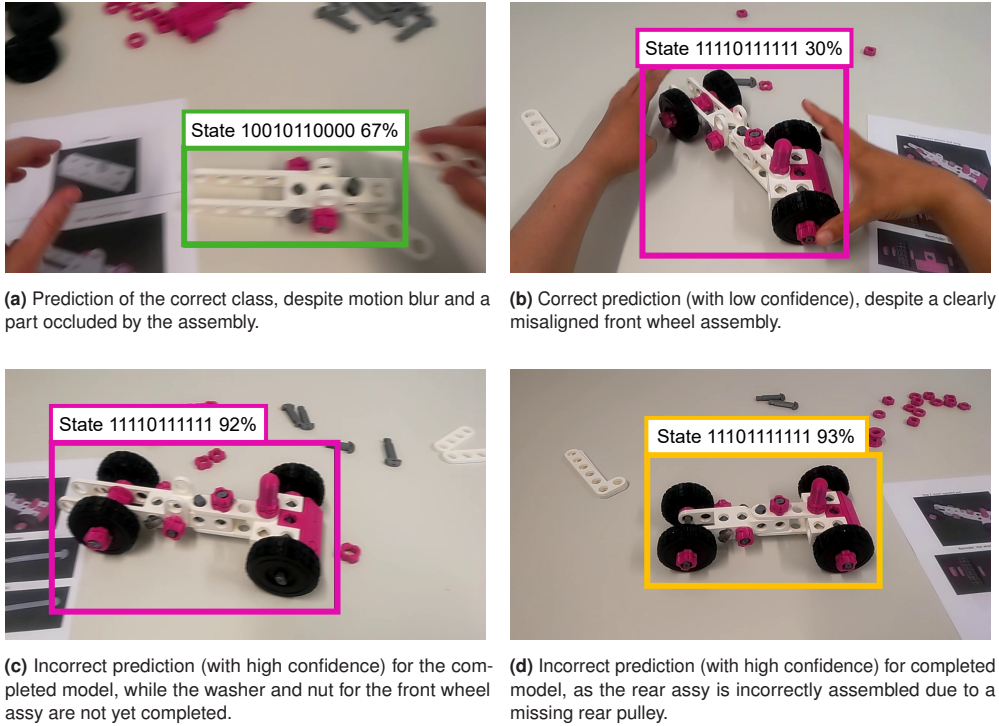


Figure 3.4. ASD predictions on the IndustReal test set, using YOLOv8-m [101].

3.5.3 Procedure step recognition benchmark

Definition. The task definition is given in Section 3.3.1.

Benchmark. For the PSR benchmark, we provide three baseline implementations relying on the ASD model output outlined in the previous section. The first baseline B1 determines for each change in detected assembly state which corresponding steps must have been completed since the last detected state, and assumes these actions to be executed correctly. The second baseline B2 accumulates the confidence for each predicted step completion (obtained from B1) over time, until a threshold T is reached, upon which an action is deemed correctly completed. Baseline 3 uses the same confidence aggregation as B2, but limits the number of possible step completions to those expected in the correct execution of the given procedure. For each baseline, the best performing ASD model from Table 3.3 is used. Additionally, each baseline is evaluated with the ASD approach trained exclusively on synthetic data. The detection pipeline (ASD + PSR) yields real-time operation at 178 fps on a V100 GPU.

The POS, F_1 , and τ metrics for all baselines are outlined in Table 3.4. The best performing approach achieves relatively high POS and F_1 scores (0.797 and 0.883 on

Model	All recordings			Recordings with errors		
	POS [-] $\uparrow$	F_1 [-] $\uparrow$	τ [s] $\downarrow$	POS [-] $\uparrow$	F_1 [-] $\uparrow$	τ [s] $\downarrow$
B1	0.570	0.779	14.9	0.480	0.698	14.4
B1-S	0.014	0.206	36.9	0.000	0.174	48.4
B2	0.731	0.860	22.3	0.636	0.784	20.2
B2-S	0.240	0.573	44.4	0.107	0.516	60.5
B3	0.797	0.883	22.4	0.731	0.816	20.4
B3-S	0.597	0.734	49.5	0.571	0.731	71.4

Table 3.4. PSR performance benchmark on IndustReal with ASD backbone, expressed by procedure order similarity (POS), F_1 score, and average delay τ . B1-3 denote the three baseline models and S indicates exclusive training on synthetic data. **Bold** text indicates best performance.

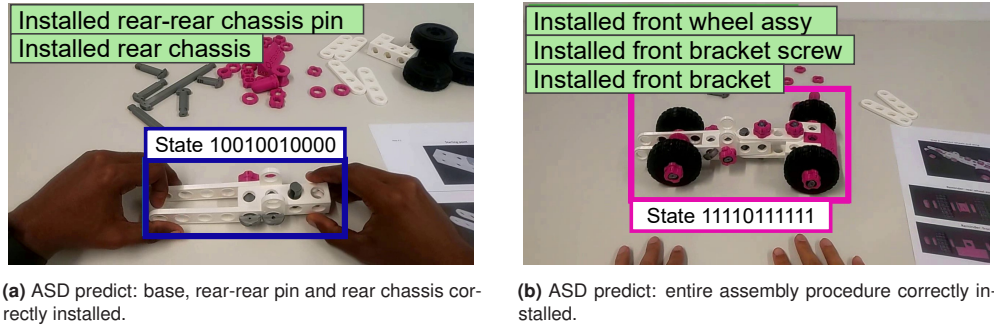


Figure 3.5. Visualization of ASD and PSR results for two cases. Bounding boxes and captions indicate the ASD predictions, text boxes indicate PSR predictions (B3). Both ASD predictions are false positives. (a) Front-rear pin orientation is incorrect, whilst the ASD predicts the absence of this pin. (b) Nut instead of screw is incorrectly used for the front brace, which is not recognized.

all recordings, respectively), due to under-representation of execution errors compared to correctly executed steps. On recordings with errors, all baselines show a distinctive decrease in performance, as shown in Fig. 3.5, thereby highlighting the need for approaches that better handle (out-of-distribution) errors. Although the performance of B3-S trained entirely on synthetic data is significantly lower than its real-world trained counterpart, it notably outperforms B2-S. This indicates that using procedural information to restrict the set of possible step completions can considerably improve the model performance.

Figure 3.6 outlines the PSR predictions on a single video in the IndustReal test set for the best scoring baseline, as well as the approach trained exclusively on synthetic data. It is observed that neither model is able to detect the error state around frame 1300, resulting in false positives for both PSR baselines. The model trained only on synthetic data wrongly classifies the second and the last state in the video,

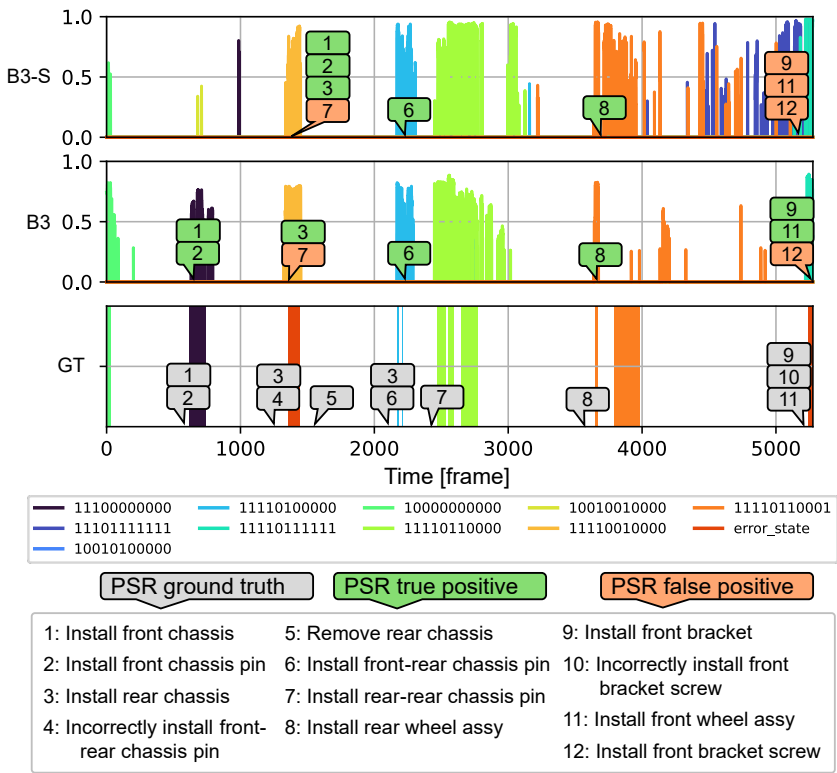


Figure 3.6. PSR predictions of B3-S and B3, together with the accompanied ASD classifications by YOLOv8-m [101], trained exclusively on synthetic data and on a combination of real and synthetic data, respectively. Predictions are shown for a single video in the IndustReal test set. False and true positive PSR predictions are highlighted.

explaining the difference in performance between B3 and B3-S outlined in Table 3.4.

3.6 Conclusion

This chapter has focused on bridging the gap left by action recognition and assembly state detection in the field of procedural activity understanding, by defining the novel *procedure step recognition* (PSR) task. In PSR, procedural information is explicitly leveraged and combined with visual perception to recognize *correctly completed* steps, even on unseen procedure execution sequences. Along with the PSR task, this chapter has presented the IndustReal dataset. IndustReal differs from existing datasets in the wide variety of procedural and execution errors and subgoal-oriented procedure

execution. Moreover, all parts are 3D printed to ensure future reproducibility, and these geometries are published to stimulate generalization and adaptation from the synthetic to the real-world domain. This aspect is crucial to scalability in industrial settings, yet absent in prior datasets. Furthermore, ACR, ASD, and PSR performance benchmarks are provided for the proposed IndustReal dataset, applying various pretraining techniques.

Findings. The three most important findings of this chapter are that, firstly, a PSR baseline based on state-of-the-art algorithms for different tasks, demonstrates promising results on procedural videos where participants do not make mistakes. Secondly, the use of CAD models to generate additional (synthetic) training data, significantly improves ASD and PSR performance. This underscores the importance of providing the object geometries, especially in procedural datasets where real-world recordings often fail to capture the full range of execution variability. Thirdly, explicitly leveraging procedure information helps to bridge the gap between models exclusively trained on synthetic data and models trained on a combination of synthetic and real-world data.

Besides these three findings, the IndustReal dataset also poses suitable challenges beyond PSR, *i.e.* for ACR and ASD. For ACR, the best-performing state-of-the-art algorithm only obtains a top-1 accuracy of 66.5%. Similarly, for ASD, the best-scoring model obtains an mAP of 0.64, leaving significant room for improvement.

Limitations. Despite the promising benchmark performance on PSR, this baseline still suffers from three notable limitations which are particularly relevant within the industrial context. First, the baseline is unable to recognize execution errors if they are not explicitly defined prior to training the model. Second, the ASD model on which the PSR algorithm relies, requires video frames where the object in the procedure is entirely visible and suffer minimal occlusions. Finally, the baseline has limited scalability within the industrial context. The best-performing model still requires real-world data, and is specific to both the procedure and the machine, thereby requiring full retraining of the models even for minor changes to task order, assembly configurations, and components (*e.g.* upgrades to parts).

In conclusion, while several limitations regarding scalability remain, the PSR task formulation and IndustReal dataset presentation are significant steps toward solving real challenges in industrial procedure activity understanding.

In this chapter, the gap in existing literature on industrial activity recognition that was highlighted in the previous chapter, was addressed by formulating the PSR task and presenting the IndustReal dataset. Although promising results are observed, issues regarding scalability have become apparent, such as recognizing procedural mistakes that are not included in the training data, and eliminating the need for model retraining. These points will be addressed in the following chapter, in which a representation learning approach is proposed to improve the generalization of an ASD algorithm to updated assembly states and novel execution errors.

Representation learning for assembly state recognition

4.1 Introduction

The preceding chapter has defined the procedure step recognition (PSR) task, with the objective of automatically recognizing correctly completed steps in videos showing procedure executions. Additionally, the accompanying IndustReal dataset has been presented. Although this research demonstrates encouraging results, its reliance on supervised learning has revealed limitations in scalability and flexibility for real industrial use. Therefore, this chapter outlines a representation learning-based approach, enabling the model to learn more generalized representations of the various states of an object throughout a procedure.

Accurately recognizing these assembly states of a particular object during a procedural action can enable monitoring of proper procedure execution and allows for dynamic follow-up instructions. Assembly state recognition (ASR) can help prevent errors and optimize the overall efficiency of procedures, *e.g.* via integration into augmented reality devices such as the HoloLens 2 [104] (Microsoft, Seattle, USA). Existing methods, such as the ASD algorithm from the previous chapter, address the problem from a classification perspective, with the supervised training of a neural network to memorize the expected object states [64], [104]–[109]. Although these works have shown decent performance for procedures with a limited amount of expected assembly states, they require the explicit definition of each possible assembly state [110]. Therefore, if any of the object states are modified, *e.g.* due to an upgraded part or a newly added state, new data must be annotated and the entire network has to be either fine-tuned or re-trained. This presents a major obstacle, and apart from being costly, it significantly hinders scalability to industrial applications. A second concern is the lack of reported performance of these aforementioned approaches on erroneous states, where the approaches remain incapable of reliably detecting erroneous assembly states, or correct but previously undefined states, that are not explicitly defined during

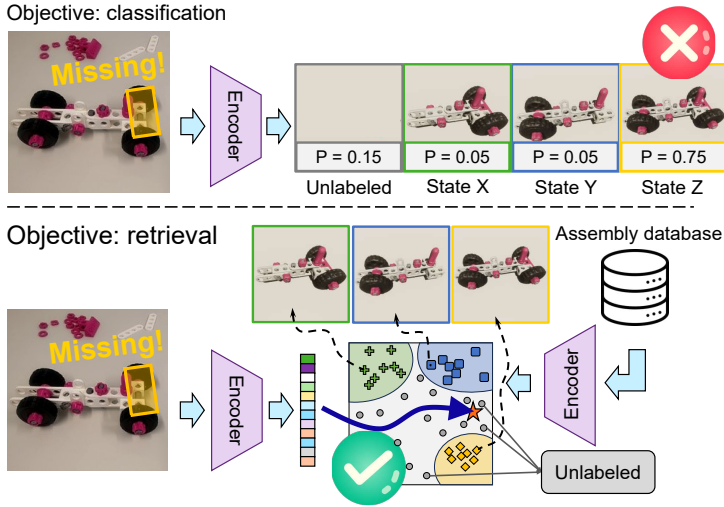


Figure 4.1. This chapter defines ASR as a representation learning approach (bottom of figure), rather than classification (top), and demonstrates that this (1) outperforms classification-based approaches, (2) avoids retraining the models after every minor update in the assembly procedure, and (3) enables distinction between correct and erroneous states.

training. Requiring explicit definitions of error states is a fundamental issue, since defining all possible errors that can occur during an assembly is infeasible¹.

Given these limitations in this existing body of literature on ASR, and the aforementioned discussion on these limitations, the research of this chapter converges to the following two research questions.

RQ4.1 Scalability to procedure updates. How can an ASR algorithm be designed, such that it does not require retraining when object states or procedures are modified?

RQ4.2 Pre-defined error states. How can erroneous assembly states be identified, without requiring the pre-definition of all potential procedure errors and what error recognition performance can be obtained by such an ASR algorithm, when it is not trained on any error assembly states?

To address these research questions, this work proposes a representation learning framework to learn a function that projects images into a meaningful embedding space, as demonstrated in Fig. 4.1. Through the use of representation learning, rather than explicitly memorizing each possible assembly state, the model is able to distinguish between states based on visual similarity. Such an approach is hypothesized to better generalize to new assembly states and (unseen) execution errors, as it highly resembles open-set classification and image retrieval tasks [111]. Furthermore, a simple

¹In our experience, the unexpected and thus undefined errors are frequently the costliest to resolve.

yet effective modification to existing loss functions is proposed, named intermediate-state-informed loss (ISIL). The ISIL modification uses unlabeled part configurations in the training data as negative samples without enforcing their image embeddings towards a single cluster. This improves clustering performance across various network architectures and contrastive loss functions, while also improving classification performance across all experiments.

The research in this chapter shows that the proposed framework, trained on a combination of real-world and synthetic images, outperforms classification-based approaches and generalizes to entirely unseen assembly states. Furthermore, the representation learning-based approaches recognize real-world assembly errors more effectively than classification-based approaches, without being trained on any data containing error states. The annotations created to perform the study on these error states are published to stimulate research on this topic. In summary, this chapter contains the following contributions.

- *Representation learning approach* to ASR that outperforms classification approaches and performs well even on unseen part configurations.
- *Intermediate-state-informed loss function modification* ISIL, which effectively leverages unlabeled data between completed assembly steps to improve both clustering and classification performance.
- *Extensive comparison on erroneous assembly states* between the representation learning framework and classification-based approaches. Moreover, new annotations for the user-intended states in the IndustReal [105] dataset are published.

The structure of this chapter is as follows. Related work on ASD and representation learning is reviewed in Section 4.2. Section 4.3 describes the methodology, and Section 4.4 presents the experimental evaluation. A discussion of the results is provided in Section 4.5, with concluding remarks in Section 4.6.

4.2 Related work

This section reviews prior work on assembly state detection and representation learning, highlighting current limitations in the detection of novel or erroneous states and discussing how representation learning can offer a more generalizable solution.

4.2.1 Assembly state detection

Assembly state detection (ASD) is generally approached as a subset of the broader object detection problem [105], [107], [109], [112]. ASD differs from ASR, as there is no set of reference images, and the objective is to classify and localize different states of the assembly process. In an assembly process, there may be many states that are

highly similar and therefore hard to distinguish individually. This issue is addressed by Stanescu *et al.* [106] using a modified object detector to include previous state detections, and in the work by Liu *et al.* [64] with specialized attention units. Another work uses a neural network trained to simultaneously perform pose estimation and state classification [104].

Despite the significant progress made on ASD, current approaches have two notable limitations. First, they focus on classification with a fixed set of assembly states, since the problem is addressed as conventional object detection. Therefore, even if the models perform well on these pre-defined states, full retraining is required for any changes in the assembly procedures. One related work does not rely on classification, however, they require multiple depth sensors arranged in a fixed set-up and a restrictive step-by-step assembly execution [113]. Second, although it is common to encounter unexpected states in complex assembly tasks (*e.g.* due to execution errors), and this is studied in robot assembly tasks [114], existing research on state recognition fails to investigate this [104], [107], [112]. None of the existing approaches generalize to new, unseen error states. We address both of the aforementioned limitations by approaching ASR as a representation learning task.

4.2.2 Representation learning

In representation learning, the objective is to learn a function that projects an input image to a meaningful embedding space. One way to implement such a function is contrastive learning, where the goal is to learn a representation of the data such that similar instances are in close proximity in feature space, while dissimilar instances are far apart [115]–[117]. This can be achieved by, *e.g.* maximizing agreement between differently augmented views of the same data example via a contrastive loss in the latent space [115], [116], or with Siamese neural networks (SNNs) [118], [119]. SNNs consist of two encoders with shared weights and have been applied to a variety of verification tasks, such as signature [118] and face verification [120]. During training, the model extracts features from two images, after which a distance metric between these features determines the similarity between them.

A commonly used loss function in representation learning is the triplet loss [121], with the objective to ensure that the projected point of an anchor sample is closer to the projected point of a positive than to a negative sample. The number of possible triplets grows cubically with the size of the dataset and a large part of all triplets will be trivial, introducing the need for hard mining. To that end, Hermans *et al.* [122] propose the batch-hard triplet loss, where the hardest positive and negative for each image within a mini-batch is calculated using the Euclidean distance. The InfoNCE [123] loss is another commonly used loss function, popular in unsupervised learning frameworks such as SimCLR [115], [116], and aims to maximize mutual information between positive samples whilst minimizing information between negatives. The supervised

contrastive (SupCon) [124] loss function extends the InfoNCE loss to a supervised setting.

A significant advantage of representation learning is its ability to generalize [119], [120]. Whereas classification and detection networks are trained to discriminate samples among previously seen/memorized categories, representation learning networks learn a correspondence task. Therefore, the model does not explicitly learn to memorize the training data, but to learn low-dimensional, discriminative features. These features can be used not only to generalize to new instances of known classes, but to entirely new classes from unknown distributions (one-shot learning) [119]. Hence, we hypothesize that the network can generalize to unseen part configurations, as well as error states. This makes the representation learning approach a viable and attractive candidate for solving the challenges encountered in ASR.

4.3 Method

The proposed representation learning approach, as outlined in Fig. 4.1, aims to extract meaningful embeddings from images of various assembly stages. At inference, the embeddings of assembly images are used as query and compared to the training embeddings, in order to recognize the state (and thereby the correctness) of assemblies. Additionally, we describe a loss function modification that can be implemented to any supervised contrastive loss function.

4.3.1 Contrastive learning framework

The general structure proposed in SimCLRv2 [116] is selected for the contrastive learning framework and used in a supervised setting. Instead of creating multiple instances of one image by applying various augmentations, M_R real-world and M_S synthetic images are sampled for N randomly selected assembly states per batch. Additionally, I_R images of unlabeled assembly images are sampled, resulting in a batch size of $(M_R + M_S) \cdot N + I_R$. These unlabeled images can contain transitions between assembly states, as further outlined in Sec.4.3.2, but can also contain images where the assembly is largely or entirely out-of-frame. No explicit domain adaptation techniques beyond standard image augmentations are employed. The synthetic images are generated with a simulator [125], using the CAD models for different assembly stages. The use of synthetic images provides more samples for each assembly state during training, which is particularly meaningful for the recognition of assembly states rarely present in real-world training data.

The objective during training is to learn a function that maps identical physical assembly states (with different orientations and backgrounds) from the input manifold $\mathcal{R}^F$ onto close points in the embedding space $\mathcal{R}^D$, while simultaneously mapping different assembly states to distant points in $\mathcal{R}^D$. This mapping function is learned by

passing images through an encoder $f(\cdot)$ that creates feature maps h_i . The encoder can be any image feature extractor, such as a ResNet [126] or vision transformer (ViT) [127]. These feature maps are then projected to the embeddings $z \in \mathcal{R}^D$ using projection head $g(\cdot)$, implemented as a three-layer multi-layer-perceptron (MLP), after which the features are normalized. Adding a projection head and defining the contrastive loss in $\mathcal{R}^D$, rather than directly using the feature maps h_i , leads to improved representations [115]. An overview of the framework is described in Fig. 4.2.

At the inference stage, images are passed through the encoder $f(\cdot)$. Given [116], we discard the last two layers of the three-layer MLP $g(\cdot)$ to create the final embeddings y_i . This is useful because using a middle layer of the projection head $g(\cdot)$ improves the performance compared to discarding the entire head. To determine the assembly state in a test image, its embedding $\hat{y}_i$ is compared to all reference embeddings y from the training data using K-Nearest Neighbor (KNN). Neighbors are determined with same similarity metric as used in the loss function during training, *i.e.* Euclidean distance (L_2) for the triplet loss and cosine similarity for models trained with an angular-based loss function.

4.3.2 Intermediate-state-informed loss

During the execution of an assembly or maintenance procedure, an object goes through various assembly states (*i.e.*, part configurations). For instance, in the procedure of attaching a wheel with 5 log nuts to a vehicle, there are $2^5 = 32$ potential part configurations. Defining all potential states is therefore frequently infeasible, resulting in *undefined* transitional states between explicitly defined states (milestones) in videos of such procedures. We define these transitional states as *intermediate states*. Discarding data containing these states during training is sub-optimal, as the configurations being undefined, does not mean that there cannot be information within these configurations, as illustrated in Fig. 4.3a. On the other hand, since the undefined states can contain any possible transition between any possible assembly state, these intermediate states are frequently not correlated to each other. Therefore, simply classifying or clustering these intermediate states into a single group forces a neural network to learn non-existing connections, which may limit the embedding capacity of a model.

To overcome this limitation, this work proposes an intuitive modification that can be applied to supervised contrastive loss functions. The proposed modification, *intermediate-state-informed loss* (ISIL), leverages the unlabeled intermediate assembly states by treating them exclusively as (hard) negative samples. By doing so, intermediate states are forced to be dissimilar to any explicitly defined assembly state, whilst not penalizing dissimilar embeddings of intermediate states, or enforcing similarity between embeddings of potentially uncorrelated training samples. This concept is illustrated in Fig. 4.3c.

The ISIL modification can be applied to different loss functions for representation

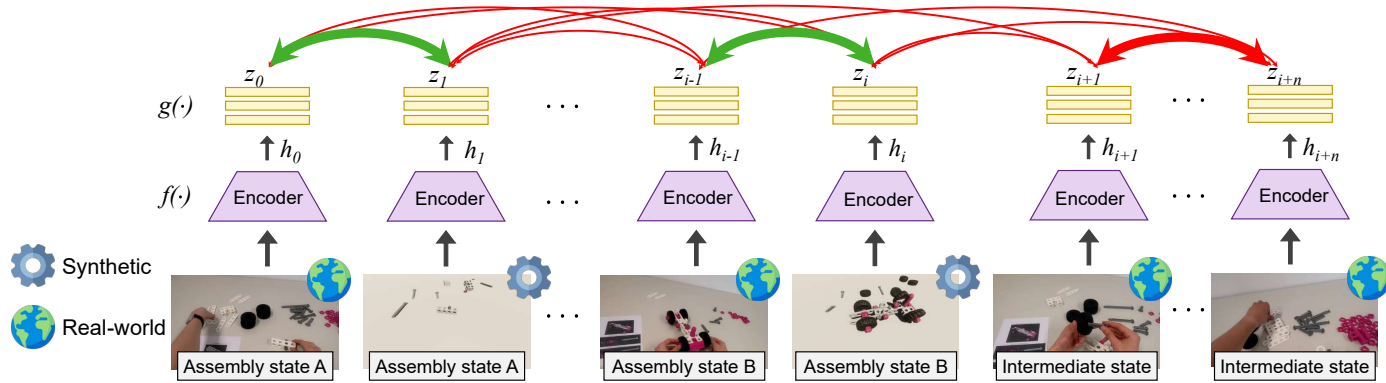


Figure 4.2. Overview of the contrastive learning framework (with the proposed ISIL modification). Each mini-batch consists of real-world and synthetic images of pre-defined assembly states and real-world intermediate states. The images are passed through an encoder $f(\cdot)$, followed by a three-layer MLP projection head $g(\cdot)$. The resulting embeddings z_i are used to calculate the contrastive loss. During inference, only the first layer of $g(\cdot)$ is used.

learning. In this work, we apply it to two common losses, namely the batch-hard (triplet) [122] and supervised contrastive (SupCon) loss [124]. The former is defined, with the ISIL modification underlined, as

$$L_{\text{triplet}} = \sum_{i=1}^N \left[m + \underbrace{\max_{j: y_j = y_i \wedge y_i \in C} \|g(f(x_i)) - g(f(x_j))\|_2^2}_{\text{hardest positive}} - \underbrace{\min_{k: y_k \neq y_i} \|g(f(x_i)) - g(f(x_k))\|_2^2}_{\text{hardest negative}} \right], \quad (4.1)$$

where $f(x)$ is the embedding function described in Sec. 4.3.1, C the set of pre-defined assembly classes, N the number of samples in the mini-batch, m the margin, x_j a positive and x_k a negative for the given anchor input x_i , with y_j , y_k , and y_i as their respective labels. The ISIL modification is enforced by $y_i \in C$, denoting that the ground-truth label of sample i is a pre-defined assembly state (therefore not an intermediate state). This excludes undefined states from the *pulling* contribution within the triplet loss, ensuring that undefined samples are not grouped towards one cluster.

Similarly, the SupCon [124] loss function with the ISIL modification is defined as

$$\mathcal{L}_{\text{SupCon}} = \sum_{i \in I} \log \frac{-1}{|P(i)|} \sum_{p \in P(i)} \frac{\exp(\frac{g(f(x_i)) \cdot g(f(x_p))}{\tau})}{\sum_{a \in A(i)} \exp(\frac{g(f(x_i)) \cdot g(f(x_a))}{\tau})}, \quad (4.2)$$

where τ denotes the temperature constant, $A(i) \equiv X \setminus \{i\}$ the set of all indices in mini-batch X , and $P(i) \equiv \{p \in A(i) : (y_p = y_i) \wedge (y_p \in C)\}$ is the set of indices of all positives for defined classes. The ISIL modification is enforced by excluding the undefined states from the set of positives indices $P(i)$, while permitting them to act as negatives in $A(i)$.

4.4 Experiments

The representation framework as described in Section 4.3.1 is implemented, as well as the intermediate-state-informed loss function modification (ISIL) for the batch-hard (triplet) [122] and SupCon [124] losses. To support our claim that representation learning is effective for ASR, we compare our method to a classification model, trained with the cross-entropy loss, where the last layer of the projection head is modified to equal the number of defined assembly states in the training data. At inference, only the first layer of projection head $g(\cdot)$ is retained, regardless of the loss used in training, which means no further modifications are required to directly compare the embeddings of a classification-trained model to those from a contrastive-trained model.

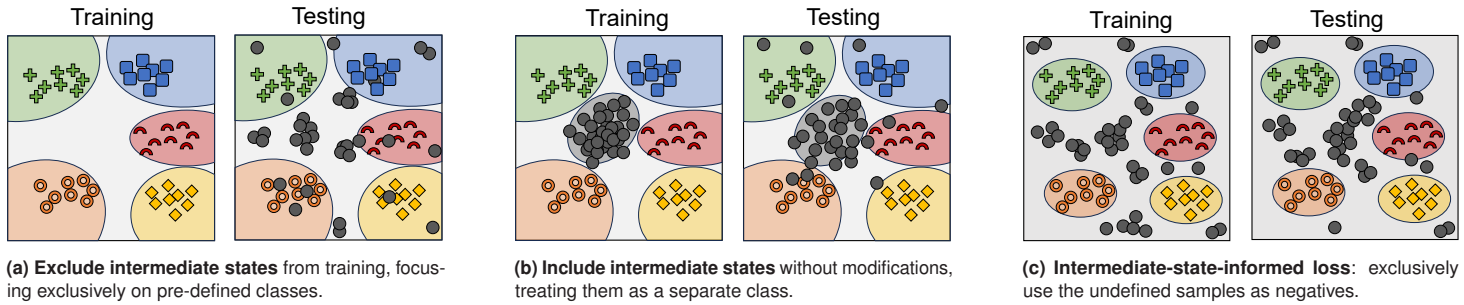


Figure 4.3. Illustration of influence of the proposed loss function that leverages images containing intermediate assembly states, *i.e.* non-defined, transitional states between pre-defined states. In (a), the intermediate states (gray) are ignored during training, not leveraging any information that these images might have. In (b), the intermediate states are grouped into a single cluster, hindering the model's capacity to capture a meaningful embedding since these states are frequently not correlated. We propose (c), an intuitive modification to loss functions that leverages intermediate states exclusively as negative samples. Dissimilar embeddings of (potentially) uncorrelated states are only penalized if they are similar to any foreground (pre-defined) class.

To evaluate the use of representation learning for ASR, the IndustReal [105] dataset is used. This dataset is selected for three reasons. Firstly, the data contain 22 annotated assembly states, making up only 13% of the total images. Therefore, significant information is discarded by training only on pre-defined states. Secondly, all parts in the IndustReal assembly are provided as 3D models. This facilitates the creation of data with entirely new part configurations, which we use to compare the generalization of our framework to the model trained for classification. Finally, IndustReal contains various execution errors, which we use to test the hypothesis that the generalization capacity of representation learning does not only benefit the recognition of novel part configurations, but also enables the verification of the correctness of assembly states. Models are trained following the IndustReal train/val/test splits [105], with the exception that we exclude erroneous assembly states during training. Therefore, the model's generalization to entirely unseen errors can be tested.

In order to quantify the performance of a trained model, we report the $F_1@1$ and $MAP@R$ scores. The $F_1@1$ metric represents the harmonic mean of precision and recall at the top of the ranked list of closest neighbors². The mean average precision at R ($MAP@R$) score is a metric used to evaluate the quality of the ranked retrieval results for a given query [111]. It calculates the mean of the average precision scores at R , where R is the number of true positives in the reference set, thereby reflecting the precision of the system across different recall levels for the top R results. $MAP@R$ is more stable than Recall@1 and takes the order of the ranking of correct retrievals into account, unlike R -Precision. As mentioned above, intermediate states are frequently not correlated to each other. Since the embeddings of two dissimilar intermediate states are allowed (and indeed desired to be) dissimilar to each other, it does not make sense to evaluate $MAP@R$ on these classes. Therefore, we report $MAP@R$ on the defined assembly states only and denote this with $MAP@R(+)$. Please note that this score is still strongly impacted by intermediate states, as these states should not appear within the clusters of defined assembly states.

4.4.1 Implementation details

The proposed representation learning approach consists of an image encoder and projection head, as outlined in Section 4.3. For the encoders, ImageNet-1k pre-trained ResNet34 [126] and ViT-S [128] are implemented and compared to each other. The last fully connected layer of the encoders is replaced with three fully connected layers to form the projection head. The first two layers maintain the feature embedding size of the encoder, whilst the last layer reduces the features onto a 128-dimensional embedding space.

²The F_1 score discussed in the previous chapter is the standard classification F_1 , whereas $F_1@1$ here specifically refers to the harmonic mean of precision and recall at the top-1 retrieved neighbor, making them distinct metrics.

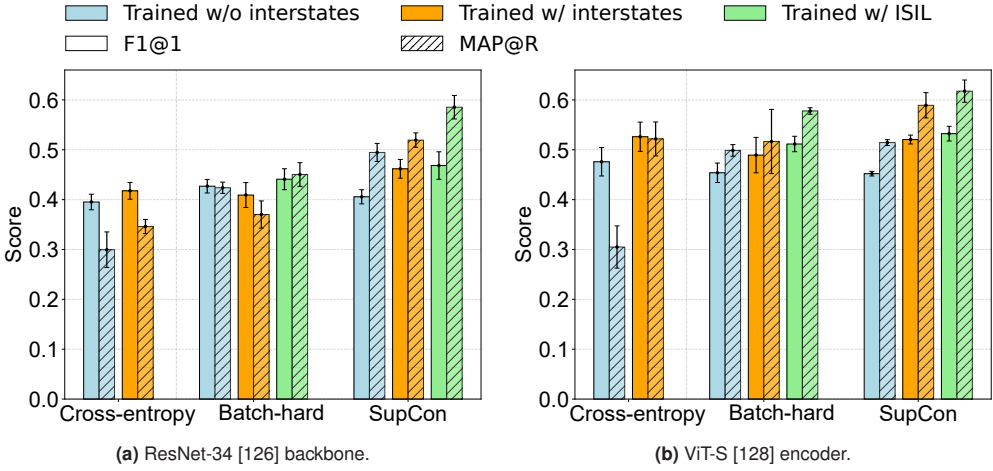


Figure 4.4. ASR performance on IndustReal [105]. The contrastive approaches outperform those trained for classification, and the proposed ISIL modification increases performance in all settings.

Each image is resized to 224×224 pixels ($W \times H$). For each batch, real-world ($M_R=8$) and synthetic images ($M_S=8$) are sampled per assembly state for $N=15$ randomly selected states. Additionally, $I_R=64$ intermediate assembly states are sampled for the loss functions that do not discard these images, resulting in a batch size of 240 or 304, depending on the use of intermediate states. The margin ($m=0.01$) and temperature ($\tau=0.07$) values for the loss functions are empirically set using the validation set.

For consistency between training cycles, the Adam optimizer with a learning rate and weight decay of 10^{-4} are selected. A 15k iteration (linear) warm-up learning rate is employed, followed by a cosine annealing learning rate with warm restarts every 40k iterations [129]. The best performing models are selected based on $F_1@1$ on the IndustReal validation set. After 100k iterations, or once a model does not improve either in $F_1@1$ or $MAP@R(+)$ on this validation set for 20k iterations, training is stopped. For every experiment configuration, five cycles with different seeds are performed. Whilst the backbones are pre-trained and thus constant between cycles, the projection layer weights and batches are generated based on the seed, and therefore different per cycle. For data augmentation during training, random Gaussian blur (kernel size 5, $\sigma=2$) and brightness, saturation, and contrast jitter of at most 0.1, 0.7, and 0.1 respectively, are applied. The framework executes in real time on a Tesla V100 GPU, obtaining approximately 150 frames/sec (fps) with either backbone.

4.4.2 ASR on IndustReal

Figure 4.4 demonstrates the results on the IndustReal [105] test set for the ResNet-34 and ViT-S backbones, trained with the cross-entropy, batch-hard, and SupCon loss

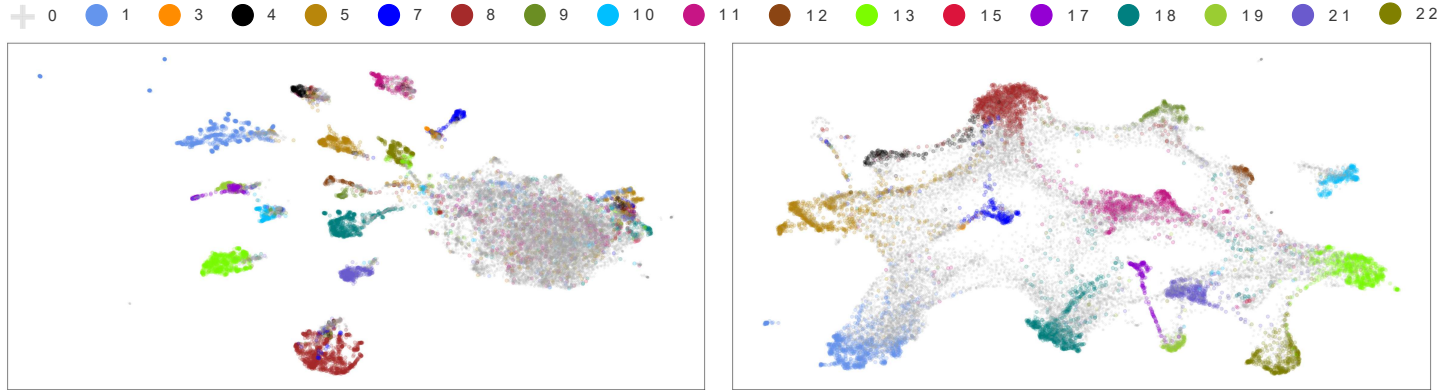
functions. The test set contains 9,659 images of assembly states, with 537 ± 413 images per state for 18 states. Furthermore, the test set contains 20,101 images containing intermediate states or background, 25% of the total unlabeled images in the IndustReal test set to significantly improve computational load. The reduction of unlabeled images is achieved by taking every fourth image with equal time spacing between the discarded images. Given the inherent class imbalance between assembly states in IndustReal, we report macro-averaged metrics. On average, the best contrastively-trained ResNet model outperforms the classification-trained ResNet by 12% on $F_1@1$, a notable result given that the metric evaluates classification rather than clustering performance. On $MAP@R(+)$, the SupCon loss improves over the cross-entropy loss by 69%. In fact, for the ResNet encoder, all configurations of the contrastive losses outperform the cross-entropy loss, regardless of whether any intermediate states are used during training. The ViT encoder trained with cross-entropy performs significantly better than ResNet trained for classification, but still scores 16% lower on $MAP@R(+)$, compared to the best contrastive-trained ViT. These results clearly demonstrate the benefit of representation learning for ASR.

Additionally, regardless of the backbone type or loss, the clustering performance is improved by 5–22% $MAP@R(+)$ when the novel ISIL modification is applied. For $F_1@1$, the ISIL modification does not significantly affect performance for the ViT. For both backbones, the SupCon loss with ISIL performs best, clearly demonstrating the effectiveness of using undefined part configurations as negative samples.

The quantitative results described above can be further explained by comparing the embedding space generated by the models, shown in Fig. 4.5, by applying dimensionality reduction using unsupervised UMAP [130]. The figure clearly demonstrates that discarding intermediate states during training, results in a large overlap of these states with defined assembly states (false positives). Clustering all intermediate states to a single group reduces the clustering quality of the remaining clusters, and results in a less meaningful embedding space. By applying the proposed ISIL modification, both the ability to discriminate between intermediate and pre-defined states, and a meaningful embedding space for intermediate representations, are retained.

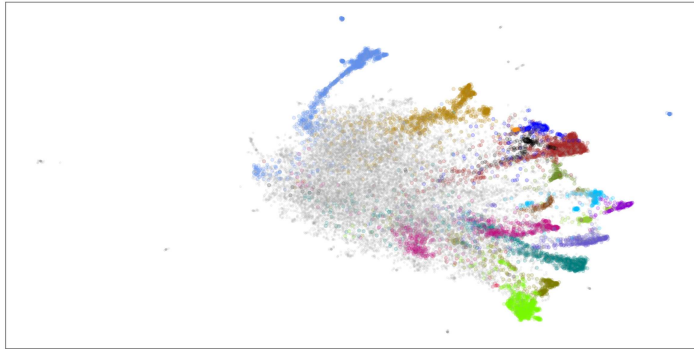
4.4.3 Generalization to unseen states

In this experiment, we test the hypothesis regarding improved generalization to entirely unseen part configurations for representation learning compared to classification-based approaches. Synthetic data is highly suitable for testing this generalization, since it provides full control over the data creation process. Specifically, we generate 18 new 3D models of assembly states and use the same data generation methodology as IndustReal [105], [125] to create 120 synthetic images per state, of which 100 are used for the reference set and 20 are kept for the query. The unseen states are modified versions of the IndustReal assembly states, *e.g.* one with screws placed differently.

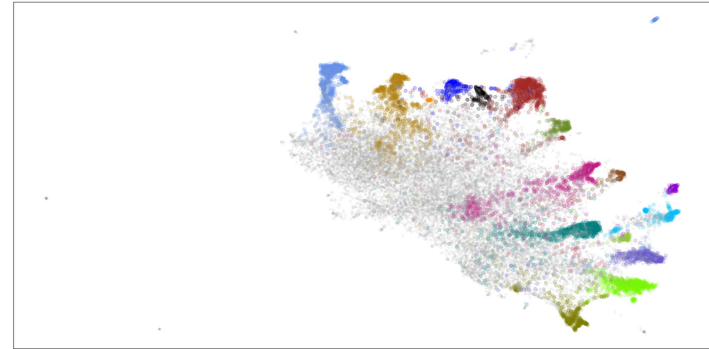


(a) **Trained with classification loss (cross-entropy).** Intermediate states are grouped into one cluster, together with a significant amount of pre-defined assembly states. Furthermore, there is no path of intermediate states through the embedding space for the transition between pre-defined classes.

(b) **Trained with SupCon loss [124] without intermediate assembly states.** The embedding space shows paths between the pre-defined states, but poorly discriminates intermediate from pre-defined assembly states, with intermediates frequently located within defined classes.



(c) **Trained with SupCon [124], clustering all intermediate states together.** The model is able to distinguish between intermediate states and pre-defined clusters, but shows significantly reduced cluster quality for defined states.



(d) **Trained with the SupCon loss [124] and the ISIL modification.** The modification retains the ability to discriminate between intermediate and defined states, whilst maintaining paths of undefined states through embedding space between pre-defined states.

Figure 4.5. UMAP [130] visualization of the embedding space created by the best performing ViT backbone for various configurations. Intermediate states are shown as gray crosses (+), and some defined-state points may appear as gray circles due to opacity.

The outcomes of this test are outlined in Fig. 4.6 and clearly confirm the hypothesis. The performance increase for contrastive approaches range from 21–53% on the classification metric $F_1@1$, with lower end of the range for ViT backbones. The increase in clustering performance ranges from a minimum of 85% to a maximum of 204% increased $MAP@R$. The ResNet backbones outperform the ViT on generalization to new assembly configurations. These significant performance gains demonstrate the benefits of representation learning for ASR, and enables systems to generalize to new states (for instance, because of an updated procedure) without requiring the entire network to be re-trained.

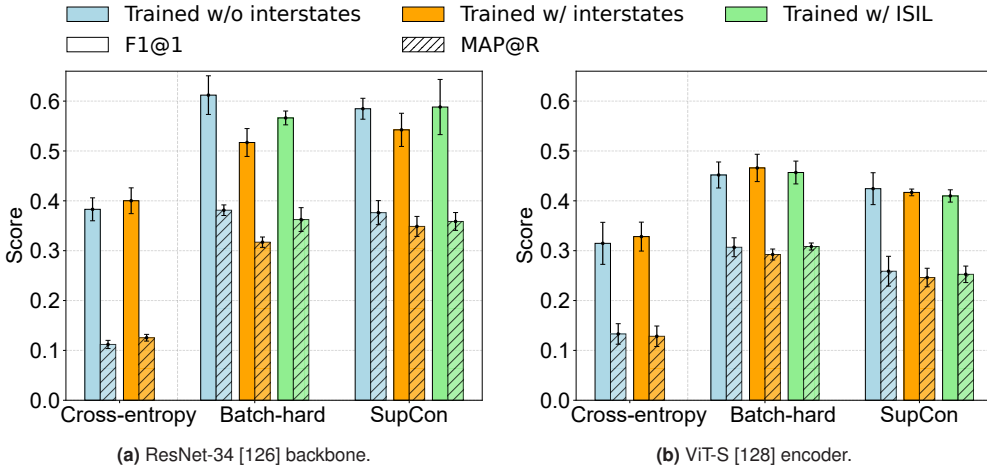


Figure 4.6. Performance of recognizing unseen states. Contrastive losses outperform the cross-entropy loss for all settings, and the ResNet backbone demonstrates significantly better generalization than the ViT.

4.4.4 Generalization to unseen error states

The performance of the proposed framework on unseen states provides a strong hint that distinction between erroneous and correct assembly states is feasible using this approach. After all, an execution error in an assembly procedure results in a new, unseen part configuration. To perform this experiment, at least two human annotators labeled the *user-intended state* and *error category* for every image in IndustReal [105] containing execution errors. The former is annotated based on the work instructions for each video, and the latter based on four error categories: missing component (I.), incorrect orientation (II.), incorrect placement (III.), and part-level errors (IV.). A description of the error categories and their occurrences in the IndustReal dataset are given in Fig. 4.7. The annotations are released with this work, facilitating future research on this topic.

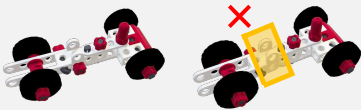
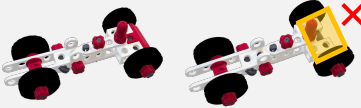
Error category	Description	Example reference – sample pair	# Frames
I. Missing (component)	Component is missing from the assembly state		1144
II. Orientation (component)	Orientation of component is wrong		398
III. Placement (component)	Component is placed at a wrong position		85
IV. Part level	Parts within components are missing/oriented wrong		763

Figure 4.7. Categorization of errors including visual examples and the number of frames in IndustReal [105] for each category. Based on whether an error is at component or part level, they are classified into four different categories.

The generalization of our framework to assembly errors is posed as a binary verification problem, where the cosine or Euclidian similarity between two correct states should be higher than the similarity between a correct and erroneous state. Our representation framework is used without any further fine-tuning, and the performance is quantified using the area under the precision-recall curve (AP). A single, error-free synthetic image is generated for every assembly state, with fixed lighting conditions and camera and object orientations. These images serve as *anchor* images. Each image with an execution error in IndustReal [105] is a *negative*, with the generated (error-free) image of the user-intended state as corresponding anchor to create the anchor-negative pair. For each negative, exactly one correctly assembled, real-world image of the user-intended state is randomly sampled to serve as *positive*, so that an equal number of positive and negative (error) pairs is sampled. In this way, the dataset becomes balanced with 50% anchor-positive and 50% anchor-negative pairs, thereby preventing bias for models tending towards predicting non-corresponding states. A randomly performing model thus achieves an AP of 0.50. Because the synthetic anchor images are always known to be defect-free, a model with an AP of 1.00 always has a higher similarity between the anchor and the error-free assembly state (positive) than between the anchor and the erroneous state (negative).

Figure 4.8 outlines the results for this experiment, where AP is reported for each error category, as defined in Fig. 4.7. The mean average precision for the best contrastive-trained ResNet is 6% higher than that of the best ResNet trained for classification.

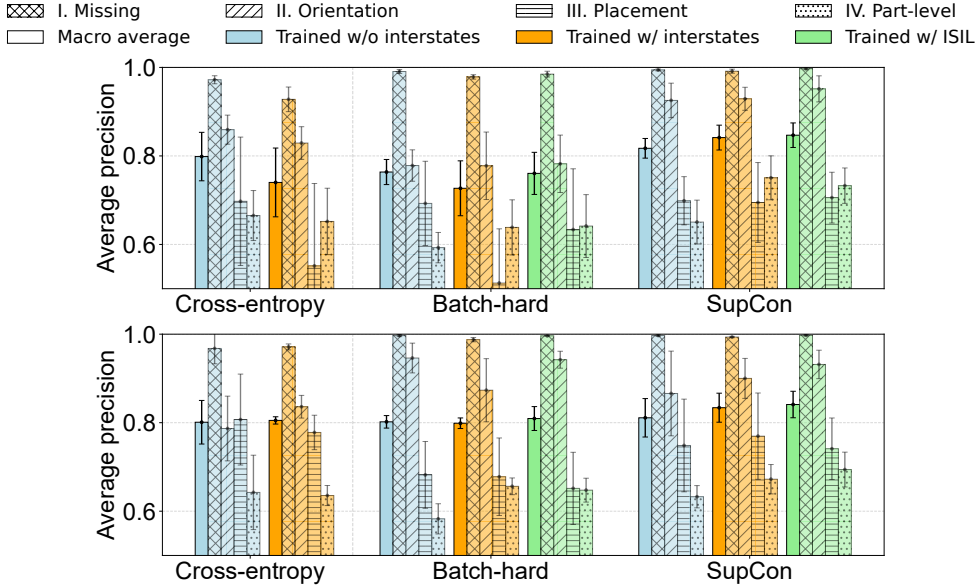


Figure 4.8. Binary verification performance of ResNet (top) and ViT-S (bottom) between an error-free anchor image and a correct (positive) or erroneous assembly state (negative). The best contrastive models outperform the classification-based models and are trained with the ISIL modification.

Similarly, for the ViT, the best contrastive-trained model score increased 5% compared to the best classification-trained model, reinforcing the benefits of representation learning with regards to ASR. The best AP for the ViT backbone is 0.006 lower than that of the best ResNet backbone. The ISIL modification improves the performance of both contrastive losses for both backbones. The results demonstrate that the proposed framework is able to accurately distinguish between correct assemblies and states with missing, incorrectly placed, and incorrectly oriented components. Furthermore, the results highlight the importance of investigating a models performance on erroneous assembly states.

4.5 Discussion

In this work, we have addressed several challenges in ASR by approaching it as representation learning, rather than a classification problem. Furthermore, we have proposed the intermediate-state-informed loss, a loss function modification that can be readily incorporated in existing contrastive loss functions. Our approach significantly outperforms classification-based approaches and generalizes to unseen part configurations. Additionally, the performance on error states is quantitatively evaluated with new error-state labels, demonstrating that the approach is able to distinguish between correct and erroneous states for various error categories, even when compar-

ing real-world to synthetic images. The performance on the fine-grained part-level error category is lowest, but can likely be improved in future work by increasing the variety of assembly states during training, which can be readily generated for the synthetic training data. To stimulate such research, the new error-state labels are published together with this work.

The proposed ISIL modification demonstrates outstanding performance on the IndustReal test set and scores highest for both backbones on the error generalizations. Nonetheless, the observed benefits of the ISIL modification on unseen synthetic assembly states are underwhelming. Grouping unlabeled states to a single cluster in feature space is expected to reduce the generalization of models, because new assembly states likely fall closely together in this cluster. Therefore, the ISIL modification is expected to show increased performance for generalization compared to models trained with intermediate states, but without the modification. However, the results do not indicate a significant difference in performance, positive or negative, between discarding and using intermediate states, regardless of the ISIL modification. This observation is likely due to the absence of intermediate states in the generalization dataset. In future work, training on a subset of the states in IndustReal [105], and using the rest of the data (including intermediate states) to test generalization, can verify this. Such an experiment can also confirm that the ISIL modification indeed creates a more meaningful embedding space.

4.6 Conclusion

This chapter has proposed, implemented and evaluated a novel approach to ASR. This framework has the potential to provide real-time assistance to numerous assembly and maintenance tasks, but requires substantial training data to learn the subtle differences between states. To learn these differences, this chapter has proposed a representation learning framework and has introduced the intermediate-state-informed loss function modification (ISIL), leveraging unlabeled transitions between assembly states by treating them as negative samples during training.

Findings. This chapter presents two main findings. First, the proposed framework is able to generalize to new, unseen assembly states, without requiring retraining or fine-tuning of the model weights. Furthermore, regardless of backbone type or loss function, the clustering performance is improved when the ISIL modification is applied. For the ResNet backbone, the proposed representation learning approach outperforms the classification-based approaches even when evaluated with classification metrics on pre-defined states³. The second finding is that the algorithm is

³This evaluation method favors classification-based approaches, because (i) generalization to unseen states is not considered and (ii) the representation learning approach is not directly optimized for classification metrics. Therefore, the out-performance of the adopted framework with a contrastive loss compared to a classification loss is remarkable.

capable of distinguishing between erroneous and correct assembly states for various error types, as demonstrated through an evaluation of the framework on images of erroneous assemblies. Performance is particularly high for missing and incorrectly oriented parts, achieving an AP of 0.997 for missing parts and 0.952 for incorrectly oriented parts, respectively. This performance is achieved without pre-defining any error type, and without training on any erroneous states.

Limitations. Although the performance improvement enabled by the ISIL modification is significant for the IndustReal test set and for error generalization, the algorithm is only able to determine whether a real-world image matches the synthetic render of an assembly state. When the two images do not match, this can be for various reasons, e.g. an error in the assembly. However, the algorithm would be of significantly higher value if it is able to pinpoint exactly *where* a discrepancy is recognized between the expectation (the synthetic render) and the real-world images. Secondly, the algorithm still relies on real-world training data to supplement the synthetic images.

In conclusion, this chapter has demonstrated that (i) a model can be used for updated assembly parts or procedure steps, without the requirement of retraining the model, and that (ii) a model can recognize errors even if they are neither pre-defined, nor represented in the training data. These two aspects are critical within the industrial domain and are valuable beyond ASR. This is demonstrated through the application of a similar representation learning approach to defect inspection with scanning electron microscopy (SEM) images, for which a European and World patent has been filed [IP2].

This chapter has addressed the limitations of existing ASR approaches revealed by the previous chapter by proposing a new method, based on representation learning. Although the potential of representation learning for ASR is clearly demonstrated, a fundamental limitation remains in pinpointing the exact location of assembly errors. Furthermore, the algorithm still relies on real-world training data, limiting its potential within high-precision, low-volume industrial applications. These challenges will be addressed in the following chapter, in which an approach for change detection is modified to segment potential errors during assembly procedures.

Error localization for industrial applications

5.1 Introduction

The preceding chapter has demonstrated that a model for assembly state recognition can recognize the presence of several types of assembly errors, without pre-defining such errors, or training on images containing execution errors. However, the usability and interpretability of the proposed system is limited, because the location of potential errors is not provided. Additionally, the algorithm requires real-world data for training. This chapter presents an error localization algorithm, trained exclusively on synthetic data and without the need to pre-define errors.

Many works on assembly state recognition perform some type of assembly state detection [105], [112], [131]–[133] or assembly quality inspection [134], which are typically performed as a supervised classification or detection problem. Though these approaches show decent performance at classifying a small number of assembly states, performance on erroneous states is not investigated. The representation learning-based approach proposed in the previous chapter, is demonstrated to be capable of recognizing *unseen* assembly errors [135], a fundamental requirement for a viable system. However, this approach lacks the (spatial) localization of errors, and therefore cannot provide easily interpretable feedback.

Another type of approach relies on anomaly detection, which requires a model to learn the typical structures of nominal data, based on *a-priori* definitions of nominal data [136], [137]. This is not feasible for many industrial procedures, since the possible presence of an error in assembly states depends on the intended state at that moment of the assembly. Accordingly, change detection has been proposed to localize progress between various assembly states through segmentation in the image [138], [139]. However, their method is applied only to non-erroneous assemblies and to a highly simplified task, with only four procedure steps in the entire assembly.

For training the aforementioned algorithms, most of the methods rely at least in part on real-world, annotated data. In many industrial cases, such data are not

available and not feasible to collect to a sufficient degree, particularly in high-precision, low-volume manufacturing, due to a high degree of object-configuration variety. In such a setting, a machine might only be assembled once, requiring the algorithm to be trained entirely on synthetic data if it has to be used during that single assembly¹.

In response to the challenges identified above, this chapter addresses the following two research questions.

- RQ5.1 *Error localization*. How can an algorithm localize the exact regions containing a potential error, rather than merely recognizing that a mistake is present somewhere in the assembly?
- RQ5.2 *Synthetic training data*. To what extent can an algorithm be trained solely on synthetic data, while still accurately recognizing errors in real assembly states, and without being trained on potential errors that it may encounter in the real world?

To overcome the fundamental limitations that arise for error localization in the industrial domain and address these aforementioned research questions, a methodology named StateDiffNet is proposed. The StateDiffNet, outlined in Fig. 5.1, pinpoints the difference between an object in different assembly states (including erroneous), using segmentation. StateDiffNet is the first error localization system that can locate errors on states that were never encountered during training, and on significantly more complex assembly configurations than related works. For example, where the closest related work trains and evaluates on the same five assembly states [138], the work in this chapter is trained on more than 10^5 unique state combinations, including states with very small differences, and tested on entirely unseen states. Given two images of an assembly object, StateDiffNet localizes through segmentation all meaningful differences between assembly states that can be inferred from its view of the object.

A core component of the proposed approach is the methodology for generating and sampling synthetic image pairs. This provides full control over the meaningful change that the system should detect, as well as the changes that the system should be invariant to, *i.e.* the expected variability resulting from aspects such as camera pose, photometry, image distortions, and shadows. The StateDiffNet is trained exclusively on synthetic data and is able to segment assembly errors, taken from real-world, ego-centric video data [105], including various error types that were never presented during model training. The proposed method can be adapted to many industrial assembly procedures, provided that a CAD model of the object is available. In summary, the main innovations of this chapter consist of the following contributions.

- The first *assembly error localization network*, applied to study error localization on assemblies with a wide variety of complex state differences.

¹For example, every produced ASML machine is unique and typically only assembled once in the factory, and then once at the customer.

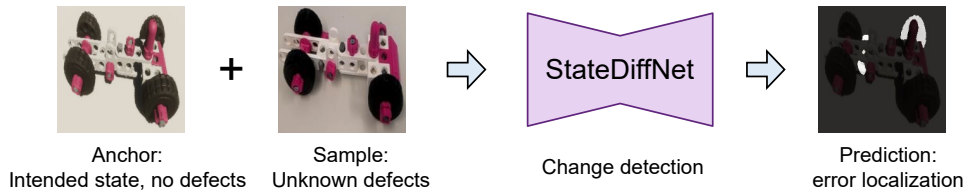


Figure 5.1. Principal view of the proposed method on an object from a complex assembly procedure [105]. The approach is evaluated on unseen assembly states and error types.

- A *technique for data generation* that, given a 3D model, creates image pairs for change detection, allowing full control of the expected (normal) variability between image pairs. The generation pipeline is made publicly available.
- *Valuable insights into the change detection mechanisms* of state-of-the-art algorithms, applied specifically to industrial applications.

This chapter is structured as follows. First, the related work on ASR and change detection are outlined in Section 5.2. Subsequently, Section 5.3 describes the proposed methodology, which is evaluated through experiments detailed in Section 5.4. The findings are discussed in Section 5.5, after which concluding remarks are provided in Section 5.6.

5.2 Related work

There are multiple existing research fields that investigate tasks similar to assembly error localization, including assembly state recognition and change detection. Relevant works from each of those fields are outlined below.

5.2.1 Assembly state recognition

Assembly (or object) state detection (ASD) is a specialized version of object detection, wherein the aim is to identify a bounding box around an object, which is labeled according to its state in the assembly process. Su *et al.* [132] and Schieber *et al.* [112] simultaneously perform ASD and 6D pose estimation, which is particularly useful for AR applications. However, they use assembly procedures of limited complexity (at most 10 parts, which can only be assembled in a limited number of ways). Two recent works focus on assembly procedures with more part complexity using object detection networks, trained on both synthetic and real-world annotated images [105], [133]. However, it is unclear whether the method scales to procedures of higher complexity, containing potentially millions (as opposed to tens) of different states. Additionally, little effort has been undertaken to investigate how these models handle erroneous

assembly states.

To address these limitations, a recent work [135] proposed a representation learning approach towards assembly state recognition. They show that such a model is able to retrieve states that are never encountered during training and classify some states as erroneous, while being trained purely on error-free data. However, the model does not provide any indication on the source of the error, thus providing limited assistance to the operator of such a system. Additionally, this prohibits the (automatic) recognition of correctly assembled parts if an error is present anywhere on the assembly. The proposed work in this paper aims to solve these limitations, while maintaining the valuable capability to generalize to assembly states entirely unseen during training.

5.2.2 Change detection

The objective of change detection is to detect areas that have incurred *meaningful* change between two images of the same scene taken at two different time instances, and potentially from different viewpoints. This commonly takes the form of binary change segmentation focused on remote sensing [140]–[142] and surveillance applications [143]–[145]. The best-performing methods for change detection are deep learning based [146], *e.g.* the Bi-temporal Image Transformer (BIT) [140]. In remote sensing applications, the two input images to the change detection network are generally taken from the same position using georeferencing and feature matching [143], simplifying a change detection model's task. To address this simplification, Sachdeva and Zisserman [147], [148] introduce a network architecture suited to detect a much more general form of change based on 'object-level' difference (irrespective of the object class), by predicting bounding boxes on both images to indicate change. Their model uses DINO [117] as a Siamese feature extractor and relies on re-projecting image features to a common coordinate system, while ignoring occluded pixels from the novel view.

Two works apply change detection to manual assemblies [138], [139], demonstrating that the BIT architecture can be used to segment change on assembly objects even under rotations. However, both apply change detection to a simplified dataset, containing images gathered from a fixed top-view camera, and a nearly constant roll, pitch, and scale for the object states. Most importantly, they do not apply change detection for the segmentation of errors, but to distinguish the differences between five highly distinctive, pre-defined object states. Therefore, their model is not capable of generalizing to unseen states, *e.g.* with assembly errors. These constraints greatly simplify their task, making it unsuitable for assembly error localization in many real industrial use cases.

The work in this chapter addresses the following limitations of related works:

- *Error localization.* The use change detection for error localization, requiring an

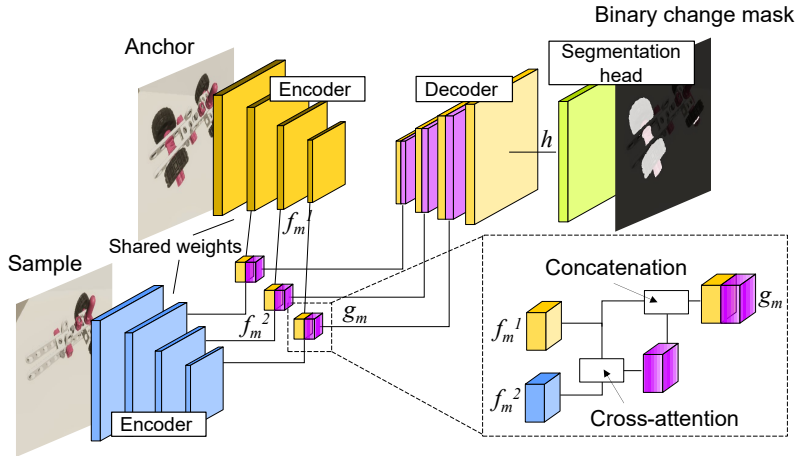


Figure 5.2. Proposed change segmentation architecture, modified from [147], consisting of a Siamese encoder, cross-attention based feature-fusion blocks as skip connections in the U-Net-style decoder, and a two-layer convolutional segmentation head.

entirely different training strategy.

- *Object complexity.* The proposed approach is evaluated on the recently introduced IndustReal dataset [105], which is significantly more complex than prior works, due to the large number of parts (36) and the high degree of visual similarity between states.
- *Generalization.* The proposed work is tested for generalization to assembly states not seen during training as well as entirely unseen error types.

5.3 Method

Given two images, an anchor and a sample image, containing objects in similar orientations during an assembly or maintenance procedure, our objective is to segment the meaningful change between these two images. This meaningful change corresponds to error segmentation when provided with a known error-free anchor image and a sample image with unknown presence of defects. This segmentation can provide valuable feedback to workers during the execution of assembly procedures. We propose an approach for error segmentation based on a modified change detection algorithm, and a novel approach for image-pair generation that provides full control over the type and degree of change that is considered meaningful.

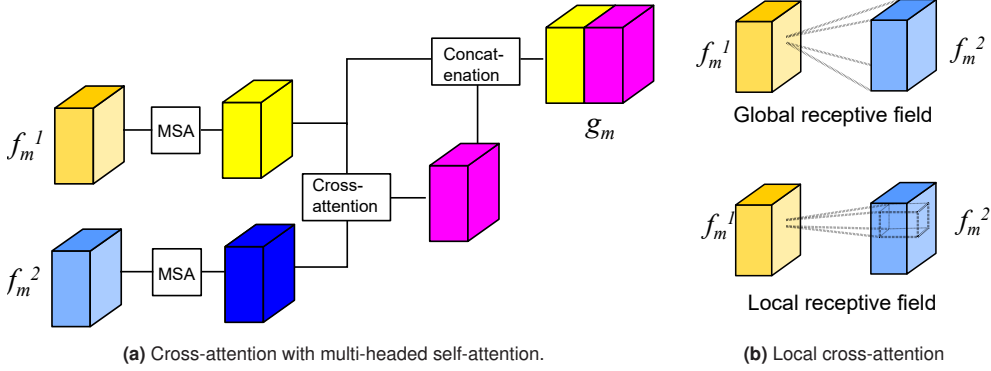


Figure 5.3. Two proposed variations to feature registration based on global cross-attention.

5.3.1 Model architecture

The architecture is modified from a state-of-the-art change detection algorithm [147], consisting of a U-Net-style Siamese encoder and decoder [149]. The original bounding-box head is replaced by a segmentation head. Furthermore, one of the decoders is removed, to make predictions on only the anchor image instead of both. The architecture of the model is depicted in Fig. 5.2. Both the anchor and sample images are independently passed through the same convolutional encoder to obtain feature maps f_m^1 and f_m^2 at multiple spatial resolutions m . The decoder is tasked with up-sampling and segmentation of the differences in features between f_m^1 and f_m^2 , to produce a feature map h at the original resolution of the sample image. For this purpose, each skip connection contains the feature map f_m^1 concatenated with the cross-attention between f_m^1 and f_m^2 , to propagate the features from the sample image to the decoder.

The use of this cross-attention module, called global cross-attention (GCA), is the architecture's key component for change detection [147]. For each feature vector $f_m^1(x, y)$ at spatial location (x, y) , the concatenated output of the cross-attention mechanism contains the sum of all feature vectors in f_m^2 , weighted by their similarity to the feature vector $f_m^1(x, y)$. Intuitively, the most similar features of f_m^2 become concatenated to f_m^1 at the same spatial location, allowing the model to predict change by identifying at which spatial locations no matching feature was found in f_m^2 , since then the output of the cross-attention is simply some random weighted average of the features in f_m^2 . However, there are some inherent fundamental flaws in this cross-attention module, some of which become particularly relevant when employed in the industrial domain. The following paragraphs discuss these flaws and two proposed ways to address them.

A Linear multi-headed self-attention:

Firstly, relying on registering each feature vector in f_m^1 with (its most similar counterpart in) f_m^2 assumes that there is a rich, descriptive and discriminative feature at each spatial location of f_m^1 . However, assembly images contain a lot of textureless, and therefore featureless, background, which can easily be ‘matched’ by the cross-attention module across images. Thus, if the sample image contains a part that is not in the anchor image, and the spot where this part is missing from the anchor image contains textureless background, it is harder to detect this part as missing from the anchor image. Inspired by the work of Sun *et al.* on feature matching [150], we propose to combine GCA with their linear multi-headed self-attention method (GCA+MSA) for enriching features in textureless image areas with global contextual information, demonstrated in Fig. 5.3a. In theory, this should enable points on textureless regions to be corresponded with the same points across image pairs, where they are described not only by their CNN-extracted local features, but also by the global information from the image. For instance, the global information can contain a point’s distance from key assembly object parts. Thus, computing the cross-attention on these globally-enriched feature vectors can enable the model to better identify changes in places where the anchor image contains background pixels. Instead of full self-attention, we use linear self-attention, since the latter decreases the model’s computation complexity at no cost to performance [150].

B Local cross-attention:

Secondly, the cross-attention is global and, although we train with some orientation differences between images, these differences are relatively minor, meaning that the same part in each image pair will have a similar spatial location. Thus, computing the cross-attention across the entire image is redundant and can introduce noise. We address this limitation by introducing a local cross-attention (LCA) network, where each feature vector in f_m^1 is given a restricted receptive field within which to find a matching feature vector in f_m^2 . This mechanism is illustrated in Fig. 5.3b. It incorporates an inductive bias derived from the task into the model architecture, since we assume that image pairs contain the assembly object with a similar pose. By searching for changes in a more local way, the model becomes less susceptible to being distracted by features in non-corresponding places. This cannot be guaranteed with the standard GCA mechanism, which is completely invariant to spatial location.

5.3.2 Data generation and image pair selection

The objective of the proposed system is to capture meaningful change, *e.g.* a missing part, while being resilient to the expected variability between two images. This variability can be caused by, *e.g.* background noise, motion blur and 6D pose differ-

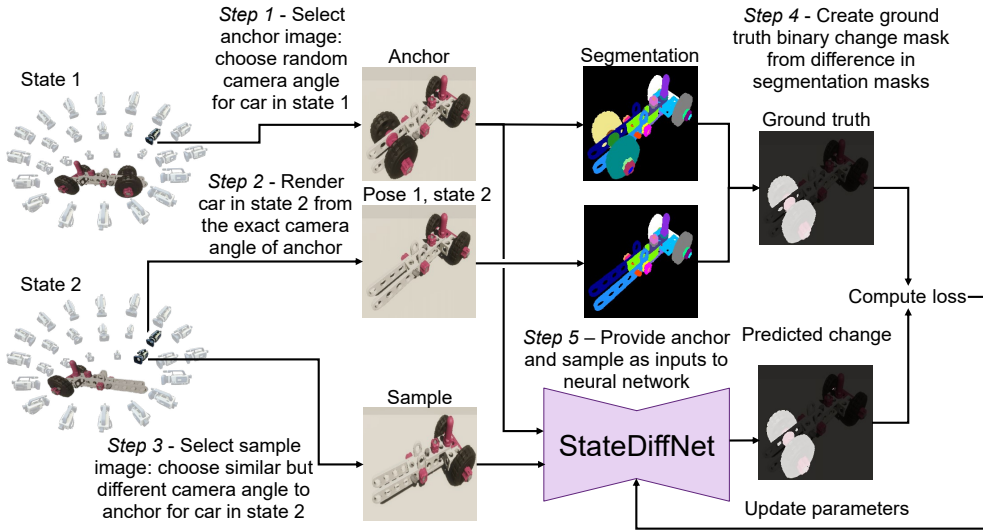


Figure 5.4. Overview of the training process. The ground-truth binary change mask of an image pair is created by taking the difference between the instance segmentation masks of the anchor and the sample image viewed from the camera angle of the anchor.

ences. Training the model entirely on synthetic data enables full control over both the meaningful change and the expected variability between images. Additionally, it allows a large quantity of training data to be gathered with many randomly selected instances of meaningful change, which we posit to be crucial for a model tasked with finding errors in states that are not in the training data. To this end, we propose an innovative data generation and selection pipeline that can be applied to any assembly object for which a 3D model is available. The process is described in detail in Fig. 5.4. Briefly, given an assembly object in many different possible states, image pairs are generated by choosing two states with a controlled amount of state difference (defined by the difference between visible parts), and generating three synthetic images with instance-level segmentation masks: the sample image, the anchor image, and the assembly state of the sample image in the pose of the anchor image. The segmentation masks of the latter two are used to construct a binary change mask, containing all of the meaningful change between both images. We add variance between images by randomizing the background planes and lighting conditions and varying the camera pose. This pipeline provides full control over the object poses, as well as the amount of part-level differences between the anchor and sample image used during training and testing.

5.4 Experiments

This section outlines our experiments. First, we provide the experimental setup and demonstrate that our model performs well at change segmentation on synthetic images with unseen object states, both for object-camera poses seen during training as well as for novel poses. Additionally, a qualitative analysis of the model’s performance on real-world test data is provided. To understand the functionality and limitations of the model, an ablation study is presented, where the approach is trained and applied to an even more challenging dataset, wherein the test set contains not just novel part configurations but entirely novel parts.

5.4.1 Experimental setup

Datasets: In this work, we use the IndustReal [105] dataset. The assembly is performed on a toy car composed of 34 parts that can be assembled following many different orders, generating potentially millions of different states in the proposed synthetic data-generation pipeline. The car’s pose relative to the camera varies along all six degrees of freedom, pieces of the car to be assembled are scattered in the background, and parts of the car are occluded by the user’s hands when being held. Based on the 3D model of the toy car, we generate a training dataset consisting of 10^6 images, containing 5,000 different states and 200 camera poses per state, selected to give a top-side view of the car. The data are generated using Unity Perception as a simulator [151] and the rendered states are created by randomly removing parts without creating disconnected parts. The data-generation pipeline is made publicly available.

Two (synthetic) test sets are used to evaluate StateDiffNet. The first has exactly the same poses as the training data, but contains previously unseen states. The second test set contains the same ‘unseen’ states, but with exclusively novel poses outside the range of the ones in the training set. We qualitatively test the model on real-world images with assembly errors from the IndustReal dataset, including error types that are entirely missing from the training and test sets, namely misplacement or wrong orientation of components. Each real-world error image is paired with a *synthetic* anchor image containing its intended state under a sufficiently similar pose to the real image.

Implementation details: Experiments are conducted using a ResNet-34 encoder [152] backbone with ImageNet-pretrained weights and a two-layer convolutional segmentation head. The cross-attention module is used as skip connection at the spatial resolutions 8×8 , 16×16 , and 32×32 , with channel resolutions 512, 256 and 128, respectively. At each of those resolutions, the GCA+MSA model introduces a linear self-attention layer according to the implementation of LoFTR [150] to both feature

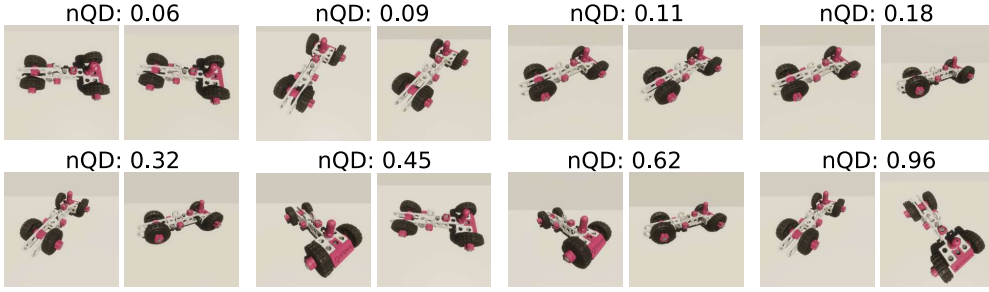


Figure 5.5. Visualization of nQD measure for orientation difference between various image pairs.

maps prior to the cross-attention layer, as shown in Fig. 5.3a. Positional encoding is added to the feature maps prior to the self-attention, which uses 8 heads and a dimension equivalent to each channel resolution. At each spatial resolution, the model trained with LCA use an empirically selected cross-image receptive field of 5×5 , 7×7 , and 11×11 , respectively.

All images are region-of-interest (ROI) cropped and resized to 256×256 pixels, and color jitter, random rotation, vertical flipping, horizontal flipping and Gaussian blurring are used as image augmentations on the training set. ROI crops are generated by adding a 10% margin to the bounding-box annotation of the object in each image and randomly translating the object therein. Each experiment uses a batch size of 64, with 15 warmup epochs and a learning rate of 10^{-5} with cosine learning rate decay to 0 over 400 epochs using the Adam optimizer [153] to minimize the cross-entropy loss. Following the convention for segmentation tasks, we report our results on the Intersection over Union (IoU) metric [140] over the *change* class. Unless stated otherwise, we set the number of differences in parts to detect during training between 1 and 6 and the threshold on the orientation difference between anchor and sample images to 0.1 norm of quaternion difference (nQD), defined as

$$\Phi(q_1, q_2) = \min\{\|q_1 - q_2\|, \|q_1 + q_2\|\}, \quad (5.1)$$

where $\|\cdot\|$ denotes the Euclidean norm and q_1 and q_2 are unit quaternions. The minimum operation is required because unit quaternions q and $-q$ denote the same rotation.

The nQD is used in this chapter to systematically control the amount of viewpoint variance between image pairs presented to the model during training and testing, as visualized in Fig. 5.5. Huynh [154] compares multiple potential metrics to quantify rotation difference. The nQD is found to be a suitable metric, since small values correspond to nearby rotations and large values correspond to distant rotations (unlike *e.g.*, the Euclidean distance between Euler angles). Moreover, the nQD metric is more intuitive and straight-forward to implement.

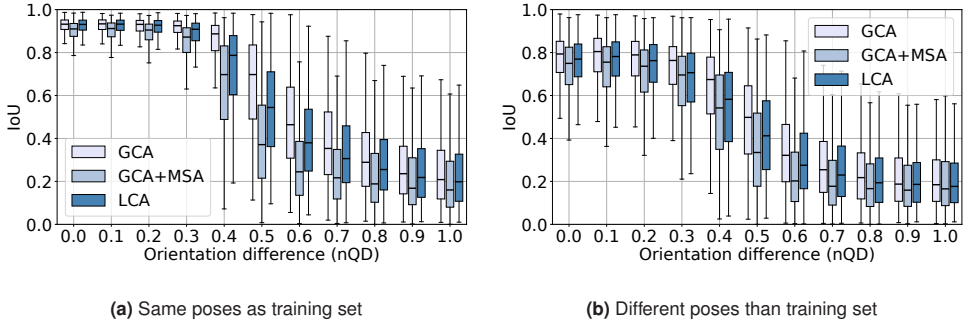


Figure 5.6. Performance of three models with varying orientation difference. The box plot shows the median and first and third quartiles.

5.4.2 Error localization performance

Synthetic test sets: The results of all three tested models, each evaluated on varying orientation differences and with 4 to 6 parts as change between the anchor and sample images, are shown in Fig. 5.6. Results for different amounts of part-changes are demonstrated in Fig. B.1 (Appendix B). It can be observed that the model with global cross-attention (GCA) performs best, especially at high orientation differences, and the model with linear MSA performs worst. The local cross-attention performs similarly to the GCA at small orientation differences, but, as expected, degrades for large orientation differences.

Nevertheless, despite being trained with only relatively small orientation differences (0.1 nQD), most of the models exhibit decent performance on image pairs with considerably more orientation differences, up to 0.3 or 0.4 nQD. When evaluated on poses that are not seen during training, as shown in Fig. 5.6b, the overall performance of all models drops by around 0.1 IoU, and the variability in performance between images (whisker size) grows considerably. This indicates that the models are overfitted on the range of poses encountered during training.

Real-world test set: The qualitative results of all three models on real-world images, without using any explicit domain adaptation techniques, are shown in Fig. 5.7. All models exhibit decent performance at detecting the assembly state errors, despite the obvious differences in visual appearance between real and synthetic images. The GCA model has the most accurate predictions, particularly because it scores less false positives compared to the other attention mechanisms. The qualitative analysis suggests that our approach to assembly error localization is able to handle missing (Rows 1-2) and mis-oriented parts (Row 3), but fails at localizing placement errors (Row 4). Furthermore, the models successfully disregard scattered background objects that do not constitute meaningful change (Rows 2-4).

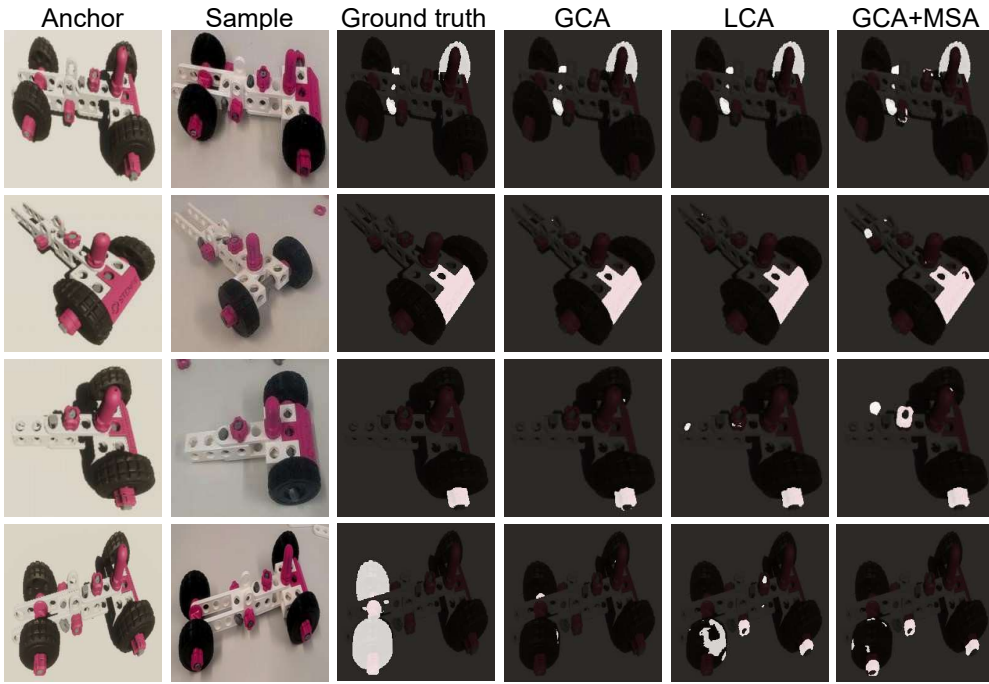


Figure 5.7. Real-world qualitative results on different assembly errors. The global cross-attention based model performs best, particularly on missing parts, components, and mis-orientations. No model is able to reliably detect placement errors (last row).

5.4.3 Ablation study

In addition to demonstrating that the StateDiffNet is capable of generalizing to new configurations, poses, and real-world data, we perform an ablation study. To this end, we investigate (1) whether the proposed system can be used out-of-the-box in scenarios where a part of the assembly is upgraded, *i.e.* how it performs on parts that were never seen as change during training, (2) the impact of orientation differences between the anchor and sample in training, and (3) the difference between a part being added versus removed between the anchor and sample. This ablation aims to investigate the GCA performance on a more complex task and to understand the limitations of the cross-attention mechanism.

For the new objective of detecting change in parts that are entirely unseen and/or never changed during training, a new training set is created, consisting of 1,000 different states of the car with the same 200 camera poses per state as used previously. The dataset is designed such that one part, specifically the front bracket, is always present, while two parts (the pulley and fourth wheel) are never present. Therefore, no change segmentation is ever obtained on those three parts during training. The same synthetic test set, which contains no limitations on the parts that are removed, is used as for prior experiments.

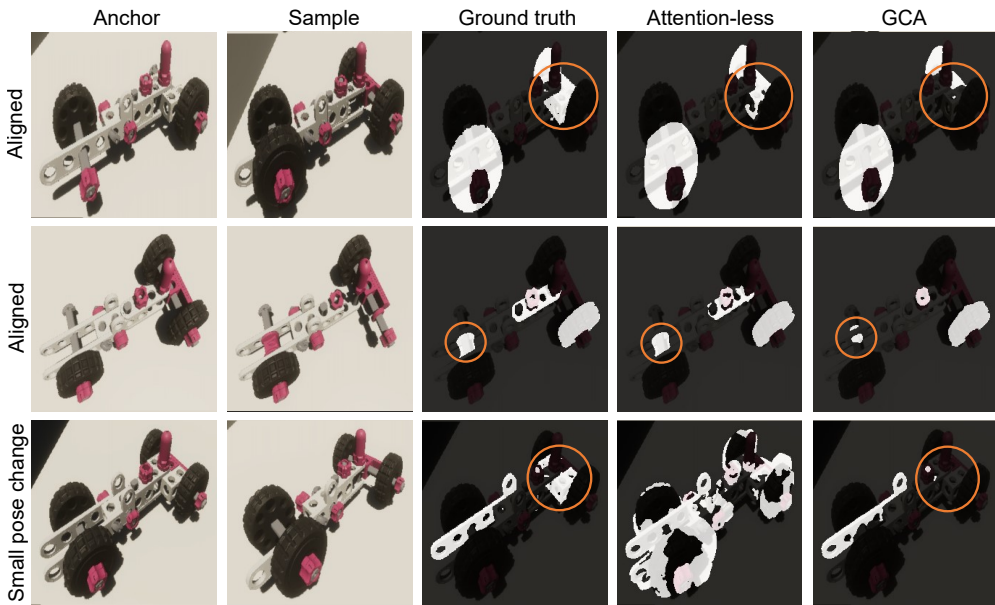


Figure 5.8. The attentionless model (fourth column) refers to the concatenation-based model, trained on perfectly aligned images, and GCA (last column) refers to the cross-attention based model, trained on small orientation differences. The orange circles highlight parts that never constitute change in the training set. The attentionless model is seemingly more capable to capture change on those parts than the GCA-based model, which particularly fails to detect any change on the front bracket (rows 1 and 3). Note that the first two rows contain perfectly aligned image pairs and the last row contains an image pair with orientation difference. In the latter case, it can be seen that the attentionless (concatenation-based) model predicts a lot of change because it is only able to compare features across images at the exact same pixel coordinates, while the GCA model is able to capture meaningful change on the parts that it encounters during training.

Perfect image alignment versus slight orientation differences: To validate the fundamental feasibility of a model that generalizes to detecting changes on unseen parts, we first test whether this is possible for two images that are perfectly registered. This is tested by training one StateDiffNet without any attention mechanism, by simply concatenating the two feature maps instead of using cross-attention, and another StateDiffNet with global cross-attention, on perfectly aligned images. The results are presented in Fig. 5.9, and demonstrate that the best performance is achieved when cross-attention is not used. This is intuitive, since the images are already perfectly aligned, rendering the use of cross-attention to register features across images obsolete and more sensitive to noise. Since features only need to be compared locally, a purely convolutional network performs best at detecting change on registered images. The network can even detect changes never encountered during training, as demonstrated in Fig. 5.8, by simply determining whether identical features are present in both images at each spatial location. We posit that this allows change to be detected at the ‘feature’ level, instead of requiring a semantic understanding of the assembly object to reliably detect change, thereby greatly simplifying the problem and enabling better

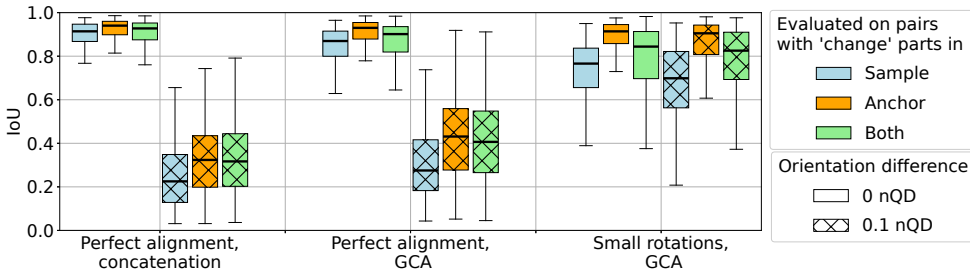


Figure 5.9. Performance on perfectly aligned and differently oriented images for three different models. The first two are trained on aligned image pairs, one with simple feature concatenation, the other using cross-attention. The third is trained on slight orientation differences. The colors represent the origin of the changes to be segmented, *i.e.* parts that are only present in the sample (blue), anchor (orange) or both (green) images, and the hatches indicate the maximum nQD encountered during testing.

generalization to unseen parts.

However, as soon as slight pose differences of the assembly object are introduced, the aforementioned models fail to meaningfully detect change, as can be seen in Fig. 5.9. Only when the model with GCA is trained on image pairs with orientation differences is it able to segment change on non-aligned images. However, we observe that this model performs worse than both the previous models on perfectly aligned images. This performance drop is caused by the model failing to generalize to new changes when input images are not perfectly aligned. This failure to generalize is clearly visible in the qualitative results shown in Fig. 5.8.

In conclusion, change detection without attention generalizes well to change in unseen parts when images are perfectly aligned. However, when the images contain some orientation difference, cross-attention is required for feature registration, removing the model's ability to generalize to this type of novel change.

Parts missing in anchor versus parts missing in sample: StateDiffNet is tasked with segmenting all the changes identified between image pairs on the *anchor* image. Thus, for changes of parts that are present in the anchor and missing from the sample image, the model needs to predict a segmentation mask over those parts. However, when the change is caused by a part missing from the anchor image, the model needs to provide a segmentation mask corresponding to where that part *would be in the anchor image*, which is a considerably more difficult task. The increased difficulty is highlighted in Fig. 5.9. Regardless of whether the model is trained on perfectly aligned images, it performs best for parts present in the anchor image. This performance difference is amplified when the model is trained on non-aligned input images: the model's segmentation performance of parts present in the anchor image drops by 2% IoU when introducing orientation change, whereas it drops by 10% when the parts are only present in the sample image. This highlights a fundamental flaw of the cross-attention mechanism: features from the sample image are only transferred to

the decoder if similar features are present in the anchor. Therefore, features from the sample image that do not match any features from the anchor image are lost.

5.5 Discussion

This section analyzes observations from the experimental results, focusing firstly on attention mechanisms, secondly on generalization limits, and finally on the formulation of error localization.

5.5.1 Global cross-attention mechanism:

This work has compared three attention mechanisms for feature registration, namely global cross-attention (GCA), local cross-attention (LCA), and multi-headed self-attention (GCA+MSA). These three mechanisms are compared, because it is not intuitive that GCA is suitable for error localization: it lacks positional encoding and intuitively *should* therefore not be able to register features from feature-less regions. However, the results surprisingly demonstrate that the GCA mechanism performs best. To understand this, we visualize the cross-attention weights of the GCA in Fig. 5.10.

The example in Fig. 5.10 demonstrates that, even if the queried area consists of mostly background, the GCA attends to the wheel in the sample image that is missing from the queried area in the anchor (Fig. 5.10d). Therefore, we conclude that the cross-attention does not match visually similar features as hypothesized and shown by [147] on their change detection dataset. Rather surprisingly, despite having no spatial encoding of the tokens in the cross-attention, it is apparently capable of leveraging contextual information (*e.g.* the parts surrounding the wheel) to find the matching wheel in the sample image. This is further evidence that, at least in our experimental setup, the cross-attention has semantic understanding of the parts and their 3D configuration, and does not merely operate at a feature-matching and registration level.

5.5.2 Memorization versus generalization

The tested models all suffer a performance drop for poses that were never encountered during training, compared to poses that lie within the training set (Fig. 5.6b). This result suggests that the models may determine whether all parts in one image are present in the other, learn missing part locations for each pose by heart, and produce the memorized segmentation mask for any parts that are changed. This is further evidenced by the worse generalization to entirely new parts when that part is missing in the anchor, compared to when that part is missing from the sample image (Fig. 5.9). Therefore, the proposed algorithm must be retrained when a part is modified or added

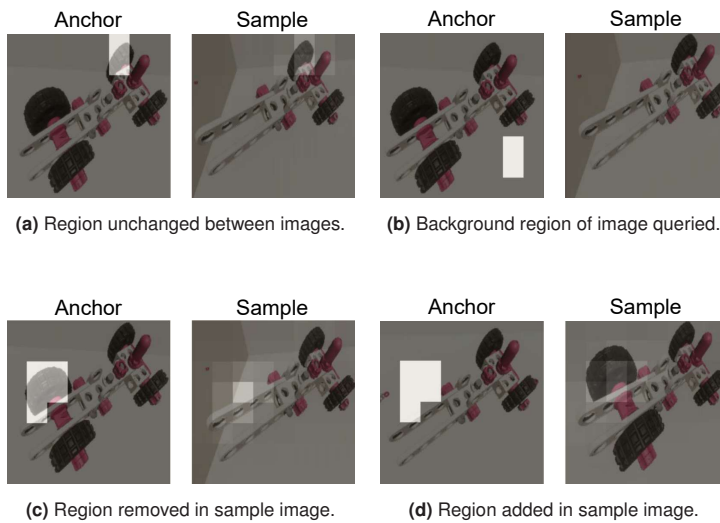


Figure 5.10. Visualization of GCA weights on a sample image for queried areas on anchor image. The GCA behaves as expected by attending to the most similar region in the sample (a). Similarly, GCA behaves as expected by not attending to any specific region when the queried background region matches many regions in the sample image (b). However, for parts present in the anchor but not in the sample (c), the GCA attends to the region where the part is expected to be located in the sample. Surprisingly, when selecting a background region in the anchor image where a part can potentially be placed (d), the GCA attends to the region in the sample image where this part is actually placed. Best viewed on a computer monitor.

to the assembly. The model only depends on synthetic data and, therefore, retraining the model only incurs computational cost (no manual annotation efforts are required). Nonetheless, these insights highlight a valuable direction for future research into the use of cross-attention for change detection.

5.5.3 Change segmentation versus change detection

In this work, error localization is addressed as a segmentation rather than bounding-box detection problem because segmentation provides a more fine-grained prediction. Therefore, segmentation can provide more feedback to operators, creating a better user interface, *e.g.* for an AR system. Nonetheless, training for segmentation rather than bounding-box prediction has the inherent flaw that the model optimizes towards precise segmentation of fine edges, forcing the model to learn the projection of missing parts by heart at the cost of generalization. To improve generalization to new parts and poses while maintaining fine-grained predictions, it can be beneficial to optimize a cost function that less focuses on edges.

5.6 Conclusion

In this chapter, error localization for assembly procedures is addressed. On one hand, anomaly detection approaches are not directly applicable, since the correctness of an object state depends on the intended state in the procedure. On the other hand, related assembly state detection approaches focus on assembly configurations with limited complexity and do not investigate error localization. Therefore, this chapter has introduced an approach called StateDiffNet, tasked with change segmentation between an image of the intended assembly state and an observed image. Since the intended states are known to be error-free, any change with respect to the sample image can be considered an error.

Findings. This chapter provides two notable findings. Firstly, it is demonstrated that with minor modifications to the algorithm, a change detection algorithm can be adapted for assembly error localization (StateDiffNet), pinpointing the exact location of potential errors. This is enabled through a data-generation pipeline that allows for the explicit modeling of the expected variability between image pairs, such as camera pose, photometry, and image distortions. Secondly, the StateDiffNet demonstrates strong performance for error segmentation on a complex assembly configuration, as shown quantitatively on synthetic data and qualitatively on real-world data. The approach is able to localize errors in unseen assembly configurations and even for entirely unseen error types. This is achieved while exclusively utilizing synthetic training data. Besides these two findings, valuable insights into the cross-attention feature-registration mechanism are presented. For example, the conducted experiments have demonstrated that this mechanism is capable of leveraging contextual information to

attend to feature-less regions. This is unexpected since the mechanism does not have any spatial encoding.

Limitations. Although the aforementioned findings are significant and directly address the fundamental limitations presented in the previous chapter, they reveal new challenges. The first challenge is that the error localization algorithm requires the object to be (i) entirely within the camera FoV and (ii) unobstructed by occlusions such as hands or tools. Such frames are rare in ego-centric video data, where the camera moves with the head of the user and hands are frequently interacting with the object. Secondly, all PSR algorithms discussed thus far in the thesis, process video frames as individual images, leveraging only spatial features. Therefore, concepts such as temporal consistency are not yet considered. Thirdly, while the performance of StateDiffNet is extensively evaluated on synthetic data, it is only qualitatively examined on real-world data. Finally, the experiments demonstrate that feature registration is the limiting factor in training a model to localize errors on entirely unseen parts and that two ideas towards enhanced feature registration do not improve performance. This aspect provides an interesting direction for future work.

In conclusion, this chapter has demonstrated that an error localization algorithm can find new, unseen errors in assembly procedures, without the need to pre-define or train on any erroneous samples. Furthermore, this can be effectively achieved by exclusively training on synthetic data. As in the previous chapter, these results are relevant beyond the domain of procedural activity recognition, as demonstrated by the application of an EP patent filing within the domain of defect identification and segmentation for SEM images [IP1].

Building on the previous chapter, this chapter has addressed the fundamental limitations to error recognition in assembly procedures by highlighting the regions where errors occur. Additionally, the proposed model relies only on synthetic training data. However, the algorithm assumes that all relevant objects remain entirely within the camera's FoV and the algorithm does not leverage any temporal features, potentially limiting its applicability. This shortcoming will be addressed in the succeeding chapter, by training a second model to learn temporal features. Furthermore, StateDiffNet has only been qualitatively evaluated on real-world data. This is addressed in Chapter 7, where the algorithm is implemented into a real-time, real-world system to provide guidance during assembly procedures.

Procedure step recognition through spatio-temporal modeling

6.1 Introduction

The preceding chapter has introduced an algorithm that localizes errors in images of objects undergoing maintenance or assembly procedures. Prior to this, Chapter 3 has demonstrated an assembly state detection algorithm for procedure step recognition. However, both methodologies require the object of interest to be unoccluded, and with all parts entirely within the image boundaries, regardless of the relevance of a part to the current step completion. Furthermore, none of the approaches is directly optimized for PSR. Instead, PSR predictions are made by inferring the completed steps between recognitions. This chapter presents a framework for training a neural network directly on the PSR labels, allowing the model to make predictions when only the relevant parts are visible.

Understanding and tracking key steps in videos of procedure execution is an active line of research [21], [49]–[54]. Procedural tasks often require multiple interdependent actions to be executed correctly, although reliably recognizing these steps remains challenging due to variable viewing angles, frequent occlusions, and flexible execution orders. Particularly, ego-centric cameras introduce dynamic perspectives and frequent hand occlusions, obscuring key objects or actions and creating gaps in visual continuity that complicate accurate procedural tracking [22], [57], [155]–[158].

Existing work on recognizing and tracking procedural videos takes multiple approaches, such as action recognition [21], [22], [49], [50], procedural activity understanding [51]–[53], keystep recognition [49], [54]–[56], and, as introduced in Chapter 3, procedure step recognition [105]. In procedural action recognition, the objective is to categorize actions performed by humans in short video clips. In contrast, the task of procedural *activity* recognition focuses on understanding longer videos. Within this, the task of *keystep* recognition aims to find key moments and their sequences in long videos of procedure executions. Crucially, the aforementioned approaches lack a measure of completeness and correctness for the recognized actions and steps. PSR

addresses this limitation by aiming to identify all *correctly completed* procedural steps in long videos, each contributing to the successful execution of a procedure. A notable limitation of the approach presented in Chapter 3 is its reliance on assembly state detections (ASD) [64], [112], [159]–[161], which only provide reliable predictions for unobstructed views of the object. Such unobstructed video frames are rare, particularly in ego-centric videos. As motivated in Chapter 2, ASD requires full visibility of all object parts to determine the assembly state, making it poorly suited for scenarios with frequent occlusions. Moreover, ASD is based exclusively on spatial information, operating on single frames. Contrarily, PSR is a temporal video understanding task, requiring reasoning over video sequences to assess which steps have been successfully completed. Yet, none of the existing PSR methods have explicitly leveraged temporal features for this purpose. Additionally, the existing approaches are only evaluated on a single dataset, namely the IndustReal [105] dataset. Therefore, the generalization to other data remains unknown.

This chapter explores the hypothesis that procedural step recognition does not require full visibility of the object, but only of the components pertinent to the current step completion. The aim is to determine whether a spatio-temporal model, trained directly on PSR labels, can address the constraints imposed by ASD-based methods. Specifically, the following research questions are addressed.

- RQ6.1 *Addition of temporal features.* How can a spatio-temporal neural network be trained directly for the PSR task?
- RQ6.2 *Partial visibility of objects.* As the algorithm is directly optimized for PSR, can it recognize step completions with a reduced delay, since it intrinsically does not have to wait for images with the assembly object unoccluded and entirely visible within the image?
- RQ6.3 *Generalization.* How does a spatio-temporal neural network for PSR perform on another dataset, and how does this compare to the approach outlined in Chapter 3?

To overcome the aforementioned limitations and address these research questions, the novel Spatio-Temporal Occlusion-Resilient Modeling for Procedure Step Recognition (STORM-PSR) is introduced. STORM-PSR is outlined in Fig. 6.1 and is a two-stream approach for PSR that combines ASD with a spatio-temporal model to recognize procedure steps even under partial occlusions. The ASD stream detects the assembly stage in each video frame whenever unobstructed images are available. In those cases, the robust ASD algorithm provides reliable predictions. However, such unobstructed frames are rare in ego-centric videos due to hand occlusions and camera motion, resulting in significant delays between the actual completion of a step and its subsequent recognition. Therefore, STORM-PSR utilizes a second stream, comprising a transformer-based temporal encoder to aggregate spatial features even when only obstructed images are available. This so-called spatio-temporal stream directly

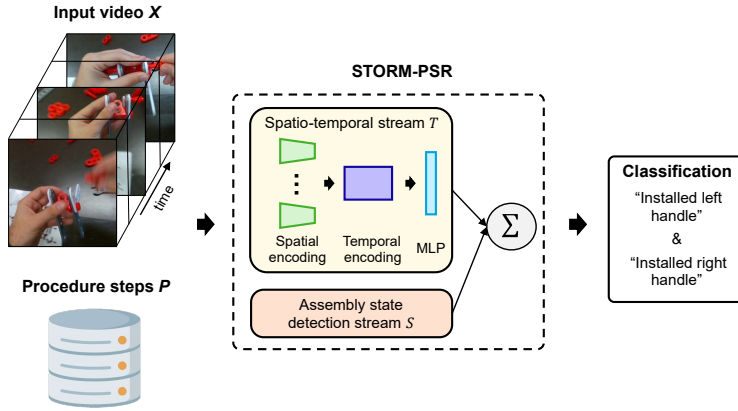


Figure 6.1. Diagram of the proposed Spatio-Temporal Occlusion-Resilient Modeling for Procedure Step Recognition (STORM-PSR) approach, using assembly state detection for non-occluded frames and temporal modeling to recognize step completions when the assembly object is only partially visible.

predicts correctly completed procedure steps, rather than inferring them between consecutive observations of assembly states. It therefore only requires visibility of the parts of the object that constitute a specific step. For example, to recognize the completion of the step ‘install front wheel’, the rear wheel, for instance, does not have to be visible.

These direct PSR predictions are facilitated through a multi-layer perceptron (MLP) head following the temporal transformer, performing multi-label classification to handle multiple step completions within a single video clip. A weakly supervised pretraining approach that leverages sparse step-completion annotations, named key-frame sampling (KFS), is also presented. KFS is inspired by the representation learning framework from Chapter 4 and enables the spatial encoder to provide meaningful representations of step completions. This is achieved by using a contrastive loss that maximizes agreement between the same step completions from different videos and occlusion levels, while minimizing the agreement with different completions. When available, synthetic data of assembly states can be readily integrated with KFS, thereby reducing reliance on annotated real-world data. For training the temporal transformer, key-clip aware sampling (KCAS) is proposed. KCAS is based on a bi-modal distribution to over-sample step completions (positives) and video frames that are shortly preceding these step completions (hard negatives). Simultaneously, KCAS under-samples potentially ambiguous frames surrounding the exact moment of completion, and less informative clips that contain no step completions. Trained with KFS and KCAS, STORM-PSR reduces the average delay (τ) between the completion and corresponding recognition of a step by 26.1% on the IndustReal dataset [57] and 11.2% on the MECCANO dataset [22], which we have annotated with PSR and ASD labels to establish the performance benchmark for this task. In summary, this chapter

contains the following contributions.

- STORM-PSR: a dual-stream approach that leverages spatio-temporal features for PSR, thereby significantly reducing the average delay compared to state-of-the-art methods.
- Key-frame sampling (KFS) for weakly supervised pretraining of the spatial encoder, and key-clip aware sampling (KCAS), a novel sampling strategy for training the temporal encoder.
- Evaluation of PSR generalization to a new dataset, enabled through the newly presented PSR and ASD annotations for the MECCANO dataset [22].

The content of this chapter is divided as follows. The related work is outlined in Section 6.2, after which Section 6.3 presents the proposed methodology. The experiments are described in Section 6.4 and subsequently discussed in Section 6.5. Finally, the conclusion is provided in Section 6.6.

6.2 Related work

This section reviews key work on procedural action understanding, representation learning, and procedure step recognition, highlighting limitations in handling occlusions and temporal reasoning that motivate the proposed approach.

6.2.1 Procedure understanding

Related work explores various approaches to understanding procedural actions, such as action recognition (AR), which focuses on classifying short video clips of procedure executions [21], [22], [50], [162], [163]. Temporal action segmentation (TAS) aims to segment actions, given a long video of procedures [164]–[166]. Other works aim to extract key moments and the sequence in which they occur in long-form videos [49], [51]–[56]. Each of the aforementioned tasks is of interest for procedure understanding, since it is particularly challenging to extract the spatio-temporal features required for acceptable performance. However, these tasks have limited applicability for use cases where a measure of completion, correctness, or quality is desirable or even required, *e.g.* for industrial procedures.

While the aforementioned approaches emphasize human execution of procedural actions, assembly state detection (ASD) focuses on the transformation and progression of an object within a procedure, by detecting subsequent assembly states [64], [112], [159]–[161]. The ASD task typically involves training an object detection network with bounding boxes and corresponding state labels, and performs well on datasets where images contain unobstructed views of assembly states. While ASD can provide a measure of correctness for assembly states [167], ASD approaches do not leverage

temporal consistency on (even briefly) partially occluded objects. Therefore, in ego-centric videos, where assembly states are often partially occluded or out of frame, ASD models are unable to recognize object states.

Such occlusions are evident in procedural datasets such as Assembly101 [21], MECCANO [22], Ikea ASM [162], and IndustReal [57]. To mitigate the impact of occlusions, Assembly101 [21] and Ikea ASM [162] provide multiple viewpoints, while operators in IndustReal [57] were instructed to occasionally provide an unobstructed view of the object. For MECCANO [22], no such instructions were provided, resulting in a highly challenging dataset with only sparse non-occluded images of the toy motorcycle assembly. Finally, only IndustReal provides CAD data to generate synthetic images of assembly states to supplement the real-world dataset.

6.2.2 Representation learning with procedural data

Videos of procedure executions exhibit continuous movements, where objects change gradually, resulting in a high similarity among consecutive frames and larger variations between more distant frames. Consequently, representation learning has emerged as a valuable tool for understanding procedural video data, since it inherently learns frame embeddings that trace the procedure progression through latent space. Bansal *et al.* [56] propose a self-supervised *Correspond and Cut* framework to learn temporally corresponding features by aligning key steps in the embedding space across different videos of the same procedure. Schoonbeek *et al.* [167] propose a representation learning framework for assembly state recognition that enhances generalization to unseen states, demonstrating that supervised contrastive learning generates meaningful embedding spaces for procedural data. For skeleton-based action recognition in videos, contrastive learning has been used to learn multi-granularity anchor-contrastive representations in a semi-supervised manner [168]. Finally, Vidal-mata *et al.* [169] propose a two-stage spatio-temporal approach for temporal action segmentation in instructional videos, employing a U-Net [170] to extract spatial features and an MLP for temporal embedding. Inspired by these works, we propose a weakly supervised representation learning method to pre-train the spatial encoder within a spatio-temporal network.

6.2.3 Procedure step recognition

This section reminds the reader about the procedure step recognition (PSR) task as defined in Chapter 3, outlines the evaluation metrics used to assess system performance, and reviews current limitations of state-of-the-art approaches based on assembly state detection.

A Task definition

The PSR task aims to detect procedure steps that are correctly completed by a person up to time t , given a sensory input $X_t = (x_1, x_2, \dots, x_t)$ (e.g., video) and $\mathcal{P} = \{p_0, \dots, p_N\}$, a descriptive set of the procedural actions to be performed [57]. The predicted set of completed procedure steps $\hat{Y}_t$ consists of both the recognized steps and their respective completion times up to time t , formulated as

$$\hat{Y}_t = \{(\hat{a}_{\sigma(0)}, \hat{t}_{\sigma(0)}), \dots, (\hat{a}_{\sigma(m)}, \hat{t}_{\sigma(m)})\}, \quad (6.1)$$

where $\hat{a}_{\sigma(i)}$ represents the predicted completion of action $a_i \in \mathcal{P}$ at time $\hat{t}_{\sigma(i)}$, and $m + 1$ denotes the total number of recognized procedure steps. The ground truth procedure steps completed up to time t are defined as

$$Y_t = \{(a_{\rho(0)}, t_{\rho(0)}), \dots, (a_{\rho(k)}, t_{\rho(k)})\}, \quad (6.2)$$

where $a_{\rho(i)}$ is the predicted completion of the action $a_i \in \mathcal{P}$ at time $t_{\rho(i)}$, and $k + 1$ denotes the total number of correctly executed steps.

B Evaluation metrics

To quantify the performance of the PSR task, three evaluation metrics are used [57]. First, the procedure order similarity (POS) measures how closely the predicted step sequence aligns with the correct order using the edit distance [95], and is defined as

$$\text{POS} = 1 - \min\left(\frac{\text{DamLev}(Y, \hat{Y})}{|Y|}, 1\right), \quad (6.3)$$

where the $\text{DamLev}(\cdot)$ is a weighted DamLev edit distance [171] function. Importantly, POS does not evaluate conformity to a canonical or “correct” step order. Instead, it measures how accurately the system recovers the order in which the user actually executed the procedure. This allows the POS metric to remain relevant even when steps are swapped or repeated, penalizing the model only if steps are recognized in an order different than the one executed by the user.

The second metric, F_1 score, calculates the harmonic mean of recall and precision for all procedure steps. A true positive (TP) is defined as the prediction of a procedure step $(\hat{a}_{\sigma(i)}, \hat{t}_{\sigma(i)})$ observed at or after the actual completion of the ground truth action $(a_{\rho(j)}, t_{\rho(j)})$, defined as

$$(\hat{t}_{\sigma(j)} \geq t_{\rho(i)}) \wedge (a_i \in Y). \quad (6.4)$$

A false positive (FP) is defined as a procedure step $\hat{a}_{\sigma(i)}$ that is detected prior to the actual completion of action $a_{\rho(j)}$, or if a_i is not completed, defined as

$$(\hat{t}_{\sigma(j)} < t_{\rho(i)}) \vee (a_i \notin Y). \quad (6.5)$$

A false negative (FN) is defined as correctly completed procedure step $a_{\rho(j)}$ that is never recognized by the model, thus is defined as

$$(a_i \in Y) \wedge (a_i \notin \hat{Y}). \quad (6.6)$$

Finally, the average delay (τ) measured in seconds, quantifies the average time between the completion and corresponding recognition of a step and is defined as

$$\tau = \frac{1}{n} \sum_{i=0}^{n-1} (\hat{t}_{\sigma(i)} - t_{\rho(i)}), \quad (6.7)$$

where n is the number of TPs in the prediction $\hat{Y}$.

Interpreting any of the three metrics in isolation can lead to a misleading assessment of system performance. For example, a model that predicts all steps as completed only at the final frame of a video might achieve a high F_1 score while exhibiting a significantly large average delay (τ) and POS scores. Therefore, a comprehensive evaluation requires analyzing the three metrics together to obtain an accurate understanding of the model's overall performance.

C State-of-the-art and limitations

The state-of-the-art approach to PSR relies exclusively on assembly state detections, indirectly achieving step recognition by inferring steps between consecutive assembly states using prior knowledge of the procedure [57]. However, ASD can only accurately recognize object states that are minimally occluded and entirely within the image boundaries, since ASD algorithms predict the state of the whole assembly object requiring a comprehensive view. Consequently, with a single image offering only partial visibility, predictions would require assumptions about the state of non-visible parts. This limitation causes significant delays between the completion and consequent recognition of steps, as unobstructed views of the object are rare, especially in ego-centric videos. Therefore, the proposed approach complements ASD predictions with a novel temporal network, optimized directly to recognize the completion of procedure steps, even under partial occlusions.

6.3 Method

This section details the proposed STORM-PSR, a two-stream framework designed to recognize completed procedure steps under occlusion by combining spatial and temporal cues.

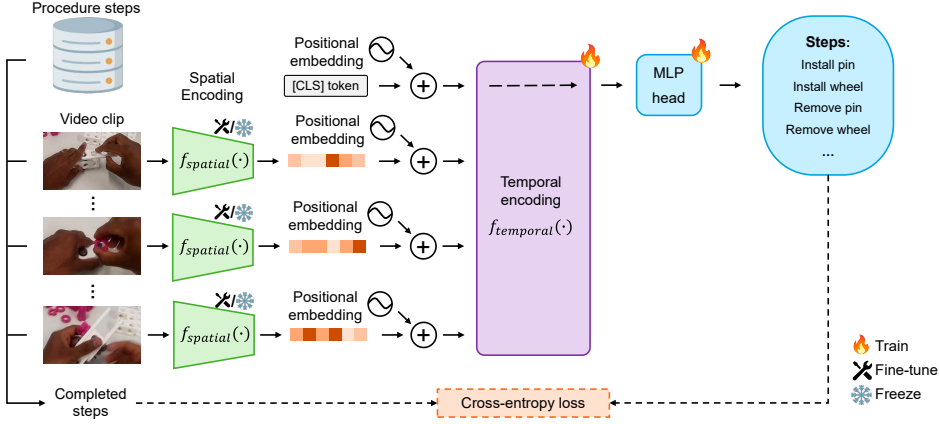


Figure 6.2. Overview of the spatio-temporal stream of STORM-PSR and its training procedure. The frames of each video clip are passed through the spatial encoder $f_{\text{spatial}}(\cdot)$ to obtain per-frame embeddings. These embeddings are combined with a learnable [CLS] token, and augmented with 1-D positional encoding before being fed into the temporal encoder $f_{\text{temporal}}(\cdot)$. The final state of the [CLS] token provides a clip-level representation, which is fed into the MLP classification head that makes the final predictions. Because multiple procedural steps can be completed within a single clip, the MLP performs multi-label classification. During the training of the temporal encoder, the weights of $f_{\text{spatial}}(\cdot)$ can be either frozen or fine-tuned. The cross-entropy loss is used to optimize the model weights.

6.3.1 Overview

This section presents the proposed approach to PSR, using the Spatio-Temporal Occlusion-Resilient Modeling for Procedure Step Recognition (STORM-PSR) framework. STORM-PSR combines an assembly state detection (ASD) stream ($\mathcal{S}$) with a spatio-temporal stream ($\mathcal{T}$), as outlined in Fig. 6.1.

Specifically, given an input video X and a set of procedural actions $\mathcal{P}$, the objective of the two-stream method, STORM-PSR ($\mathcal{F}$), is to map X and $\mathcal{P}$ to $\hat{Y}$, an ordered set of tuples containing each correctly completed procedure step $p \in \mathcal{P}$ and its moment of completion $t \in [0, T]$, where T is the total duration of video X , such that

$$\hat{Y} = \mathcal{F}(X, \mathcal{P}). \quad (6.8)$$

Each stream produces a probability distribution over every procedure step $k \in \mathcal{P}$, denoted by $\hat{y}_{S,k} = \mathcal{S}(X, \mathcal{P})$ and $\hat{y}_{T,k} = \mathcal{T}(X, \mathcal{P})$ for the assembly state and spatio-temporal streams, respectively. The prediction probability $\hat{y}_k$ for the correct completion of step k is then computed as the average with equal contribution from both streams,

$$\hat{y}_k = 0.5 \cdot \hat{y}_{S,k} + 0.5 \cdot \hat{y}_{T,k}. \quad (6.9)$$

As outlined in previous work, the ASD stream $\mathcal{S}$ performs well on video frames where the objects are entirely visible, but fails fundamentally when any parts are

occluded or outside the image boundaries. We propose to mitigate the issue of occlusion by complementing the ASD stream with a spatio-temporal stream, leveraging temporal consistency when parts are occluded.

6.3.2 Supervised training of spatio-temporal model

Frames showing full visibility of all assembly parts are rare, yet previous work relies entirely on such frames [57]. To avoid this, we propose directly optimizing a spatio-temporal model to recognize the correct completion of procedure steps. By directly recognizing correctly completed procedure steps, only the parts relevant to each step need to be visible. Additionally, the temporal model learns to leverage temporal continuity, enabling it to use information from components previously visible, even when temporarily occluded.

The overall structure and training methodology for the spatio-temporal model are illustrated in Fig. 6.2. The temporal model includes a pre-trained spatial encoder $f_{\text{spatial}}(\cdot)$, a temporal encoder $f_{\text{temporal}}(\cdot)$, and a classification head. The temporal encoder comprises a transformer architecture [172]. The frames of a video clip X_i are processed by the spatial encoder $f_{\text{spatial}}(\cdot)$ and the first layer of its projection head $g(\cdot)$ to obtain meaningful embeddings, following the approach in [173]. Analogous to BERT [174] and the vision transformer [128], these embeddings are then combined with learnable 1D positional encodings [128]. A classification token, [CLS], is added at the beginning of the spatial representation sequence to aggregate global information about the clip. After propagating this input sequence through the temporal encoder, the output of the [CLS] token is fed into the classification head. The classification head is implemented using an MLP with one hidden layer, designed for multi-label classification at the clip level. The output dimension of the classification head corresponds to the number of unique assembly components in the procedure $\mathcal{P}$, where each component is defined as the object part(s) necessary to complete a procedure step correctly.

Similar to the video transformer network (VTN) [175], the two encoders are trained separately for distinct objectives. The spatial encoder is pre-trained in a weakly supervised manner to learn meaningful spatial features from similar assembly states, as outlined in the following section. Subsequently, the temporal encoder is trained to capture temporal dynamics, by minimizing the multi-label binary cross-entropy loss $\mathcal{L}_{CE}$ to identify which components (if any) have undergone *correctly* completed procedure steps in the video clip X_i . The loss function $\mathcal{L}_{CE}$ is defined as

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C [y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log(1 - \hat{y}_{ij})], \quad (6.10)$$

where N is the number of samples in the batch, C the number of steps to be completed, $y_{ij} \in \{0, 1\}$ the binary ground-truth label for sample i and class j , and $\hat{y}_{ij}$ the predicted

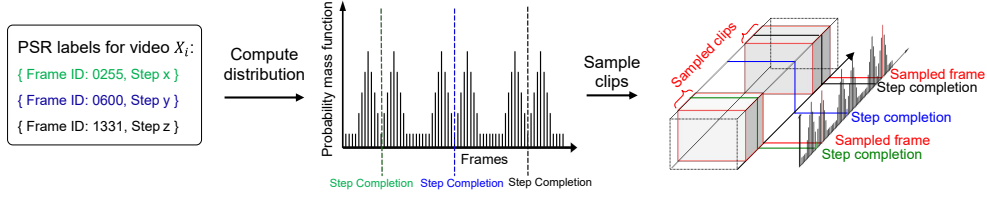


Figure 6.3. Principle view of generating the probability mass function with bi-modal sampling used in key-clip aware sampling (KCAS). The KCAS over-samples hard-negative clips (directly before a step completion occurs) using the first Gaussian of the bi-modal distribution, and positive clips (directly after the step completion) with the second Gaussian. Potentially ambiguous frames at the exact moment of step completions are sampled at a reduced rate, as well as frames where no steps are completed. The long tails of the distribution deliberately suppress background clips far from any completion, preventing the mini-batch from being dominated by uninformative content.

probability. A target value of zero is assigned to each procedure step that was not correctly completed within the clip, and a target value of one is assigned if it did.

Similarly to [57], a temporal filter is used to only consider predictions that are consistent across multiple frames. To this end, the confidence for each predicted step completion is accumulated over time until it reaches a threshold T , at which point the step completion and time-stamp are added to $\hat{Y}$. On frames where no assembly state is recognized, the accumulated confidence scores for all incomplete steps decay at a fixed rate.

6.3.3 Key-clip aware sampling

Since most video clips do not contain step completions, uniformly sampling input clips for each mini-batch of the spatio-temporal model is inefficient. To address this, we propose using key-clip aware sampling (KCAS), which increases the sampling probability around step completions in each video.

Specifically, KCAS creates a probability mass function, formed by two Gaussian distributions around each step completion time. Let $T = t_1, t_2, \dots, t_m$ be the set of timestamps corresponding to the completions of procedure steps in video X_i , where t_j represents the completion time of step j , and δ be the separation between the two Gaussian distributions for each step completion. The probability mass function $p_i(x)$ for sampling a clip ending at frame x is then defined as

$$p_i(x) = \sum_{t_j \in T} [g(x | t_j - \delta, \sigma) + g(x | t_j + \delta, \sigma)], \quad (6.11)$$

where $g(x | \mu, \sigma)$ is the probability mass function of a Gaussian distribution with mean μ and standard deviation σ . To ensure that $p_i(x)$ is a valid probability mass function, we normalize it by dividing $p_i(x)$ by the sum over all values of x .

A bi-modal distribution is chosen over a single Gaussian because of potential ambiguity regarding the exact frame of completion for a procedure step, e.g. the

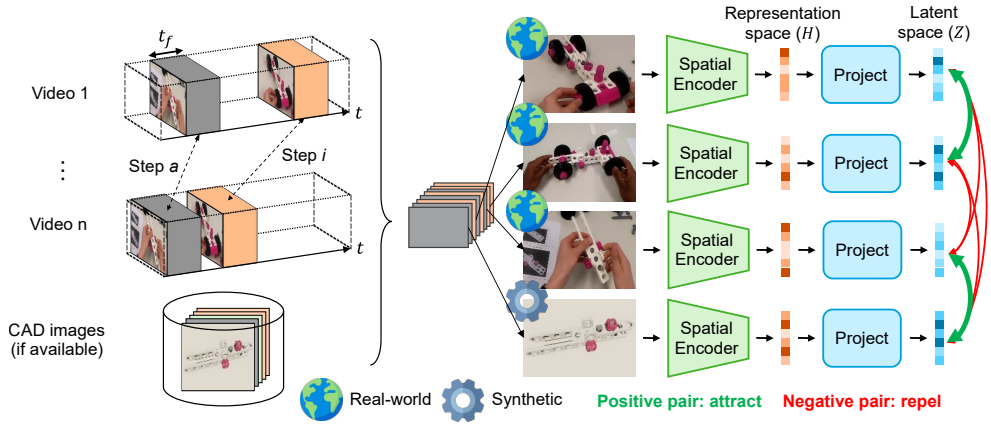


Figure 6.4. Overview of the key-frame sampling (KFS) weakly supervised pretraining. Each mini-batch consists of real-world images sampled around the time stamps of the PSR labels from different videos, with a sampling window t_f . When CAD data are available, synthetic images may be included. The sampled images are passed through a spatial encoder $f_{\text{spatial}}(\cdot)$, comprising a ViT-S [128] architecture, into representation space H , after which a three-layer MLP $g(\cdot)$ projects the embeddings into another space Z , on which the supervised contrastive loss [178] is calculated.

moment when a nut is tightened to the desired level. We pose that those clips prior to completion can act as hard-negative samples, since they contain important information on the appearance of nearly completed steps. Therefore, as demonstrated in Fig. 6.3, KCAS explicitly samples *fewer* video clips ending in the ambiguous moment, and *more* clips that end prior to and after the step completions. For each index sampled from the probability mass function, a temporal window of w frames prior to this index is created, from which N_w frames are sampled. To reduce computational cost, a stride can be applied to limit the number of sampled frames ($N_w \leq w$) [176], [177].

6.3.4 Weakly supervised training of spatial encoder

The goal of weakly supervised training of the spatial encoder is to encourage it to learn occlusion-robust state representations from limited supervision. Rather than requiring dense frame-level assembly state annotations, we leverage PSR labels as weak signals to group visual observations that correspond to the same procedural step, despite differences in viewpoint, occlusion, or tool presence. This enables robust feature learning from real-world observations, with synthetic data serving as an optional supplement that can enhance training.

The spatial encoder $f_{\text{spatial}}(\cdot)$ is pre-trained with contrastive learning, outlined in detail in Fig. 6.4. The approach, inspired by [167], encourages $f_{\text{spatial}}(\cdot)$ to project instances of the same procedure steps, observed from different angles, videos, and occlusion levels, into similar embedding representations. Each image is processed by $f_{\text{spatial}}(\cdot)$, generating spatial feature embeddings h . For $f_{\text{spatial}}(\cdot)$, an ImageNet-21K

pre-trained ViT-S [128] model is employed, of which the last layer is replaced by a one-layer MLP to project the output into a 128-dimensional vector. The embeddings $h \in H$ are then projected into a new embedding space z using a nonlinear projection head $g(\cdot)$, implemented as a three-layer multi-layer perceptron (MLP), as described in [173].

The supervised contrastive loss [178] ($\mathcal{L}_{\text{SupCon}}$) is employed to maximize the agreement of embeddings $z \in Z$ between different views and videos for identical assembly states, whilst minimizing the agreement of embeddings from different states, and is defined as

$$\mathcal{L}_{\text{SupCon}} = \sum_{i \in I} -\log \frac{1}{|P(i)|} \sum_{p \in P(i)} \frac{\exp(\frac{g(f(x_i)) \cdot g(f(x_p))}{\tau})}{\sum_{a \in A(i)} \exp(\frac{g(f(x_i)) \cdot g(f(x_a))}{\tau})}, \quad (6.12)$$

where $A(i)$ the set of all indices in mini-batch X , and $P(i) \equiv \{p \in A(i) : (y_p = y_i)\}$ is the set of indices of all positives, and τ is the temperature constant.

Mini-batches are created through key-frame sampling (KFS), leveraging PSR labels to gather images surrounding procedural step completions, as described in Algor. 1. These PSR labels denote the actual completion of steps, regardless of the visibility of irrelevant parts of the assembled object within the frame. In contrast, assembly state detection annotations are present only when the entire object can be observed from a single image, requiring the entire object (including the irrelevant parts) to be largely unobstructed and within the frame. Thus, pretraining on the PSR labels encourages similar assembly states to be grouped based on the relevant visible parts that constitute change.

KFS is weakly supervised, because it relies on sparse procedural step annotations rather than dense, per-frame supervision of assembly states. Specifically, the duration of visibility for the parts constituting a step completion is not annotated, so KFS must assume a fixed temporal window, denoted by t_f and expressed in seconds, after each step completion. As a result, the sampled frames are coarsely annotated, meaning there is no guarantee that they capture the actual parts undergoing change, except for the first frame ($t_f=0s$). This t_f is the main hyperparameter of KFS. When the value is increased, more variability in occlusion levels and viewpoints are provided, potentially leading to richer feature embeddings. However, this may introduce redundant or noisy samples that do not constitute meaningfully to the step completion.

Further hyperparameters of KFS are the number of real-world samples per step completion in each mini-batch (N_{sample}). Additionally, if synthetic data (e.g. CAD models) are available, the hyperparameter that determines the number of generated images N_{syn} can be set to readily incorporate the synthetic images in the spatial encoder training.

Algorithm 1 Key-frame sampling

Input: Video dataset X , PSR labels Y , temporal window per step t_f , real-world frames per state N_{sample} , assembly states N_{state} , synthetic images per state N_{syn}

Output: Mini-batch (x, y)

```

1:  $x \leftarrow []$ 
2:  $y \leftarrow []$ 
3:  $\text{FPS} \leftarrow 10$  ▷ constant
4: for  $\text{stateID} \leftarrow 1$  to  $N_{\text{state}}$  do
5:    $\text{images} \leftarrow []$ 
6:    $\text{syn\_images} \leftarrow []$ 
7:    $\text{indices} \leftarrow \{i \mid Y_i = \text{stateID}\}$ 
8:   for all  $i \in \text{indices}$  do
9:     append  $X[i : i + t_f \cdot \text{FPS}]$  to  $\text{images}$ 
10:    append  $X[i : i + N_{\text{syn}}]$  to  $\text{syn\_images}$ 
11:   end for
12:   for  $i \leftarrow 0$  to  $(N_{\text{sample}} + N_{\text{syn}})$  do
13:     if  $i < N_{\text{state}}$  then
14:        $x[i] \leftarrow$  image from  $\text{images}$ 
15:        $y[i] \leftarrow \text{stateID}$ 
16:     else
17:        $x[i] \leftarrow$  image from  $\text{syn\_images}$ 
18:        $y[i] \leftarrow \text{stateID}$ 
19:     end if
20:   end for
21: end for
22: return  $(x, y)$ 

```

6.3.5 Assembly state detection stream

As outlined in Fig. 6.1, STORM-PSR combines the spatio-temporal stream ($\mathcal{T}$) with an assembly state detection stream ($\mathcal{S}$). This ASD stream comprises the object detection-based PSR baseline from [57], based on a YOLOV8-m backbone [101]. Specifically, this stream indirectly recognizes procedure steps $\hat{y}_S$ in X by determining which steps in $\mathcal{P}$ are required to transform the previously observed assembly state s_{i-1} to the newly recognized state s_i .

While the spatio-temporal stream enhances robustness to occlusion, clean frames – in which the entire object is visible – still provide valuable and reliable assembly state information. By integrating both streams, STORM-PSR effectively leverages the strengths of each, using spatial cues when available and relying on temporal consistency when visibility is limited.

Method	IndustReal [57]			MECCANO [22]		
	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$
IndustReal [57]	<u>0.797</u>	<u>0.891</u>	21.0	<u>0.354</u>	0.545	<u>99.8</u>
Spatio-temp. stream	0.497	0.506	14.2	0.206	0.247	120.3
STORM-PSR	0.812	0.901	<u>15.5</u>	0.377	<u>0.497</u>	88.6

Table 6.1. Performance of STORM-PSR and its two streams for procedure step recognition on the IndustReal [57] and MECCANO [22] datasets. **Bold** text indicates the highest performance and underlined text indicates the second best.

6.4 Experiments

In this section, the two-stream framework for procedure step recognition is implemented and evaluated on the IndustReal [57] and MECCANO [22] datasets. Both datasets contain videos of construction-set toy assemblies recorded from an ego-centric perspective. A comparison on specifically these two datasets is interesting for three reasons. First, operators in IndustReal are instructed to periodically provide unobstructed views of the toy car, which is not the case in MECCANO. Therefore, MECCANO contains significantly more occluded frames, making it a more challenging dataset for PSR. Second, the MECCANO dataset is 17% longer (in hours) than IndustReal, but includes approximately half of the step completions and annotated ASD frames. Finally, IndustReal offers synthetic data for all assembly states, providing additional data for the pretraining. Hence, although the datasets are similar, STORM-PSR faces distinct challenges with each dataset.

6.4.1 Annotation of MECCANO

Because MECCANO does not include the required ASD and PSR annotations, we have manually annotated the dataset for both tasks, according to the same procedure as [57]. These new labels are made publicly available to stimulate research on PSR. For the MECCANO toy motorcycle, we define 11 assembly states consisting of 17 components. The dataset contains 431 correctly executed procedure steps (25 ± 4.0 steps per recording) and 30 incorrect step completions (1.5 ± 0.9 per recording). For ASD, 6,948 images are annotated with bounding boxes in the train set, 3,053 images in the validation set, and 3,656 in the test set. Details of the definition of the components and assembly states in the MECCANO dataset, as well as statistics on the annotations, are provided in the supplementary material.

6.4.2 Implementation details

Both the spatial and the temporal encoder are optimized using SGD [179] with an initial learning rate of 10^{-3} , momentum of 0.9, five warm-up epochs, and a cosine-annealing learning rate scheduler. All input images are resized to 224×224 pixels. The encoders are trained separately for 100 epochs, after which the spatial encoder is unlocked with learning rate 10^{-5} , to jointly converge further for 75 epochs.

For the spatial encoder, pre-trained with KFS, the number of frames eligible to be sampled after a step completion is empirically set to $t_f=2$ seconds and the number of states per batch is set to the total number of unique states in the dataset. Per state, $N_{\text{sample}}=16$ images are sampled if no synthetic data are used, otherwise N_{sample} and N_{syn} are set to 8. A margin of 0.01 and temperature of 0.07 are set for the SupCon loss, and the best epoch is selected based on the precision on the validation set.

For the temporal encoder, 6 stacked self-attention layers, 8 attention heads, and an MLP output size of 4,096 is used. The batch size B and temporal window size w are both set to 256. Within this window, $N_w=64$ spatial embeddings are sampled with equal spacing. For KCAS, the standard deviation σ is set to 45 frames, and δ , the separation between the two Gaussian distributions, is set to 80. The best checkpoint is selected based on the F_1 score on the validation set.

As in [57], predictions are accumulated over consecutive video frames until a cumulative confidence score of T is reached. Confidence scores for prediction-less frames are decayed at a rate of 25%. For PSR inference with the two-stream approach, the thresholds $T=6.0$ for IndustReal and $T=1.0$ for MECCANO are established based on empirical results on the validation set. STORM-PSR operates at 75.1 frames/second on a NVIDIA A100 GPU.

6.4.3 Procedure step recognition results

Table 6.1 presents the performance of STORM-PSR across both datasets. STORM-PSR reduces the average delay (τ) by 26.1% for IndustReal and 11.2% for MECCANO, outperforming the state-of-the-art baseline. Simultaneously, STORM-PSR achieves state-of-the-art performance on IndustReal for the POS and F_1 metrics. On the MECCANO dataset, STORM-PSR improves the POS (+6%) but reduces the F_1 score (-9%).

Additionally, the impact of key-frame sampling (KFS) as well as key-clip aware sampling (KCAS) on the temporal stream performance is highlighted in Table 6.1. Without KFS pretraining of the spatial encoder, the temporal encoder is not able to learn meaningful spatio-temporal features and fails to make any PSR predictions. With KFS pretraining, but without KCAS, the temporal encoder is able to learn from the meaningful spatial features. However, a clear improvement in performance is noted when KCAS is used to effectively over-sample positive and hard-negative clips. On both datasets, both the POS and F_1 improve 14% to 79% when KCAS is used. Nonetheless, the temporal stream alone does not outperform the object detection

stream on either dataset. This is likely because the datasets contain relatively few step completions, compared to the number of ASD-annotated frames. Additionally, spatio-temporal features are more challenging to learn due to increased dimensionality, the need to learn temporal coherence, and computational complexity [19], [180], [181]. These factors lead to more false positives for the temporal stream compared to the ASD stream. Combining the object-detection and temporal streams achieves state-of-the-art performance by leveraging the strengths of both approaches.

A qualitative example of predictions from both individual streams and their combined output (STORM-PSR) is provided in Figure 6.5, along with images illustrating typical occlusions types. This example clearly demonstrates how the two streams complement each other. When the temporal stream predicts with high confidence, while the ASD stream predicts with lower confidence (*e.g.* due to occlusions), procedure steps can still be recognized. The STORM-PSR accumulates the confidences from both streams (reducing false positives), allowing steps to be recognized when either stream is highly confident, or both streams maintain moderate confidence over several frames.

6.4.4 Assembly state detection on MECCANO

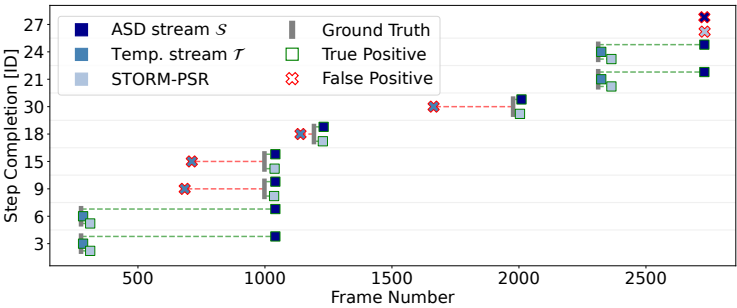
The assembly state detection stream S , using a YOLOv8-M model, is also evaluated for ASD on the MECCANO dataset [22]. The model obtains a mean average precision (mAP@50) of only 0.120, compared to 0.838 for the best model on the IndustReal dataset [57]. The latter is trained on real-world data, supplemented with images generated using the provided CAD models. Nonetheless, even without leveraging these synthetic data, an mAP@50 of 0.753 is still achieved. The performance gap between IndustReal and MECCANO clearly highlights the difficulty of recognizing entire assembly states in videos with frequent occlusions, since limited visibility of the object parts significantly impairs ASD models.

6.4.5 Ablation studies

The following ablation studies isolate and quantify the contributions of key components within the proposed framework, including the sampling strategies (KFS and KCAS), temporal modeling backbones, sampling distributions, temporal receptive field, and key-frame sampling parameters. These experiments provide insight into design choices and their impact on procedure step recognition performance.

A Sampling techniques and different backbones for temporal modeling

To evaluate the contributions of our sampling strategies (KFS and KCAS), as well as the choice of temporal backbone, a three-way ablation is reported in Table 6.2. We compare long short-term memory networks (LSTM) [182], temporal convolutional



(a) PSR predictions from the ASD and spatio-temporal streams, as well as STORM-PSR, which combines the both streams.



(b) Two video frames demonstrating the completion of steps 3 and 6.

Figure 6.5. Qualitative demonstration of STORM-PSR on the IndustReal test set. The ASD stream fails to predict Steps 3 and 6, until it encounters unobstructed frames, resulting in a delay of 765 frames. The temporal stream recognizes Steps 3 and 6, but predicts false positives (with lower confidence) for other steps.

Model	Sampling		IndustReal [57]			MECCANO [22]		
	KFS	KCAS	POS ($\uparrow$)	F_1 ($\uparrow$)	τ [s] ($\downarrow$)	POS ($\uparrow$)	F_1 ($\uparrow$)	τ [s] ($\downarrow$)
LSTM			0.000	0.000	-	0.000	0.000	-
LSTM	✓		0.180	0.303	33.2	0.047	0.146	222.8
LSTM	✓	✓	0.204	0.365	40.9	0.037	0.248	121.6
TCN			0.000	0.000	-	0.000	0.000	-
TCN	✓		0.124	0.185	25.4	0.000	0.000	-
TCN	✓	✓	0.195	0.414	49.4	0.134	0.181	153.7
Transformer			0.000	0.000	-	0.000	0.000	-
Transformer	✓		0.356	0.419	24.4	0.180	0.138	283.7
Transformer	✓	✓	0.497	0.506	14.2	0.206	0.247	120.3
STORM-PSR			0.467	0.511	62.6	0.269	0.358	<u>102.9</u>
STORM-PSR	✓		<u>0.766</u>	<u>0.892</u>	30.0	<u>0.329</u>	<u>0.459</u>	135.8
STORM-PSR	✓	✓	0.812	0.901	<u>15.5</u>	0.377	0.497	88.6

Table 6.2. An overview of the impact that different backbones, KFS, and KCAS have on the performance of the temporal model. The evaluated backbones consist of a long short-term memory (LSTM), temporal convolutional network (TCN) and transformer temporal model. Additionally, the performance of STORM-PSR (with the transformer backbone) is reported without KFS and KCAS, with only KFS, and with both. **Bold** text indicates the highest performance and underlined text indicates the second best.

networks (TCN) [183], and the transformer backbone. Each backbone is trained (i) without key-frame sampling (KFS) or key-clip aware sampling (KCAS), (ii) with KFS only, and (iii) with both KFS and KCAS. Finally, the end-to-end STORM-PSR results are reported for the same three settings to show how improvements in the temporal stream translate to the full two-stream system.

The impact of key-frame sampling (KFS) and key-clip aware sampling (KCAS) on the temporal stream performance is highlighted in Table 6.2. Without KFS pretraining of the spatial encoder, the temporal encoder is not able to learn meaningful spatio-temporal features and fails to make any PSR predictions. With KFS pretraining, but without KCAS, the temporal encoder is able to learn from the meaningful spatial features. However, a clear improvement in performance is noted when KCAS is used to effectively over-sample positive and hard-negative clips. On both datasets, both the POS and F_1 improve 14% to 79% when KCAS is used.

Among the evaluated temporal backbones, the transformer consistently outperforms the LSTM and TCN across all metrics when paired with both key-frame and key-clip aware sampling. On the IndustReal dataset, the transformer achieves the highest POS (0.497), F_1 (0.506) score, and the lowest delay (14.2s), with the LSTM and TCN backbones performing between 18% and 250% worse. On MECCANO this trend is less pronounced due to the generally decreased scores on that dataset.

Sampling	IndustReal [57]			MECCANO [22]		
	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$
Uniform	0.356	<u>0.419</u>	24.4	0.180	0.138	283.7
Gaussian	<u>0.419</u>	0.382	<u>22.4</u>	0.212	<u>0.175</u>	<u>143.7</u>
KCAS	0.497	0.506	14.2	<u>0.206</u>	0.247	120.3

Table 6.3. Comparison of two baseline sampling approaches and the novel key-clip aware sampling (KCAS). For uniform sampling, clips are randomly selected, regardless of the corresponding class label. For Gaussian sampling, all clips with at least one step completion are regarded as positive samples, which are over-sampled accordingly. KCAS uses a bi-modal sampling distribution, to explicitly over-sample (i) hard negative samples directly before step completions and (ii) positive samples, while under-sampling the negatives.

These results indicate that the transformer is better suited for capturing long-range temporal dependencies required for recognizing procedure step completions under partial occlusions, compared to the LSTM and TCN.

B Key-clip sampling for temporal modeling

A second ablation study evaluates the impact of the KCAS probability mass function by comparing uniform, Gaussian, and bi-modal sampling methods. In this experiment, the same spatial encoder weights are used for all three sampling methods. After training the temporal encoder, both encoders are jointly fine-tuned for 75 epochs. As shown in Table 6.3, the results confirm the hypothesis that sampling before and after key moments performs best, with the bi-modal distribution achieving the best performance across most metrics. This outcome is in line with expectations, since the temporal encoder is exposed to more positive samples during training compared to uniform sampling. In comparison to Gaussian sampling, KCAS selects fewer ambiguous clips by lowering the probability at the moment of step completions, while increasing hard-negative samples. This encourages the temporal encoder to learn subtle yet critical differences that constitute step completions.

C Temporal receptive field

The temporal encoder is tasked with recognizing (potentially occluded) steps based on spatial features within a given temporal window of size w . This final ablation study examines the effect of the size of the temporal window on the performance for the PSR task. Furthermore, the study compares temporal self-attention using a transformer to temporal aggregation of spatial features with a simple MLP, an LSTM, and an TCN. These models are trained with the same spatial encoder for different

Model	IndustReal [57]			MECCANO [22]		
	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$	POS $\uparrow$	$F_1 \uparrow$	$\tau \downarrow$
MLP-16	0.226	0.346	42.2	0.168	0.166	162.7
MLP-64	0.285	0.357	34.0	0.202	0.130	172.3
MLP-256	0.407	0.330	23.0	0.249	0.156	<u>115.5</u>
LSTM-16	0.121	0.238	27.2	0.190	0.189	194.5
LSTM-64	0.200	0.286	30.2	0.173	0.207	258.9
LSTM-256	0.277	0.470	30.3	0.156	0.225	132.4
TCN-16	0.119	0.144	34.2	0.172	0.113	163.9
TCN-64	0.203	0.235	25.6	0.185	0.130	186.1
TCN-256	0.265	<u>0.502</u>	43.1	0.195	0.151	127.8
Tr.-16	0.228	0.218	38.1	0.207	<u>0.212</u>	125.7
Tr.-64	0.304	0.364	45.6	<u>0.289</u>	<u>0.212</u>	137.0
Tr.-256	<u>0.406</u>	0.514	<u>25.3</u>	0.351	0.163	108.5

Table 6.4. Comparison of PSR performance of a multi-layer perceptron (MLP), long short-term memory (LSTM), temporal convolutional network (TCN) and transformer (Tr.) temporal model. Each model is trained with three receptive fields, namely 16, 64, and 256 frames, which is indicated behind the model abbreviation (*i.e.* Tr-16 is the transformer model with a window size of 16 frames).

temporal window sizes w , with KCAS. All frames within the window are sampled ($w=N_w$), except when $w=256$, where $N_w=64$ is obtained by applying a temporal stride of 4 to reduce the computational load. Finally, the same spatial encoder is used for all experiments and is not fine-tuned together with the temporal encoder.

The results of this test, presented in Table 6.4, demonstrate that increasing the temporal window size w improves performance. This is in line with expectations, since a larger temporal window offers more context to identify step completions. Moreover, the attention-based temporal encoder outperforms the aggregation-based MLP with the same temporal window, although the difference is not substantial.

D Time after step completions to sample for KFS

The time after a step completion from which to sample real-world frames, denoted as t_f , is an important hyperparameter in the weakly-supervised key-frame sampling (KFS) strategy. It determines the diversity and quantity of visual representations available during contrastive pretraining of the spatial encoder. A larger t_f introduces more variability in occlusion levels, viewpoints, and context, potentially leading to richer and more robust feature embeddings. However, excessively increasing this value may introduce redundant or noisy samples that dilute the discriminative signal of step completions. This ablation study investigates how varying t_f affects the

t_f [s]	IndustReal [57]			MECCANO [22]		
	POS $\uparrow$	F_1 $\uparrow$	τ $\downarrow$	POS $\uparrow$	F_1 $\uparrow$	τ $\downarrow$
0.5	0.308	0.511	43.3	<u>0.193</u>	<u>0.241</u>	192.8
1.0	0.311	<u>0.514</u>	48.0	0.159	0.199	<u>114.7</u>
2.0	0.406	<u>0.514</u>	25.3	0.351	0.163	108.5
4.0	<u>0.350</u>	0.526	<u>39.8</u>	0.127	0.233	252.2
8.0	0.346	0.508	52.7	0.126	0.298	150.4

Table 6.5. Ablation study on the time after a step completion to sample from for the weakly-supervised training of the spatial encoder.

downstream performance of the spatio-temporal stream and the overall STORM-PSR model, aiming to identify an optimal trade-off between training efficiency and feature quality.

Table 6.5 shows the results for this ablation study. The results show the intricate balance between the three metrics used to evaluate PSR. Nonetheless, for both datasets, a t_f of 2 seconds results in the highest POS score and the lowest average delay τ .

6.5 Discussion

This work aims to reduce delays and enhance robustness of procedure step recognition algorithms by enabling step predictions even when parts of the assembly are occluded. This is achieved using a two-stream framework called STORM-PSR, where one stream focuses on robust state detection under minimal occlusions, while the second directly predicts steps based on spatio-temporal features. STORM-PSR is evaluated on two datasets, IndustReal [57] and MECCANO [22]. For the latter, PSR and ASD labels were initially unavailable, so these were manually annotated, and are published alongside this work. On IndustReal [57], STORM-PSR achieves state-of-the-art performance, and on MECCANO it establishes a new benchmark.

STORM-PSR significantly reduces the delay in recognizing the completion of steps. While the delay reduction is in line with expectations, a larger improvement for the temporal stream of STORM-PSR on the MECCANO dataset was expected due to its high number of occluded step completions. In fact, a 26% delay reduction (τ) is observed on IndustReal, compared to only 11% on MECCANO. Simultaneously, STORM-PSR improves the performance on both other metrics (F_1 score and POS) compared to the baseline. On the other hand, STORM-PSR on the MECCANO dataset only improves the POS metric, while reducing the F_1 score by approximately 0.05. On this point, the interaction between the three metrics is important to highlight,

because the average delay τ can only be calculated on true positive recognitions of the model [57]. Therefore, while the benefit of STORM-PSR on IndustReal is readily observed (since all metrics improve), the benefit on MECCANO is less pronounced and more complex to understand. First, spatio-temporal models generally require a larger amount of data compared to models relying exclusively on spatial features, without the complexity of temporal learning [19], [181]. The MECCANO and IndustReal are of approximately the same total video length, but MECCANO contains only 1.1 procedure steps are completed per minute of video data, compared to 2.2 steps per minute in IndustReal. We hypothesize that this higher data sparsity in MECCANO poses challenges for spatio-temporal learning, disproportionately affecting the spatio-temporal stream (and thereby STORM-PSR) compared to the ASD stream, that only relies on spatial features. This difference is further amplified by the lack of synthetic data for MECCANO, which is used in STORM-PSR on IndustReal during pretraining of the spatial encoder. Second, the reduction in delay obtained by MECCANO may occasionally lead to earlier predictions that slightly precede true step completions, counting them as false positive predictions and thus negatively impacting F_1 score. Finally, the frequency and nature of occlusions in the MECCANO dataset differs from that in the IndustReal dataset on three important aspects. Firstly, operators in IndustReal, contrary to MECCANO, are asked to provide unobstructed views with some frequency, resulting in more frequent images without occlusions. Secondly, MECCANO participants use a tool, whereas the participants in the IndustReal dataset only use their hands, creating a different type of occlusion. Thirdly, the IndustReal dataset is recorded with a projected virtual box that outlines the FoV of the cameras. Therefore, participants at least somewhat consistently keep the object of interest within this FoV. Contrarily, videos in the MECCANO dataset frequently do not contain any frames with the object visible for the entire length of the temporal window.

To address this data sparsity in the MECCANO dataset without additional video recordings and labor-intensive annotations, MECCANO could be further extended with synthetic data. Given the commercial application of the MECCANO parts, using CAD models for creating synthetic data might not be feasible. Instead, approaches such as neural radiance fields (NeRFs) [184] may allow for synthetic view generation based on real-world captures, providing an alternative to CAD-based synthesis. Additionally, video augmentations can be used to increase data variability [185], [186].

Further experiments may explore larger temporal windows, as the largest tested in this study spans 256 frames, corresponding to video clips of 25.6 seconds in IndustReal and 21.3 seconds in MECCANO. Considering the time between step completions in both datasets, increasing this window size may be beneficial. To use larger temporal windows without encountering computational limitations, frames can be sampled at a reduced rate. Alternatively, a selection procedure can be applied to retain only the most informative frames (*e.g.* through attention mechanisms), since not all frames are equally valuable [187], [188], and approaches that consider the entire video se-

quence can be explored [189]. The computation cost of STORM-PSR compared to the baseline, consisting only of the ASD stream, is significantly higher. Specifically, the ASD stream operates at 284.8 FPS on an A100 GPU, while STORM-PSR with the transformer backbone operates at 75.1 FPS. With the LSTM backbone, STORM-PSR achieves 72.9 FPS, and with a TCN backbone it operates at 75.2 FPS. This trade-off between computational complexity and improved recognition delay is an important factor for practical applicability. Although STORM-PSR requires more computation, its ability to substantially reduce recognition delay provides a clear advantage in scenarios where timely predictions are essential.

Another interesting direction of future research is regarding the fusion of the two streams. In the present work, a linear late-fusion approach is adopted to reduce the risk of overfitting, particularly given the limited size of the two evaluated datasets. Introducing learning-based fusion techniques can allow the model to weight the two streams differently to optimally exploit the complementary strengths of either stream. Especially in the context of larger-scale data, such adaptive fusion mechanisms may further improve performance. Another promising direction for future work lies in adapting existing methods originally developed for related tasks to the PSR setting. While these methods were not designed for PSR and require significant adjustments, they can offer comparative baselines and help contextualize the strengths and limitations of STORM-PSR.

6.6 Conclusion

This chapter has proposed and tested STORM-PSR, a novel approach to procedure step recognition, based on spatio-temporal features, that is directly optimized for the PSR task. The STORM-PSR framework combines assembly state detections, that are robust when objects are entirely visible and unoccluded, with a temporal model to predict step completions. This is the first work to directly optimize for PSR, rather than inferring step completions from observations of entire assembly states. To effectively train the temporal model, we introduce key-frame and key-clip aware sampling strategies.

Findings. STORM-PSR is evaluated on an existing PSR dataset, obtaining state-of-the-art performance (a POS of 0.812 and F_1 score of 0.901), and on a new dataset where it establishes a benchmark (a POS of 0.377 and F_1 score of 0.497). More importantly, the average delay between a step completion and the subsequent recognition of that step is reduced by 26% on the IndustReal dataset and by 11% on the MECCANO dataset. The results demonstrate that waiting for unobstructed views of entire assembly states leads to significant delays between the completion and corresponding recognition of procedure steps. This delay can be reduced by using a spatio-temporal model capable of learning from visible assembly parts, without requiring assumptions about occluded parts. Nonetheless, ASD-based predictions remain valuable when the full object is observable, as shown by the superior performance of the two-stream approach

combining ASD with temporal modeling.

Limitations. Although the experiments outlined in this chapter show a significant improvement in prediction delay, two notable limitations remain. First, the proposed method does not explicitly model the full procedural sequence due to its restricted temporal receptive field. Second, to learn the spatio-temporal dynamics, the model requires a significant amount of real-world training examples, which are cumbersome to collect and annotate. Therefore, the significant reduction in prediction delay comes at the cost of scalability and real-world applicability.

In conclusion, this chapter has demonstrated that leveraging temporal features and optimizing directly for the PSR task can significantly improve the prediction delay between the actual completion and corresponding recognition of a procedure step. This reduction in delay is achieved by eliminating the need for unobstructed views of the entire assembly object at the cost of scalability. In future work, this reduced scalability can be mitigated, *e.g.* by using a video-frame selection procedure that retains only the most informative frames for processing by the spatio-temporal network.

This chapter has addressed the fundamental limitation of the preceding chapters: the assumption that the object of interest is clearly and fully visible with sufficient frequency in ego-centric assembly videos. This assumption is addressed by directly training a spatio-temporal model for the PSR task.

The current and all preceding chapters have focused on the algorithmic aspects of industrial automated assistive systems, without explicitly evaluating end-user experiences with such systems. The following chapter addresses this shortcoming by implementing a real-time AR-guidance support system and investigating the effects that such a system has on its users.

The impact of imprecision in error localization on AR guidance

7.1 Introduction

Up to this point in the thesis, the potential value and demands for automated support during procedural actions have been outlined, as well as the technical algorithmic aspects to achieve this. Despite the clear need for these technical advancements, the experience of the end-users remains unexplored. Therefore, this chapter implements the algorithms proposed in the earlier chapters into a real-time AR-guidance support system, and investigates the effects that such a system has on its users.

Augmented reality (AR) for industrial assembly assistance, coupled with computer vision (CV)-based error detection predictions, has gained momentum, particularly since the release of the HoloLens 2, as a promising tool to reduce errors, automate progress logging, and enhance overall quality metrics [66], [67], [190]. Despite its promise, the adoption of industrial AR is hindered by workers' resistance [191]–[193]. This might differ depending on the type of expertise and background of the worker, which is known as expert-characteristics-related biases [192], [194]–[196]. Past research has shown that domain experts often display overconfidence when performing their tasks and are more reluctant to use AR [196]. In contrast, AR and virtual reality (VR) experts appear to exhibit lower resistance toward immersive systems¹ [192], given their role in driving innovation and recognizing the AR's potential [195]. However, domain experts are more skeptical because they are acutely aware of the risks related to their work tasks [194], [196]. Thus, it can be assumed that biases differ among various types of experts, influencing their performance and attitudes toward AR/VR. Despite the importance of such cognitive biases toward AR, the systematic literature review (SLR) in this chapter reveals that no studies to date have explicitly highlighted expert-related cognitive biases. Also, prior research that involved experts as participants

¹Immersive technologies create distinct experiences by merging the physical world with a digital or simulated reality.

often uses the term “experts” without adequately differentiating between *types* of experts. Additionally, many studies lack sufficient details about the backgrounds of expert participants, including the methods used to assess their expertise.

Moreover, previous research on automated error detection algorithms, which are integrated within AR guidance systems, has largely focused on bespoke solutions, with limited emphasis on scalability, as outlined in the previous chapters of this thesis. For instance, the CV models must be either re-trained when the procedure order is modified or parts are updated [105], [112], [131]–[133], or potential execution errors need to be previously defined [197], [198]. The algorithms presented in the preceding chapters of this thesis address this problem, but lack thorough real-world evaluation [167], [199]. Besides this, the current body of literature does not investigate the optimal tuning of industrial assistive systems regarding the trade-off between avoiding missed errors (false negatives: FNs) and minimizing false alarms (false positives: FPs) for user trust and comfort. While literature on medical and automotive assistive systems suggest that a high rate of FPs frustrates users and under-warnings risk errors [200]–[202], these trade-offs have not been systematically studied in industrial AR, particularly across diverse user groups.

Summarizing the aforementioned aspects, our objective is to introduce an AI-enabled AR assistive system into the workplace for various types of experts. Such a system inherently requires a trade-off between FPs and FNs, while the experts themselves have their own biases. The interaction between this trade-off and the experts’ background in this setting is not studied. Motivated by this gap, and to contribute towards the external validity understanding of AR studies, this chapter addresses the following three research questions.

- RQ7.1 *System performance in a real setting.* Can the algorithms presented in this thesis be integrated into a real-time AR system to provide advanced support during a procedure and what is the real-world system performance?
- RQ7.2 *System feedback trade-off.* Should the system feedback be tuned to over-warn/under-warn users about potential mistakes, and how does this relate to the different user groups?
- RQ7.3 *User experience.* How do users, having different background expertise, experience such an AR assistive system and are they receptive to using it in the workplace?

To investigate these research questions, this chapter examines the effects of CV-driven AR-based assembly guidance on performance, trust, and acceptance among three distinct user groups: (i) novices, (ii) manufacturing experts, and (iii) AR/VR experts. Manufacturing experts are recruited as domain experts, as the task revolved around assembling mechanical assets. Specific focus is placed on how different types of system imperfections, *i.e.* under-warning and over-warning for errors, impact the participants’ assembly correctness alongside perceived workload and flow. Through

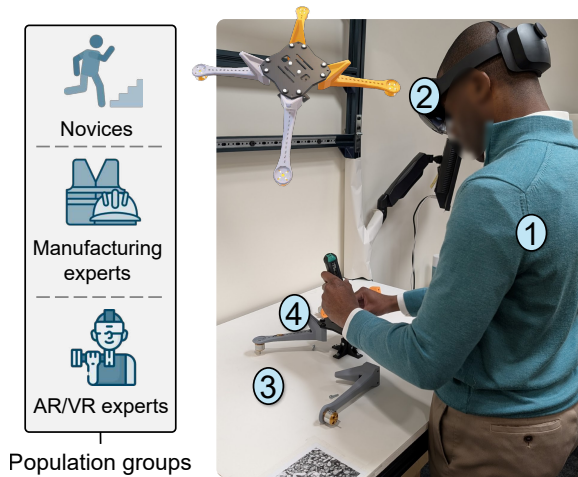


Figure 7.1. Illustration of the use of an AR assembly guidance system by a user. The user (1) wears the head-mounted display (2) while working at an industrial workstation (3) to assemble a quadcopter (4).

this study, insights are gained into how system reliability and user expertise interact in shaping both the effectiveness and perception of AR-guided assembly tasks. To this end, the state-of-the-art error localization algorithm from Chapter 5, trained exclusively on synthetic images and scalable to unseen errors and assembly configurations, is integrated into a real-time system and evaluated on actual assembly data.

To summarize, the three main contributions of this chapter are as follows.

- *User study on the cognitive biases across population groups.* Different user groups are compared with a specific focus on expert bias, as defined through the presented SLR. To the best of our knowledge, no study has investigated cognitive biases — particularly expert-specific biases — across different types of experts, nor has any study examined how these experts' performances and attitudes differ or overlap.
- *Research on the impact of system imperfections.* The effect of incorrect warnings (false positives) and missed errors (false negatives) on assembly outcome and user perception is systematically assessed, providing new insights into the balance between caution and intrusiveness.
- *Real-world system performance quantification.* The real-world performance of the scalable error localization method presented in Chapter 5 is quantified, highlighting the feasibility of a fully AI-driven approach for industrial AR.

This chapter is structured as follows. First, the related work, including the SLR, is outlined in Section 7.2. Subsequently, the AR assembly guidance system is described

in Section 7.3, after which the user study is presented in Section 7.4. The quantitative and qualitative data from the user study are provided in Section 7.5 and Section 7.6, respectively. The results are discussed in Section 7.7, after which conclusions are drawn in Section 7.8.

7.2 Related work

First, the state-of-the-art in error detection methods for passive and active AR systems is outlined. Next, two SLRs are presented, addressing the definition and differentiation of experts, along with expert-related challenges in industrial AR studies.

7.2.1 Error detection and task verification in AR

Automated AR assistance has been increasingly studied for industrial tasks and can be broadly divided into active and passive systems. The former provides static visual instructions without real-time error verification [203], [204]. Marino *et al.* [205] confirmed the effectiveness of such passive AR systems, yet observed diminishing returns for experienced workers already proficient without AR instructions. While passive methods enhance basic task guidance, they delegate error detection entirely to the user, thus lacking the dynamic responsiveness necessary for comprehensive assembly support. To address this limitation, recent CV developments have enabled active AR systems capable of real-time error detection and dynamic user feedback. For instance, Stanescu *et al.* [197] proposed a CAD-based, state-aware method for detecting assembly errors and actively prompting corrections. Similarly, Fang *et al.* [206] and Zhang *et al.* [198] introduced systems that validate task execution step-by-step, significantly reducing assembly errors. An overview of these works is provided in Table 7.1.

Error detection methods in industrial assembly tasks broadly fall into two categories, namely, data-driven and CAD-model-based approaches. Traditional data-driven methods require extensive datasets depicting correct assemblies, as well as all possible erroneous states to *memorize* assembly states and errors. This type of algorithm is limited in scalability and generalization since each new assembly scenario demands explicit error-state definitions and necessary retraining on these updated data. Many existing active AR assembly guidance systems either depend on pre-defined assembly states and error conditions [198], [206], [207] or utilize fixed-camera setups that restrict operator movement [206], [208]–[210].

Conversely, recent CAD-model-based techniques avoid these restrictions by *comparing* CAD renders of correct assemblies to real-time captured RGB images [132], [167], [197], [199]. One such approach, StateDiffNet [199], achieves comparisons with a global cross-attention mechanism, detecting structural discrepancies without requiring exact camera poses or prior exposure to specific assembly configurations or error

Author	Assembly Asset	Task	# Parts	Device	Cues	A	B	C	D	E
Frandsen <i>et al.</i> [208]	Machinery	Alignment	n/a	HHD	Text, 3D	0	n/a		✓	✓
Stanescu <i>et al.</i> [197]	Toy car/IKEA	Assembly	5/9	HMD	Text, 3D	0	n/a	✓	✓	
Lehman <i>et al.</i> [199]	Toy car	Assy., main.	36	HMD	Image	0	n/a	✓	✓	✓
Bohus <i>et al.</i> [211]	Generic	Assy., main.	n/a	HMD	Text, audio, 3D	0	n/a	✓		
Yu <i>et al.</i> [212]	Machinery	Assy., main.	8	HMD	Text	0	n/a	✓	✓	
Ojer <i>et al.</i> [209]	PCB	Assembly	20	Monitor	Image	21	Nov.			
Zhang <i>et al.</i> [198]	Gearbox, controller	Assembly	8/32	HMD	Text, 3D	32	Nov.	✓		
Fang <i>et al.</i> [206]	Gearbox	Assembly	5	HMD	3D	30	Nov.		✓	
Zhang <i>et al.</i> [210]	Gearbox	Assembly	4	Monitor	Text, 3D	30	Nov.		✓	
Seeliger <i>et al.</i> [213]	Factory	Inspection	1	Monitor, HMD	3D	35	Nov.	✓	✓	✓
Tadeja <i>et al.</i> [204]	3D Printer	Maintenance	16	HMD	Text, audio, image, 3D	3	Exp.	✓		
Fortuna <i>et al.</i> [203]	Automotive	Inspection	21	HHD	3D	33	Exp.	✓		
Marino <i>et al.</i> [205]	Valves	Assembly	21	HHD	Text, 3D	16	Both	✓		
Wang <i>et al.</i> [207]	Electronics	Assembly	8+	HMD	Text, image, 3D	48	Both	✓	✓	
Ours	Quadcopter	Assembly	36	HMD	3D	24	Both	✓	✓	✓

Table 7.1. Detailed overview of industrial AR guidance: (A) user study participants; (B) experience level of the users in study; (C) non-fixed camera perspective; (D) automatic quality control (checking presence and correctness of parts), (E) scalability (no real-world data requirement, able to detect errors that are not encountered during training); HHD: hand-held device, HMD: head-mounted device.

types, greatly enhancing its scalability and generalization capabilities. However, these algorithms have not undergone rigorous validation with real-world data. Therefore, this study integrates the state-of-the-art CAD-based StateDiffNet model [199] into a real-time AR system to quantify the real-world performance of an advanced error detection method.

7.2.2 Cognitive phenomena in expert participants

Previous work has shown that AR/VR reduces experts' productivity rather than improves it [214]. Furthermore, some experts found the transition to AR-supported workflows cumbersome. Experts are significantly slower and experienced more cognitive load when using AR instructions compared to without any [215]. To better understand the literature on expert resistance towards AR, this chapter presents two SLRs following the standards of the preferred reporting items for systematic reviews and meta-analyses (PRISMA) [216] to identify and collect relevant articles.

The first SLR explored definitions and distinctions of experts in industrial AR literature, focusing on the *ACM Digital Library* and *IEEE Xplore* databases, yielding 21 papers to be included in the review, with an additional 11 studies identified through forward and backward search. The second SLR examined expert-related challenges that hinder the future adoption of industrial AR and investigated how cognitive biases in experts influence system evaluation. This SLR utilized the *ACM Digital Library*, *IEEE Xplore*, *Scopus*, and *ScienceDirect*, resulting in 18 papers included in the final review. Further details on the SLRs and their search strings can be found in Appendix C.

When the term "experts" was used, the majority of the studies referred to domain experts (58%), followed by AR experts (19%), AR/VR experts (17%) and human-computer interaction (HCI) experts (6%). The classification of domain experts was applied when a study referred to technical experts, task-specific experts, or experts related to the employees' work domain. The average experience level of expert participants was a minimum of 2.73 years (SD=1.57), however, 66% of the investigated studies does not report the participants' experience levels. The average population size of expert participants was $\mu=10.62$; $\sigma=10.55$; median=7, excluding one outlier, and was detailed in 28 of the 32 studies. The average experts' age was $\mu=34.52$; $\sigma=7.01$; median=33.9, as calculated on only one-quarter of studies where the ages were provided. Gender distribution was reported in 53% of the studies, with an average of 84% male and 16% female participants. Non-binary genders were not mentioned. Overall, the majority of studies lacked sufficient details about the expert participants' backgrounds, including the experience levels that qualify them as experts.

In the subsequent SLR, numerous expert-related challenges that might hinder the future adoption of AR are identified. The adoption process has been perceived by professionals as a threat, in particular in terms of job and competence loss and paternalism [191], [192]. Also, loss of routine has been identified as a threat, which

refers to workers' expectations of having to change their familiar and traditional work routines due to AR implementation [192]. Zong and Guan [217] highlighted the limited familiarity of industrial domain experts with AR capabilities. Further, experienced domain workers, familiar with procedures, are more confident in independent task performance and often perceive AR instructions as unnecessary, unlike novices who rely on the system [196]. Another challenge is the importance of customizing AR to the user's skill level [195], [218]–[224], as excessive elaboration or too many steps may distract experts [218], [219], [222]–[224]. Experienced users favor more complex visual aids, such as CAD models, while novices prefer simpler cues, such as videos or photographs [221], [225]. This can be reasoned with the expertise reversal effect [226]. Experts already established domain-specific schemata in their long-term memory (LTM), making externally provided support more effortful in integration to their working memory (WM). Reconciling the externally provided guidance with existing knowledge from the LTM can potentially cause cognitive overload [227]. Hoffman *et al.* [228] attributed AR resistance to the expertise reversal effect.

In terms of expert characteristics-related biases, domain experts show greater confidence in performing tasks without any guidance, leading them to bypass or ignore AR instructions and draw from their own knowledge instead [196]. This led them to complete tasks faster, but also causes the overlooking of AR cues, reducing precision in nuanced tasks. In contrast, less experienced users followed AR instructions more closely [196]. Further, Stadler *et al.* [195] compared differences in domain experts, novices, and AR-experienced users. Here, AR's simplistic, shape-based, and color-based feedback was beneficial for domain experts but insufficiently detailed for novices, aligning with experts' advanced mental schemata of avoiding unnecessary information [226]. AR experts showed reduced workload but increased task time due to exploratory behavior in AR-based tasks [195]. In addition, Schein *et al.* [192] found that resistance factors were rated lower among workers with prior AR experience, suggesting that AR experience may reduce prejudices and concerns. However, experienced users might also become more aware of specific risks, such as privacy concerns or user tracking, once they fully understand the technology [192]. Presumably, AR/VR experts are more likely to have a positive bias toward AR, given their role in innovating and recognizing its potential [195]. Conversely, domain experts might exhibit a more negative bias, being acutely aware of risks related to their work tasks [194], [196].

Only two studies [195], [228] have explored expert-related cognitive phenomena in AR, namely the expertise reversal effect [226], while no study explicitly studied expert-related cognitive biases in industrial AR studies. The SLR in this chapter shows that expert users are not a homogeneous group—different levels and kind of expertise likely affect how users respond to AR guidance with automated error detection and task verification. This gap is outlined and addressed in this study, which involves three diverse groups of participants with varying expertise.

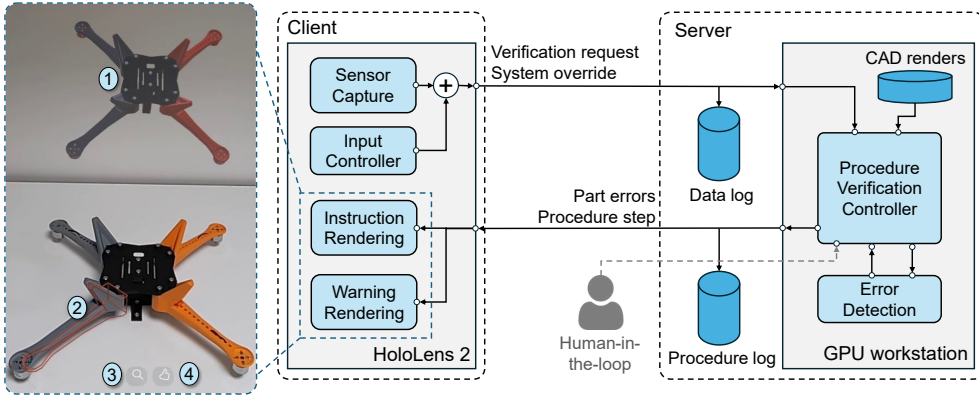


Figure 7.2. Overview of the AR assembly guidance system. The HL2 serves as the client device to record sensor data, handle user inputs, and render 3D graphics. While a GPU workstation is used to off-load computationally heavy error detection. The guidance consists of (1) a virtual 3D model of the next procedure step, (2) visual outline of parts that the system recognizes as incorrect (in this case the incorrect installation of a longer arm), (3) a verification button, and (4) a warning-overwrite button that can be used to dismiss system warnings.

7.3 Augmented reality assembly support system

This section outlines the hardware and software components of the AR guidance system used to support assembly tasks. It first details the system architecture and operator interaction, after which the integration of automated error detection using a modified version of StateDiffNet is described.

7.3.1 Apparatus and guidance system

The Microsoft HoloLens 2 (HL2) head-mounted display (HMD) is used to facilitate the user study (see Fig. 7.1 (2)). The HL2 is a see-through AR device that includes four visible-light tracking cameras, a depth camera, an inertial measurement unit (IMU), and a front-facing RGB camera capable of capturing images at a resolution of 1280×720 pixels at 30 frames per second (fps) [229]. On-device computations are performed using a Qualcomm Snapdragon 850 processor combined with a custom-built Holo-Graphic Processing Unit (HPU). The HL2 provides a diagonal field of view (FoV) of approximately 50° [14]. An HMD is selected over handheld alternatives to enable hands-free interaction and provide an immersive experience.

The AR-based assembly guidance system follows the client-server architecture depicted in Fig. 7.2. On the client side, the sensor data capturing, human-operator input processing, and rendering tasks are executed. The logging of sensor data and computationally intensive processing of video frames is offloaded to a GPU-enabled

compute server in real time. Such an approach mitigates the computational constraints of deploying advanced CV algorithms directly on the HL2, which significantly increases computation time [230]. The LAN-based communication between client and server takes place via HTTP using the Transmission Control Protocol (TCP), implemented using Flask [231]. The client side of the AR guidance system is developed using the *Unity* game engine [232], *Mixed Reality Toolkit 3* (MRTK3) [233], and *Vuforia Engine* [234] to align the virtual and physical world upon initialization of the system. The server component is developed with Python and runs on an i9-13900H CPU and an RTX 4060 GPU.

The AR instructions are provided by projecting a 3D model of the next assembly step in front of the operator. No animations or textual descriptions are provided to the operator to maintain a minimalistic interface and to allow the operator to execute the procedure step as deemed appropriate. The interaction between the operator and the AR system is handled through two buttons. The first button activates the assembly verification process by sending a signal to the server to initialize error detection. The second button allows the operator to ignore the system feedback. Both buttons require a double press to prevent accidental activation. To compensate for the delay in the video stream to the server, a delay of 0.5 s is added to the procedure verification controller, ensuring up-to-date frames are used in the error detection.

Components identified as incorrectly assembled by the system are highlighted by rendering an outline representing the correct shape, orientation, and location of the part, as demonstrated in Fig. 7.2. This technique is prominent within the industrial domain, as it provides a non-intrusive way to compare the current assembly with the objective [235], [236], and was reported to have the highest perceived usefulness among both expert and novice operators [203].

Furthermore, to investigate the impact of different types of impreciseness of the error detection, the AR system is also capable of deterministically providing imprecise warnings. To this end, a human-in-the-loop option is incorporated into the system to allow manual annotation of errors made by the operator when desired (see Fig. 7.2). The system can also operate fully autonomously, without requiring human input.

7.3.2 Error detection

For the error detection, the StateDiffNet [199] is adopted, which operates by comparing images of the procedure execution to rendered CAD models representing the correct assembly configuration. This comparison is achieved through a global cross-attention mechanism that implicitly aligns and contrasts the two images at various stages in an encoder, after which a decoder and segmentation head are used to segment areas of meaningful difference. Specifically, the CAD render is used as the *anchor* image, and the real-world image serves as the *sample* image. Subsequently, StateDiffNet predicts which pixels in the anchor are meaningfully different in the sample image. Because

the approach is comparative, it does not rely on memorizing specific component configurations and can generalize to new, unseen errors [199]. This is particularly relevant given the wide range of mistakes that operators can make during procedure executions. Furthermore, the algorithm can be trained entirely on synthetic data, eliminating the need for labor-intensive real-world data collection.

The StateDiffNet is modified in three ways to integrate it into the AR guidance system. Firstly, the data generation pipeline is extended to include the quadcopter model and random rotation of parts which are expected to rotate during assembly (*e.g.* propellers). This is achieved by randomly rotating these components for each different camera pose in the dataset, ensuring that for identical camera angles, the parts maintain a consistent rotation, which is required to compute accurate ground truth segmentation maps [199]. Secondly, the model output (*i.e.* a pixel-wise prediction of change on the CAD render) is translated into a part-level change prediction by extracting part locations from the semantic maps generated with each render and averaging the pixel-level prediction within these regions. A threshold of $T_e=0.5$ is used to classify a part as erroneous. Finally, the original implementation compares a single anchor to a single sample image, resulting in noisy predictions that require temporal filtering [199]. To mitigate this, multiple comparisons are performed, selecting $N_a=4$ anchors and $N_s=4$ sample images for a total of $N_a \times N_s$ comparisons for every verification.

Using these modifications, a synthetic dataset for StateDiffNet training is generated. The training set contains 1,511 unique states of physically feasible assembly configurations, each captured from 200 different poses and viewpoints, yielding $\sim 3\text{M}$ training images. The validation set contains 377 states and 100 viewpoints. The encoder and decoder are 34-layer ResNet [126] models (ImageNet [16] pretrained), trained with a batch size of 16 pairs of 512×512 (W $\times$ H) pixel images, for a maximum of 50 epochs using the cross-entropy loss function and the Adam optimizer [237]. Image pairs are matched into training samples if two conditions are met: (1) a maximum nQD of 0.15 in the object poses, and (2) a difference of at most six parts between them.

7.4 User study

Most industrial AR studies use student samples [198], and even when experts are involved, their backgrounds are often poorly documented. To improve research validity, the work in this chapter distinguishes three user groups: novices, manufacturing experts (providing task-specific expertise), and AR/VR experts (contributing technology-specific expertise). This differentiation helps clarify how different types and levels of expertise influence study outcomes, thereby strengthening the population validity of industrial AR research [238].

Expertise	N	Gender Male/Female ratio	Age	
			μ	σ
Novices	8	62.5% / 37.5%	24.13	2.03
Manufacturing experts	8	75.0% / 25.0%	35.38	8.83
AR/VR experts	8	87.5% / 12.5%	29.00	5.61
Total	24	75.0% / 25.0%	29.50	7.53

Table 7.2. Demographic profile of the study participants.

Expertise	Self-rated assembly skills [0-4]		Years of assembly experience					Years of AR/VR experience	
	μ	σ	0	<1	1-3	3-5	>5	μ	σ
Novices	1.75	0.89	62.5%	12.5%	12.5%	0.0%	12.5%	0.31	0.44
Manufacturing	2.88	1.25	0.0%	12.5%	25.0%	12.5%	50.0%	0.94	0.78
AR/VR	1.88	0.83	12.5%	0.0%	25.0%	25.0%	37.5%	5.69	2.76
Total	2.17	1.09	25.0%	8.3%	20.8%	12.5%	33.3%	2.31	2.93

Table 7.3. Experience profile of the study participants.

7.4.1 Participants

Based on the mean interaction effect $f=.351$, reported in prior work [239]–[242], and assuming $\alpha=.05$ and power=.80, the power analysis indicates that $N=24$ participants ($n=8$ per group) suffices. This alignes with the SLR results ($avg=7$ experts per study).

Accordingly, eight participants per group ($N=24$) are recruited using purposive sampling. A minimum relevant experience threshold of 2.73 years is set for the expert participants, derived from the average observed in the SLR. In contrast, the novices are university students without the minimum experience requirement.

Participants self-rated assembly skills on a scale from 0 to 4 (poor to excellent) and are asked to indicate their years of assembly experience, with response options of 0, <1, 1–3, 3–5, and >5 years. Participant demographics are provided in Tab. 7.3.

The novices are university students who self-rated their assembly skills and AR/VR experience as the lowest among the three groups (see Tab. 7.3), whereas manufacturing experts report having professional and academic engineering backgrounds as well as a substantial amount of professional work experience (see Tab. 7.3). They rated their assembly skills the highest among the groups. A Kruskal–Wallis analysis confirmed a significant group effect ($\chi^2(2)=7.690, p<.021$) with post hoc pairwise comparisons (Bonferroni-adjusted) demonstrating significantly more assembly experi-

ence than novices ($p < .032$). All AR/VR experts also report engineering backgrounds and significantly more years of AR/VR experience than novices and manufacturers (see Tab. 7.3), which is confirmed with a Kruskal-Wallis test ($\chi^2(2)=16.864$, $p < .001$). Hereinafter, the novices are referred to as N1-N8, manufacturing experts as M9-M16, and AR/VR experts as X17-X24.

7.4.2 Experimental Design

The user study employs a mixed design with one between-subjects factor *expertise level* at three levels (novice, manufacturing expert, AR/VR expert) and one within-subjects factor *warning signal* at two levels (FP and FN). Expertise level is simply referred to as expertise, and FPC (false positive condition) and FNC (false negative condition) are used to denote warning signals. Each participant performs both task variants in a counterbalanced order, and the two quadcopter configurations are counterbalanced as well. This allows for the investigation of differences in warning signals during assembly while also exploring the impact of differing population groups to determine whether these tasks should be assessed with the target group or, as is typically done, with non-representative participants from academic backgrounds, typically using students or academics as participants [198], [243]. The null hypothesis (H_0) states that manufacturing experts, AR/VR experts, and novice users do not differ significantly in their reactions to AR system warnings, while the non-directional H_1 states that manufacturing experts, AR/VR experts, and novice users differ significantly in their reactions to AR system warnings.

To ensure precise differentiation between the FPC/FNC, the automatic error recognition is complemented with human annotations. This allows for the manual control of every warning shown to participants, ensuring consistency across conditions. For the FNC, 50% of the errors made by participants are not provided as warnings. For the FPC, one part, halfway through the procedure, is consistently highlighted as erroneous.

7.4.3 Manual assembly task and procedure

The study is initiated by asking participants to read and sign a consent form and complete a demographic questionnaire, followed by the Big Five personality questionnaire [244], measuring the five dimensions: extraversion (E), agreeableness (A), conscientiousness (C), neuroticism (N), and openness to experience (O)². A precise script is followed, briefing every participant in the same manner. They are instructed to mimic real workplace situations by stating, "Do not try to break the system. Work with it as you would in a real workplace." Next, participants are asked to assemble

²Although Big Five data were collected, the results were not analyzed for the work presented in this thesis.

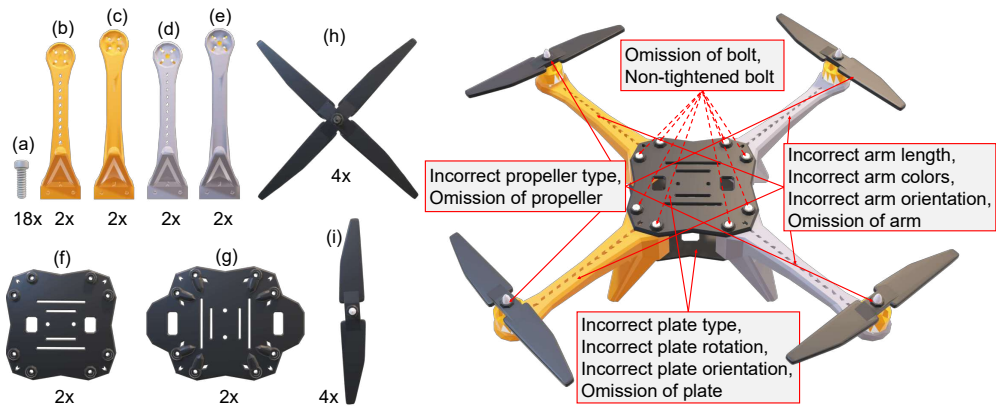


Figure 7.3. Overview of the parts and one assembly configuration. The potential mistakes that participants can make are outlined. The parts in the assembly are bolts (a), short orange arms (b), long orange arms (c), short gray arms (d), long gray arms (e), square plates (f), wide plates (g), quad rotors (h), and dual rotors (i).

a modified version of an open-source quadcopter³ adapted for controlled error insertion, as demonstrated in Fig. 7.3. Participants assemble the model on a real-life, industry-grade assembly mobile workstation⁴ (see Fig. 7.1 (3)). During each trial, the task completion time and errors are recorded and all system data is logged. After each trial, the participants complete the *Raw Task Load Index* (RTLX) to evaluate users' perceived workload [245], the *System Usability Scale* (SUS) to evaluate the perceived usability of the AR guidance [246], and the *Flow Short Scale* (FSS) to evaluate flow and anxiety experienced during the assembly [247]. After each trial, participants are also asked whether they find the system acceptable for their everyday work, offering a binary response option. Finally, semi-structured interviews are conducted, with the main interview themes inspired by the *Technology Acceptance Model* (TAM) [248] to assess the attitudes towards AR adoption among the different participant groups.

7.5 Quantitative data

Task performance and perception differences are analyzed between the FPC and the FNC to determine how an incorrect warning or the absence of a warning affects task execution. Participant groups are also compared to assess the expertise influence on error-fixing behavior and interaction effects (see Table 7.2). Although the group sizes align with typical expert samples, caution is advised when interpreting the between-subject statistical results due to relatively small sample sizes.

³Garud-500 Drone: <https://thingiverse.com/thing:2453393>

⁴Vention: <https://vention.io/>

First, the metrics of the user study are discussed, which are task completion time, error counts, cognitive load, and system acceptability. After that, the performance of the error detection system is evaluated.

7.5.1 Task completion time

Descriptive statistics showed that, for the FPC/FNC, the assembly task in the FNC ($\mu=795.50s$, $\eta_p^2=.285$, $\sigma=395.46s$) consumes more time than in the FPC ($\mu=768.38s$, $\sigma=283.19s$). Also, novices have overall the highest task completion time ($M=817.94s$, $\sigma=173.22s$), compared to manufacturing experts ($\mu=792.00s$, $\sigma=293.33s$), while AR/VR experts have the lowest task completion time ($\mu=735.88s$, $\sigma=278.25s$). A 2×3 mixed analysis of variance (ANOVA) with the between-subjects factor *expertise* and within-subject factor *FPC/FNC* is conducted. The ANOVA reveals that the FPC/FNC does not significantly affect task completion time ($F(1, 21)=0.075$, $p=0.787$). Also, no significant difference in task completion time across expertise are shown ($F(2, 21)=0.218$, $p=0.806$).

A Linear Mixed Model (LMM) is executed to account for fixed and random effects. Task completion time is log-transformed to account for normal distribution violation and treated expertise, FPC/FNC, gender, overall skill in assembling, and the overall experience with AR/VR as fixed effects. A significant interaction is observed between the FPC/FNC and expertise ($F(1, 12)=4.778$, $p=0.049$), which suggests that the effect of the FPC/FNC on task completion time depends on expertise (see Fig. 7.4). While manufacturing experts and novices take longer under the FNC (novices: $\mu=853.50$, $\sigma=430.47$; manufacturing experts: $\mu=873.50$, $\sigma=462.82$) than under the FPC (novices: $\mu=782.38$, $\sigma=201.42$; manufacturing experts: $\mu=710.50$, $\sigma=255.09$), AR/VR experts show the opposite pattern, with faster completion times under the FNC (FPC: $\mu=812.25$, $\sigma=389.37$; FNC: $\mu=659.50$, $\sigma=287.05$). Therefore, domain experts and novices are slowed down more by FNs than by FPs, whereas AR/VR experts are quicker with FNs.

A significant interaction is also observed between the FPC/FNC and overall assembly skill ($F(3, 12)=4.873$, $\eta_p^2=.549$, $p=0.019$). Similarly, at a lower self-rated assembly skill level, the FPC tends to lead to a lower task completion time, whereas at higher perceived skill levels, the pattern reverses. This crossover supports the conclusion that perceived skill level moderates the impact of the FPC/FNC (FPC: assembly skill rated as fair: $\mu=865.56$, $\sigma=338.24$ and assembly skill rated as excellent: $\mu=819.67$, $\sigma=131.07$; FNC: assembly skill rated as fair: $\mu=1019.56$, $\sigma=439.70$ and assembly skill rated as excellent: $\mu=505.67$, $\sigma=138.15$).

7.5.2 Error counts

When examining the effectiveness of receiving a warning versus not receiving it on error correction, a Wilcoxon Signed-Rank test reveals that the percentage of fixed

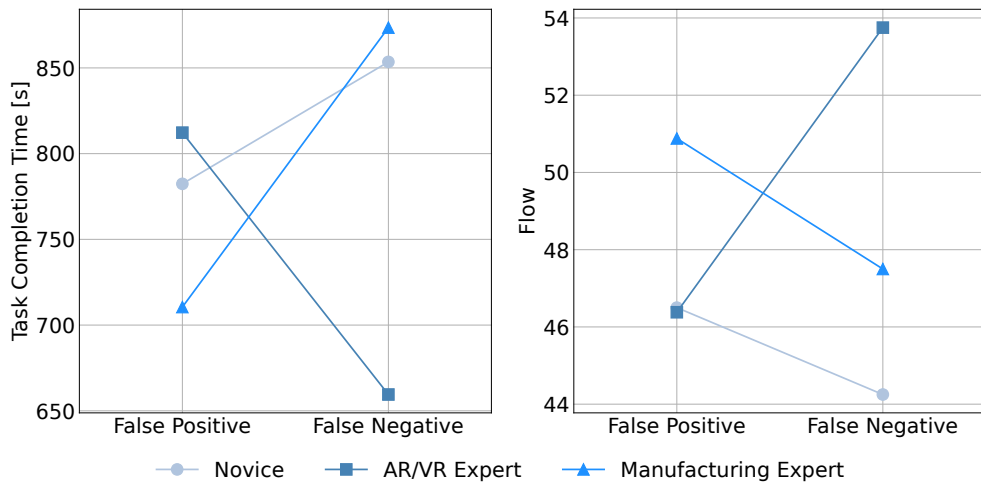


Figure 7.4. The interaction plots show the significant interaction between warning signals (FPC/FNC) and expertise level (expertise) on task completion time (left) and flow (right).

errors is significantly higher after receiving a warning ($\mu=0.74$, $\sigma=0.42$) than after not being warned ($\mu=0.28$, $\sigma=0.38$, $Z=-2.77$, $p=0.006$).

The probability of fixing an error after a warning for the FPC/FNC is calculated. Specifically, the number of times a mistake is fixed following a received warning is divided by the total number of warnings. An LMM indicates no significant effect of the FPC/FNC (FPC: $\mu=0.646$, $\sigma=0.478$; FNC: $\mu=0.75$, $\sigma=0.452$; $p=0.894$). The probability of fixing an error also does not differ significantly across expertise groups ($p=0.149$). Descriptively, novices shows the highest probability of fixing an error ($\mu=0.88$, $\sigma=0.248$), followed by AR/VR experts ($\mu=0.75$, $\sigma=0.463$) and manufacturing experts ($\mu=0.58$, $\sigma=0.495$).

The overall effectiveness of warnings is also evaluated, derived by comparing the probability of fixing an error when warned with the probability of fixing an error when not warned. This results in three dichotomous options: (i) smaller, (ii) equal, and (iii) larger. A chi-squared test of independence is performed to evaluate the relationship between the effectiveness of warnings and expertise. The relationship between these variables is not significant ($\chi^2(4)=6.262$, $p=0.180$, $N=20$). Still, the descriptive results indicate that the effectiveness of warnings is considerably higher for novices compared to the other groups, since for 85.7% of novices the effectiveness of warnings compared to non-warnings is higher. In contrast, the effectiveness of warnings is higher for only 50% of the AR/VR experts and 42.9% of the manufacturing experts. Manufacturing experts also make on average the most errors ($\mu=5.63$, $\sigma=1.92$), followed by novices ($\mu=4.00$, $\sigma=1.69$), while AR/VR experts make the fewest errors ($\mu=3.38$, $\sigma=1.85$). A

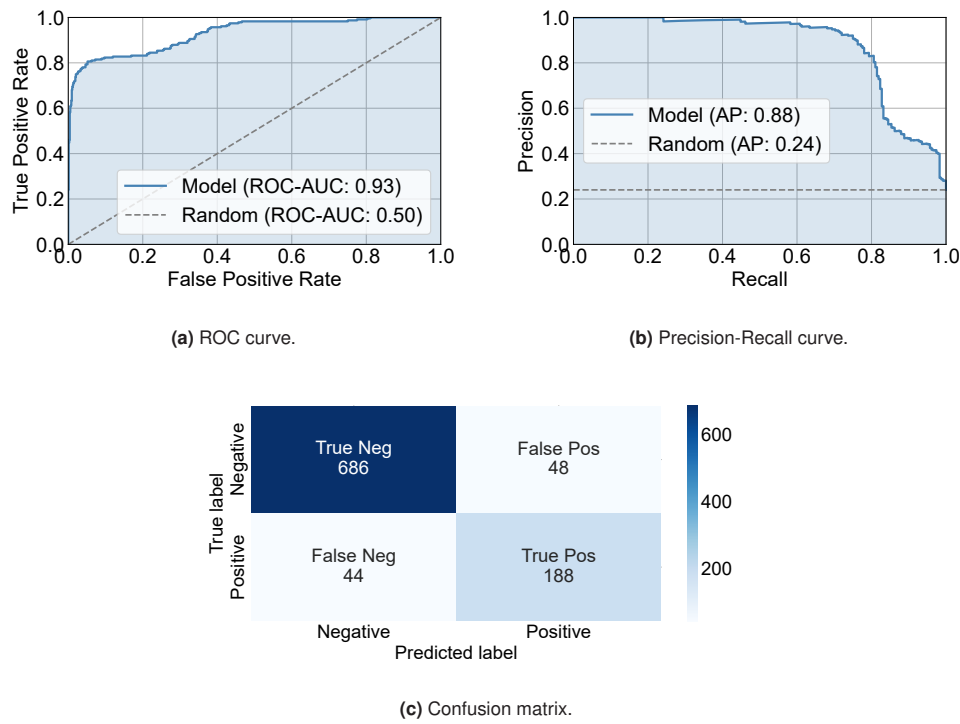


Figure 7.5. Summary of CV model's error recognition performance across different evaluation metrics. For the receiver operator characteristic (ROC) and the precision-recall curves, the performance of a model with random predictions is also indicated.

generalized linear mixed model (GLMM) based on Poisson distribution reveals no significant main effect on the difference in errors between expertise ($F(2, 37)=1.019$, $p=0.371$), nor between the FPC/FNC ($F(1, 37)=0.180$, $p=0.673$).

A GLMM with a binary logistic regression is also used to determine whether there is a difference between the FPC/FNC or expertise in terms of assembly task accuracy (binary: correct or incorrect). The GLMM shows that the FPC/FNC is a significant predictor of final model correctness ($F(1, 37)=7.604$, $p=0.009$). Assembling the asset in the FPC is more likely to be correct ($\mu=0.67$, $\sigma=0.482$) compared to the FNC ($\mu=0.29$, $\sigma=0.464$). In terms of expertise, the GLMM shows no significant differences ($F(2, 37)=2.249$, $p=0.120$). However, there is a trend suggesting that AR/VR experts are more likely to achieve a correct final assembly ($\mu=0.63$, $\sigma=0.50$), followed by novices ($\mu=0.50$, $\sigma=0.52$) and manufacturing experts ($\mu=0.31$, $\sigma=0.48$).

7.5.3 Raw Task Load Index (RTLX)

The RTLX is analyzed using a 2×3 mixed ANOVA. The ANOVA results indicates that the FPC/FNC has a significant main effect on the RTLX measure, $F(1, 21)=6.78$, $\eta_p^2=0.244$, $p=0.017$. The participants experience a significantly higher workload in the FPC ($\mu=40.90$, $\sigma=16.27$) than in the FNC ($\mu=32.57$, $\sigma=23.55$). For expertise, no significant main effect on the RTLX measure is shown $F(2, 21)=0.322$, $p=0.728$. Also, the interaction between the FPC/FNC and expertise is not significant ($F(2, 21)=2.34$, $p=0.121$). A trend toward an interaction effect, similar to the pattern observed in other significant interaction effects, is still noteworthy. A lower task workload in the FNC compared to the FPC is observed for AR/VR experts (FPC: $\mu=42.08$, $\sigma=20.56$; FNC: $\mu=24.27$, $\sigma=12.35$; mean difference: -17.81), whereas the difference is notably less for manufacturing experts (FPC: $\mu=41.56$, $\sigma=17.52$; FNC: $\mu=35.83$, $\sigma=23.55$; mean difference: -5.73) and novices (FPC: $\mu=39.06$, $\sigma=11.59$; FNC: $\mu=37.61$, $\sigma=14.77$; mean difference: -1.45).

A one-way ANOVA is conducted to analyze differences between expertise on the average six sub-scales of the RTLX: frustration, mental demand, physical demand, temporal demand, performance, and effort. One data point is excluded as it fell outside the 95% confidence interval for frustration scores among AR/VR experts. After this exclusion, a statistically significant difference between expertise in frustration scores is found ($F(2, 20)=4.90$, $p=0.018$). A Tukey post hoc test reveals that the frustration scores are significantly lower for AR/VR experts ($\mu=22.86$, $\sigma=12.94$, $N=7$, $p=0.022$) compared to novices ($\mu=52.81$, $\sigma=16.67$, $N=8$). There is also a statistically significant difference between expertise for physical demand ($F(2, 21)=5.066$, $p=0.016$). A Tukey post hoc test reveals that the physical demand scores are also significantly lower for AR/VR experts ($\mu=11.250$, $\sigma=8.02$, $N=8$, $p=0.016$) compared to manufacturing experts ($\mu=33.75$, $\sigma=19.09$, $N=8$). A similar interaction trend can be observed for frustration, with AR/VR experts exhibiting a notably lower workload in the FNC compared to FPC (mean difference: -28.48), while manufacturing experts show a difference of only $+0.625$ and novices of -1.875 . However, the 2×3 mixed ANOVA reveals no significant interaction between expertise and FPC/FNC condition for frustration ($F(2, 20) = 1.98$, $p = 0.164$).

7.5.4 Flow short scale (FSS)

The FSS consists of 13 items, each rated on a 7-point Likert scale. The overall flow score can be divided into fluency and absorption. The FPC/FNC reveals a similar overall flow (FPC: $\mu=47.92$, $\sigma=6.51$, FNC: $\mu=48.50$, $\sigma=8.76$). The 2×3 mixed ANOVA reveals no significant main effect for the FPC/FNC for overall flow ($p=0.623$). Also expertise shows no significant main effect for overall flow experience ($p=0.355$). However, a significant interaction between the FPC/FNC and expertise in flow experience is revealed ($F(2, 21)=8.53$, $p=0.002$), showing a moderate effect size ($\eta_p^2=0.448$). For

novices and manufacturing experts, the flow experienced in the FNC is lower (FPC: novices: $\mu=46.50$, $\sigma=4.00$, manufacturing experts: $\mu=50.88$, $\sigma=8.06$; FNC: novices: $\mu=44.25$, $\sigma=9.04$, manufacturing experts: $\mu=47.50$, $\sigma=8.72$), whereas for AR/VR experts the flow experienced in the FNC is higher (FPC: $\mu=46.38$, $\sigma=6.59$; FNC: $\mu=53.75$, $\sigma=6.43$) (see Fig. 7.4).

For the sub-scale fluency, a 2×3 mixed ANOVA reveals a significant interaction between the FPC/FNC and expertise ($F(2, 21)=6.59$, $\eta_p^2=0.386$, $p=0.006$). A similar pattern is observed: manufacturing experts and novices have slightly lower fluency in the FNC compared to the FPC (manufacturing experts: $\mu=5.31$, $\sigma=1.23$ vs. $\mu=5.04$, $\sigma=1.21$; novices: $\mu=4.75$, $\sigma=0.75$ vs. $\mu=4.50$, $\sigma=1.40$). In contrast, AR/VR experts report higher fluency in the FNC ($\mu=5.67$, $\sigma=0.85$) than in the FPC ($\mu=4.67$, $\sigma=0.85$).

A 2×3 mixed ANOVA for the sub-scale absorption also reveals a significant interaction effect ($F(2, 21)=4.708$, $\eta_p^2=0.310$, $p=0.020$). Manufacturers and novices experienced slightly lower absorption in the FNC (manufacturing experts: $\mu=4.75$, $\sigma=0.52$ vs. $\mu=4.31$, $\sigma=0.65$; novices: $\mu=4.50$, $\sigma=0.42$ vs. $\mu=4.31$, $\sigma=0.58$), whereas AR/VR experts exhibit higher absorption in the FNC ($\mu=4.94$, $\sigma=0.78$) than in the FPC ($\mu=4.59$, $\sigma=0.64$). Hence, the effect of the FPC/FNC on flow experience seems to depend on expertise.

7.5.5 System acceptability questionnaires

For the SUS, the FPC/FNC reveals similar system usability ratings (FPC: $\mu=75.33$, $\sigma=15.02$; FNC: $\mu=74.89$, $\sigma=18.03$). The Friedman test is applied due to violations of normality, which reveals no significant difference in system usability $\chi^2(1)=0.053$, $p=0.819$. AR/VR experts have the highest SUS score ($M = 79.84$, $SD = 12.11$), followed by novices ($M = 75.16$, $SD = 10.82$) and manufacturing experts ($M = 68.59$, $SD = 21.91$). A one-way ANOVA shows that the average usability score also does not differ significantly for expertise ($F(2, 21)=1.03$, $p=0.374$). Regarding the additional binary rating on system acceptability, the majority of participants perceive the system as acceptable for their everyday work regardless of the FPC/FNC (FPC: $\mu=0.75$, $\sigma=0.44$, FNC: $\mu=0.79$, $SD=0.42$). A GLMM based on binary logistic regression is used to test for differences between the conditions, which yields no significant main effect for the FPC/FNC $F(1, 37)=0.147$, $p=0.703$ nor for expertise ($F(2, 37)=0.147$, $p=0.223$).

7.5.6 Error detection system

Dataset To thoroughly evaluate the performance of the modified StateDiffNet on real-world data, all sensor data is recorded during the user study. Subsequently, to ensure the consistency and reliability of the data, each video frame and associated annotation generated during assembly verification is manually reviewed by a human annotator. The annotator is explicitly instructed to label all parts as erroneous if their correct installation could not be visually confirmed. Additionally, annotations are guided by the assumption that errors would be most relevant for parts manipulated

during the current assembly step. Thus, parts not involved in the immediate assembly step are neither annotated as correct nor erroneous, even if visible in the video frame. For example, a part correctly installed during the first assembly step remains unannotated during the second step unless the instructions explicitly require modifications to this part. This strategy ensures that the evaluation focused specifically on verifying assembly actions relevant to each individual step, rather than repeatedly confirming previously installed components.

Using this annotation strategy, a total of 310 verification events from 24 participants are reviewed. From these verifications, 119 (38%) contain at least one erroneous part that is modified or newly installed in that specific check. In total, the verifications consider 966 modified parts, of which 232 (24%) are incorrectly installed. The significant proportion of execution errors captured in the dataset makes it particularly suitable as an evaluation set for assessing the real-world performance of error detection. The dataset is made publicly available.

System performance The error detection performance is evaluated using the ROC-AUC [249] and AP [250] metrics. These metrics are suited for evaluating binary classifiers with imbalanced datasets [250], as is the case here, given that 24% of the samples are positive (erroneous). The results are averaged over five training runs, each initialized with a different random seed to account for variability due to stochastic training. Fig. 7.5 shows the ROC and AP curves of one model, as well as a confusion matrix for the selected threshold ($T_c=0.5$). StateDiffNet achieves an ROC-AUC of 0.93 and AP of 0.87 on the real-world data despite being trained exclusively on synthetic data. Qualitative results are shown in Fig. 7.6, illustrating typical predictions and errors.

Additionally, Fig. 7.7 shows part-specific ROC-AUC and AP for the different assembly parts, including the frequency of samples within those components. The model performs well on the quadcopter arms, achieving an ROC-AUC score of 0.99 for the long and short arms. However, the system performs significantly worse on the propellers and the bottom plates, which can be partially explained by the limited field of view (FoV) stemming from operators frequently positioning the camera close to the assembly during verification. Contrarily, the reduced performance on the bottom plates (ROC-AUC score of 0.58 and AP of 0.44) originates from their upside-down installation requirement. This error type is visually challenging to detect even for human annotators, as demonstrated through examples in Fig. 7.6.

Furthermore, the impact of the number of CAD model renders, N_a , and the number of frames from the video feed, N_s , on the performance of the verification system is outlined in Table 7.4. The results demonstrate that increasing the number of CAD model renders also increases performance, particularly for the AP score. Increasing the number of videos from the video stream, *e.g.*, to avoid evaluating on frames with extreme motion blur, does not notably improve performance. Therefore, because the

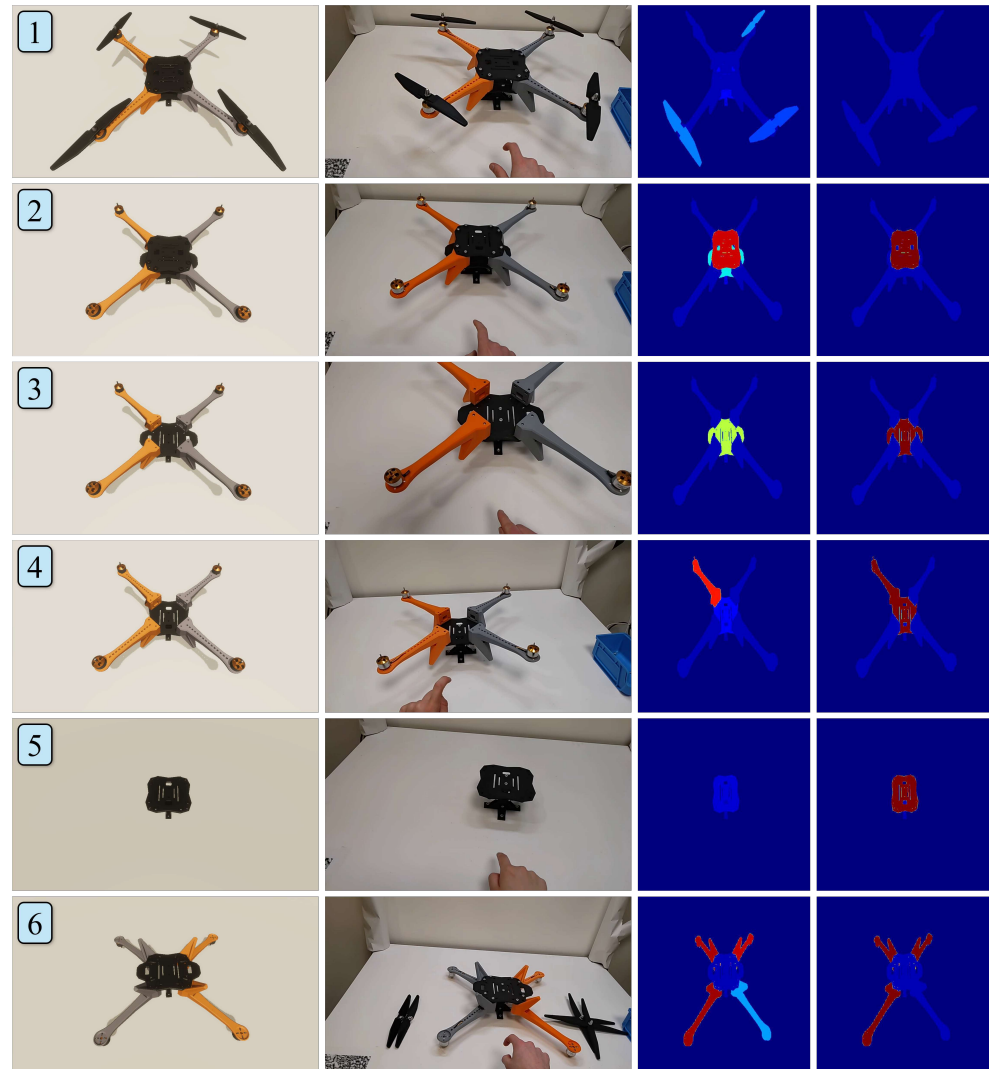


Figure 7.6. Qualitative results of the error recognition algorithm. The left column shows the CAD render of the intended assembly state, and the second column shows a frame from the HL2 video feed. The third column shows a heatmap of the error recognition prediction, where blue signifies no error and red indicates a high probability of error. Finally, the right-most column shows the ground-truth annotation. Rows 1 and 2 show correct predictions, where either no error is present (1) or the error is correctly retrieved (2). In rows 3-5, the bottom plate is mounted upside-down, which the algorithm does not adequately recognize (and is indeed difficult to see at this resolution even for human observers). Note that the incorrect arm selection in row 4 is still correctly recognized. Finally, row 6 shows a correct prediction (no errors), even though there are parts scattered close to the assembly object.

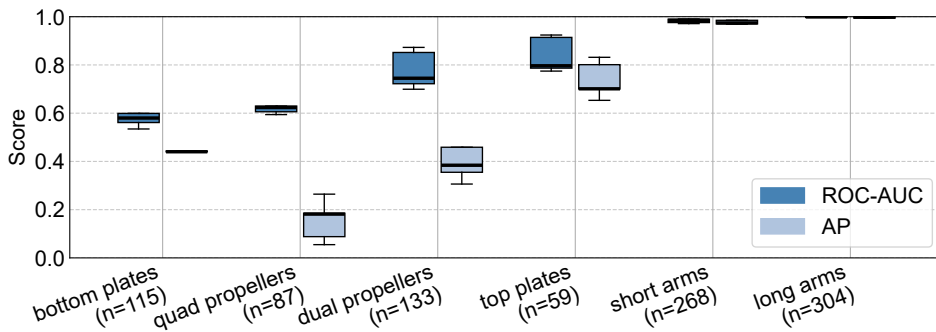


Figure 7.7. Error recognition performance across component categories, with part verification frequency per category (n). The box plot shows the median and inter-quartile range.

computation time increases approximately linearly with N_a and N_s , evaluating only half of the video frames ($N_s=2$) reduces computation time by 50% while maintaining the same performance. During the user study, the time between the verification request (button click by user) and the moment when feedback is provided is 1.83 ± 0.13 s, which includes the 0.5s delay to compensate for video latency.

7.6 Qualitative data

The semi-structured interviews following the experimental phase are analyzed using a thematic analysis approach [251]. Many participants like having the hologram as a reference for the physical model, which could be looked at while actively assembling something (N1, M11, X17, X21, X22, X24). N5, N8, and X24 point out the benefits of receiving automatic warnings when something is done incorrectly. Moreover, N7, X18, and X19 particularly like the clear and simple instructions, which avoids lengthy explanations and too many sub-steps.

At the same time, several participants (N2, N3, N6, N7, M10, M13, M16, and X19) mention that the instructions lack detail and that additional textual aid would be helpful. N4, M14, X23, and X24 criticize the HMD's restricted FoV, which aligns with prior observations [252]. For example, N4 states, (...) *it was off-putting that a slight head movement caused the AR visuals to disappear*. Additionally, M9, M12, and M14 criticize the button pressing, while N1, N2, M16, X19, and X24 raise concerns about the low resolution of the holograms. Moreover, participants N2, N8, X18, X23, and X24 stress that the error warnings are inadequate because they do not specify what exactly is wrong.

Seven participants (N1, N3, N6, M15, X19, X20, X24) explicitly state that they

N_a	N_s	ROC-AUC [-] $\uparrow$	AP [-] $\uparrow$	Computation time [s] $\downarrow$
1	4	0.922 $\pm$ 0.005	0.844 $\pm$ 0.031	0.373
2	4	0.924 $\pm$ 0.005	0.864 $\pm$ 0.015	0.567
3	4	0.924 $\pm$ 0.003	0.869 $\pm$ 0.010	0.832
4	1	0.925$\pm$0.002	0.868 $\pm$ 0.012	0.373
4	2	0.925$\pm$0.002	0.872 $\pm$ 0.007	0.567
4	3	0.925$\pm$0.002	0.873$\pm$0.007	0.832
4	4	0.925$\pm$0.002	0.873$\pm$0.007	1.112

Table 7.4. Error recognition performance for varying numbers of anchor images N_a (CAD model renders) and sample images N_s (video feed frames) for five model training cycles with unique seeds. **Bold** indicates best score.

trust the system, reasoning that it accurately points out their genuine mistakes, which quickly increases their confidence in its feedback. N6 mentioned, (...) *most of the time it's the human mistake. I thought I did something wrong.* M15 adds that a single FP does not affect their trust, stating, "I'd rather it was telling me more mistakes than fewer." In contrast, four expert participants (M9, M12, M14, and X17) state that they do not trust the system's feedback at all, anticipating from the outset that it would be wrong. As M14 explained, "At the time while I assembled it, I thought I was right and it was wrong. I couldn't really see how I was wrong, so I overrode it." Thirteen participants (N2, N4, N5, N7, N8, M10, M11, M13, M16, X18, X21 - X23) mention that their trust in the system increased over time. However, N4, N5, and X18 note that they trust the system only up until the occurrence of a FP. N2 and N8 generally trust the feedback but mention that they expect to be somewhat misled during the study, as they are familiar with experimental manipulations.

7.6.1 Influence of assembly experience

When asked about how their prior experience influence their current AR-guided assembly performance, N1, N5, N8, M11 - M15 indicated that their prior assembly experience is particularly beneficial, as it makes the screwing process more manageable in terms of the order of screwing, screw alignment, and using a screwdriver. Also, theoretical knowledge of manufacturing could increase confidence during assembly (M16).

While this expertise is perceived as helpful in the AR-guided assembly, some manufacturing experts also express that their expertise hinders them. For example, M15 and M16 are aware from past experiences that systems have flaws, which influences them

to skip over warnings more quickly. M15 explains that *I've used similar manufacturing aiding systems, and I know they can make mistakes. So I was fairly happy to skip over it.* M15 further notes that someone with less experience might be more focused on identifying the mistake, spending longer trying to figure it out. Both M12 and M15 mention that their extensive assembly experience leads them to expect that parts would be physically grouped together. M15 mentions that this assumption is detrimental when parts are not organized as expected. M12 also expressed a preference for a more open-ended process over a strict, step-by-step system.

7.6.2 Influence of AR/VR experience

Regarding AR/VR experience, X18, X19, X21, and X24 note that it helps them work more effectively with the limited FoV. They are accustomed to having to move their head rather than just their eyes. For example, X21 remarks, "I know that I can't just move my eyes up; I'd have to move my entire head to see it, instead of wondering why it's not showing up." Additionally, X17 - X19, X21, X22, and X24 mention that the AR/VR experience is advantageous in understanding that delays might occur when operating the virtual buttons. In general, AR/VR experience appears to foster a more accepting attitude toward the system's flaws and potential delays, as expressed by X17, X19, X21 - X23. They state that their awareness of the system's flaws makes them more patient with the system, but also more aware of its lack of detail, such as misalignment and drifts. While novices might assume that any misalignment is a fault in the alignment itself, the AR/VR experts recognize that alignment issues are typical in such systems and, therefore look for other potential mistakes.

Similar to manufacturing experts, AR/VR experts also indicate that their expertise could be a hindrance. Four of them (X17, X18, X20, and X21) mention that they tend not to take the actual warnings seriously because they believe that the system is prone to errors, as they are used to AR systems having inherent issues. For instance, X17 reports *I ignored the safety warning because I expected the system to be wrong. (...) In my experience, these systems can sometimes be wrong.* X18 and X20 mention that their awareness of AR-specific rendering issues leads them to believe that the system is inaccurate in its location registration, causing it to issue a warning that, in fact, correctly identifies another error. They tend to assume that any error is due to the system rather than an actual mistake on their part. X24 reports to be more skeptical with the system due to their extensive experience with AR/VR.

Additionally, two AR/VR experts (X18, X22) mention that the FP is particularly frustrating and distracting. X22 reports frustration due to the FP appearing despite its prior dismissal, whereas X18 adds that their mental effort is significantly higher during the trial with the FP compared to the trial without it and even broke their trust in the system. X18 clearly expresses their preference by stating that (...) *these intentionally added untrue mistakes bothered me a lot, and it took me a lot of time to decide*

whether to trust my own intuition, or to override a system decision or to just check again.

7.7 Discussion

The presented results emphasize the importance of warnings for assembly quality, *i.e.*, the percentage of errors fixed is higher when a warning is provided compared to when no warning is given. This observation remains in line with past research [202], [253], [254]. Furthermore, the findings indicate that even over-warnings (FP) contribute to improved assembly accuracy, whereas under-warnings (FN) lead to more inaccurate assemblies. However, the over-warning did not slow down the assembly, as both conditions did not significantly differ in assembly time. Furthermore, the probability of fixing an error after a warning is not significantly affected by a FP or FNs. This demonstrates that the over-warning condition is linked to a better assembly quality without trading off higher assembly time [254]. The over-warning improves task accuracy at the cost of mental workload, as participants experience significantly higher workload in the FPC than in the FNC. This likely happens as false warnings are seen as distractors [255], which, when increasing, leads to people decreasing their response rate [256], [257].

7.7.1 Influence of expertise on the impact of warnings

Prior work suggests that certain user groups might be more susceptible to the “cry wolf” effect, which describes that users become more likely to ignore a warning after one has turned out to be a FP [257]. Similar effects are observed in this user study, as the impact of over and under-warning on both performance and subjective experience is moderated by the user’s expertise. AR/VR experts show faster completion times under the FNC, whereas novices and manufacturing experts finish their tasks slower under the same condition. This reversed pattern is also observed in measures of overall flow, fluency, and absorption. Fewer warnings may support a smoother, more engaging workflow for those with AR/VR expertise. In contrast, novices and manufacturing experts exhibit lower flow-related measures under the FNC, possibly due to increased uncertainty in the absence of warnings. Based on these results, the null hypothesis H_0 is rejected and the H_1 is supported.

These quantitative findings are supported by qualitative insights. For instance, M15 states their preference for more rather than fewer warnings. Opposed to that, AR/VR experts (X18, X22) stress that the FPs are frustrating and distracting, with X18 noting, “I think people might do things easier without being always warned about their mistakes,” indicating a preference for fewer warnings. Moreover, half of the AR/VR experts express that they tend to not take the feedback of the system seriously because, based on their experience, they believe that the system is prone to errors, as they are used to AR systems having bugs.

Novices: N1-N8 Overall, novices are quicker in the over-warning condition and report higher flow, as shown by significant interaction effects. Also, a non-significant trend indicates that novices are more likely to fix an error after receiving a warning, with the overall highest effectiveness of warnings. This observation aligns with past literature, which reports that less experienced users tend to follow AR instructions more neatly, albeit at a slower pace [196]. Furthermore, Nourani *et al.* [258] demonstrated that novices are more susceptible to automation bias, causing the users to tend to over-trust system predictions [259].

Manufacturing experts: M9-M16 Manufacturing experts exhibit a more complex pattern of behavior. On one hand, they are more likely to override alarms. This overconfidence is highlighted by M16, who reports feeling overly confident in assembling due to their theoretical manufacturing knowledge. Three manufacturing experts (M9, M12, M14) explicitly state that they do not trust the system's feedback, often anticipating that the AR system is wrong even when it is correct. Such overconfidence bias has been previously documented, where experts ignore system warnings by relying on their own knowledge [196]. This behavior can lead to reduced precision in nuanced tasks, as reported by M12 and M15. They explain that their prior assembly experience leads them to expect items to be physically grouped together, which results in them grabbing parts without verifying details, thereby increasing errors. This is known as inter-domain rigidity, which occurs when experts work outside their primary field [260].

On the other hand, the performance of manufacturing experts in the over-warning condition is superior; they are quicker and maintain a higher level of workflow efficiency. Although these experts might over-write warnings, they also benefit from having warnings. This can be explained by manufacturing experts' limited familiarity with AR/VR [217]. The AR-guided assembly process is novel for them, making these experiences similar to those of novices. Despite their overconfidence, in opposition to AR/VR experts, they do not possess the practical skills required to tackle the AR-guided assembly correctly, as underlined by the statistically significant higher physical demand scores for manufacturing. Moreover, their expertise may allow them to differentiate between essential and non-essential warnings rapidly, which could contribute to faster assembly time in their over-warning condition, despite the higher rate of alarm overrides.

AR/VR experts: X17-X24 Expert rigidity is also observed in AR/VR experts [260], [261]. More than half of them report not following the AR feedback due to their previous AR/VR experience with faulty systems, hence are less inclined to trust the warnings (X17: "I ignored the safety warning because I expected the system to be wrong. In my experience, these systems can sometimes be wrong"; X20: "I put it down to potentially something wrong with the system (...) and that's from my past

experience with things not working perfectly"). Their experiences of faulty systems seems to be very stable, a key characteristic of expert schemata [262]. This inflexibility of thought can prevent more experienced people from abandoning their usual problem representations (*i.e.* that systems are usually wrong) to adapt to new situations (*i.e.* this system provides correct feedback now) [263], [264]. Moreover, experts are more likely to notice usability issues in their domain compared to novices [194], [196]. X18 and X20 notice minor drift and attributed it to the system being wrong rather than themselves. This is linked to confirmation bias [265], *i.e.*, a cognitive tendency where individuals favor information that confirms their pre-existing beliefs while dismissing contradictory evidence. If a user already expects the system to be inaccurate, as many AR/VR experts did based on past experiences, they are likely to disregard warnings that conflict with these preconceptions.

Apart from being more skeptical, AR/VR experts tend to be faster and more accurate compared to other user groups. They exhibit the lowest task completion times and make the fewest errors while reporting significantly lower frustration and physical demand. This superior assembly performance can be attributed to their established domain-specific schemata for AR-guided assembly in LTM, which enhances performance in a familiar context. Providing detailed external guidance to learners who already have prior knowledge can impose an additional WM load, as they must reconcile new instructions with existing information [227]. Instructional formats that benefit novices are not necessarily suitable for more experienced learners, leading to the expertise reversal effect [227]. For experienced AR users, detailed external guidance actually slows performance and reduces flow compared to a system with fewer warnings, avoiding redundant information processing in the WM. Perceived flow is highest for AR/VR experts when having less warnings, which aligns with the flow theory [266]. The flow theory posits that individuals achieve optimal performance when the challenge of a task perfectly matches their skill level. For AR/VR experts, the FNC provides that optimal challenge, while over-warning likely falls below their skill level, leading to disengagement or annoyance.

The literature also suggests that AR experts tend to exhibit fewer biases when using AR [192]. Resistance factors are rated lower among workers with prior AR experience, indicating that it may reduce prejudices and concerns [192]. It is assumed that AR experts are more likely to have a positive bias toward AR, given their role in innovating and recognizing its potential [195]. In contrast, domain experts might exhibit a more negative bias, being acutely aware of risks related to their work [194], [196]. This study shows that this is a too simple assumption. Although AR/VR experts experience significantly lower frustration, and almost all of them express that they have an increased level of patience and acceptance toward system flaws, indicating a more positive bias. However, further analysis reveals that AR/VR experts are also firmly convinced that the system's feedback has those flaws and reason with past experiences with faulty systems, even though the human-in-the-loop system is

accurate (aside from FP and FN manipulations). Thus, these results suggest that while AR/VR experts may be more patient with the system, they remain susceptible to cognitive biases that lead them to dismiss the seriousness of its feedback, which is in particular heightened when an FP is present.

Computer vision system This work helps to quantify the performance of a state-of-the-art error detection algorithm, which, while scalable to unseen assembly configurations and error types, has not been previously incorporated into a real-time system nor extensively tested on real-world data [199]. This chapter demonstrates that this algorithm, with minor modifications, can be integrated into a system that achieves an ROC-AUC score of 0.93 and AP of 0.88 and a total computation time of 1.83 ± 0.13 s. Ablation studies on the number of anchor and sample images show that this computation time can be reduced by 0.55s without compromising performance. The TCP client-server communication presents a significant latency in sending video frames, typically between 0.5s and 0.7s, which is compensated by adding a 0.5s delay to the procedure verification controller to ensure up-to-date frames are processed by the error detection algorithm. This delay can be reduced to approximately 0.1s by using HL2-tailored communication platforms, *e.g.* HL2SS [97], eliminating the need for additional delays. Combining these two aspects, the computation time can be reduced by 57%.

The performance of StateDiffNet is analyzed under varying numbers of consecutive verification frames (N_s) from the HoloLens 2 (HL2) to assess the impact of image noise (*e.g.*, motion blur) on its predictions. The results show that across different values of N_s , the ROC-AUC remains stable, and average precision (AP) improves marginally by 0.005, indicating robustness to image noise. However, incorporating frames over a longer temporal horizon (*i.e.* since the previous verification) could further enhance performance, but is not explored. For instance, parts that are out of view during the current verification may have been visible in earlier frames. To avoid substantial computational overhead, future approaches could selectively sample a small subset of past frames for evaluation. Such frames are likely to capture greater visual diversity, such as close-ups or alternate viewpoints, offering complementary information to the error detection algorithm.

7.8 Conclusion

This chapter has focused on integrating the error localization algorithm proposed in Chapter 5 into an AR assembly guidance system. The performance of this system on real-world data is quantified. Additionally, this chapter has addressed the user experience and acceptance of such an assistive system, aspects that have not been investigated in the preceding chapters of this thesis. A comparison is made between

three user groups with varying expertise levels and domains, each interacting with an AR interface tuned to either over-warn or under-warn.

System findings. The experiments presented in this chapter demonstrate strong performance of the error localization algorithm, achieving an ROC-AUC of 0.93 and an AP of 0.88. This performance is achieved without (a) training on any real-world data, without (b) explicit domain adaptation techniques, and without (c) pre-defining error states. The error recognition system is able to identify errors on the quadcopter arms with near-perfect scores and can localize other error types as well. Furthermore, this is achieved with a computation time of 1.83 ± 0.13 s, which can be reduced without an impact on the performance by another 57%.

User study findings. The three most important findings from the user study are as follows. Firstly, it is demonstrated that over-warning leads to significantly more accurate assemblies than under-warning. Although over-warning significantly increases cognitive load, it does not significantly increase task completion time. Secondly, the effect described in the first finding is mitigated by the expertise of the user. Novices and domain experts perform better and experience increased flow when over-warned, whereas AR/VR experts exhibit similar patterns when under-warned. This suggests that the tuning of feedback to users depends on the background of specific user groups. Therefore, AR system design should consider tailoring warning frequency to different types of users. Thirdly, cognitive biases such as overconfidence and confirmation bias, as well as effects like expertise reversal, influence whether users follow instructions and prefer over-warning or under-warning conditions. Not only domain experts but also AR/VR experts tend to ignore warnings based on past experiences with inaccurate outputs.

System limitations. Regarding the error recognition system, two notable limitations remain. Firstly, the rotating propellers demonstrate to be challenging for error localization. The data generation pipeline is modified to include rotating parts, and performance on the synthetic data demonstrates that the algorithm is capable of accurately segmenting change on such parts. Nonetheless, the generalization of recognizing these propellers in real-world data remains limited. Secondly, the experiments show that the error localization algorithm is not capable of recognizing bottom or top plates being installed upside down. This type of assembly error is difficult even for human annotators to spot without taking historical frames into account. Therefore, future work should consider video frames from throughout the procedure step, not just around the moment upon which a user asks the system for validation. A system capable of that could also provide feedback without being prompted by the user.

User study limitations. For the user study, relevant limitations can also be noted. Due to the difficulty of recruiting manufacturing and AR/VR experts for a time-consuming user study, a sample size of 8 per user population is selected. While the trends between user groups in this chapter are valuable, a larger sample size could demonstrate statistical significance. Moreover, the artificial task setting of a user study

may not fully reflect the real-world industrial usage, particularly due to the short-term exposure, since participants interacted with the system in a single session.

In conclusion, the presented results (i) demonstrate the real-world potential of an error localization system trained exclusively on synthetic data of correct procedure executions, and (ii) underscore the importance of selecting the appropriate testing group in industrial AR studies to ensure ecologically valid evaluation outcomes.

Conclusions and outlook

This thesis has investigated automated AI-based support for operators executing industrial procedures within the high-precision, low-volume (HPLV) industry. It has presented contributions on computer vision algorithms and systems, including industrially-relevant benchmarks and datasets, scalability of object recognition algorithms, and temporal modeling techniques for procedure step recognition. Furthermore, the thesis has contributed to the research area of human-computer interaction through an extensive user study. The first section, Section 8.1, summarizes the results of the individual chapters. Section 8.2 revisits the research questions posed in Chapter 1 and provides a discussion of the answers to these questions. Finally, Section 8.3 presents an outlook on future directions.

8.1 Conclusions of the individual chapters

Chapter 2 has provided technical background information for this thesis and highlights a distinct research gap that lies between action recognition (ACR), assembly state detection (ASD), and instructional AR systems. The chapter has also motivated that for an algorithm to support procedure executions in the HPLV industry, a fundamental capability of this algorithm should be automated logging of procedure actions in (near) real time. This relieves engineers of their tedious administrative duties, and simultaneously enables extensions to automated warnings for forgotten steps, updated procedures, and safety-critical steps. Through a proof-of-concept framework, the chapter has formulated the important requirements for automated support systems in the HPLV industry that has set the research direction for the subsequent chapters.

Chapter 3 has defined the new task of procedure step recognition (PSR), along with a set of three industrially-relevant evaluation metrics. Specifically, the objective of PSR is to recognize *correctly* and *completely* executed procedural steps, and to explicitly leverage procedural information to overcome the lack of large-scale, real-world data in the HPLV industry. Along with PSR, the IndustReal dataset has been presented. This dataset differs from existing data in the wide variety of procedural and execution errors, subgoal-oriented procedure execution, and the availability of CAD models for all parts in the procedures. Benchmarking performance has been reported for PSR,

obtaining a POS of 0.797, F_1 score of 0.883, and average delay τ of 22.4 seconds. ACR and ASD benchmarks are also reported.

Chapter 4 has proposed, implemented and evaluated a novel approach to assembly state recognition (ASR). The proposed framework uses representation learning and ISIL, a novel loss function modification that improves the generalization and scalability of ASR by leveraging unlabeled intermediate assembly states. The experiments show that this approach is able to generalize to new, unseen assembly states, without requiring retraining or fine-tuning of the model weights. For these experiments, the clustering performance of the model is improved by 5-22% when the ISIL is applied, regardless of backbone type or loss function. Finally, comprehensive evaluation on assembly states with errors demonstrates that the algorithm can recognize the presence of several error types (obtaining an average precision of 0.997 and 0.952 on missing and incorrectly-oriented parts, respectively), even though the model has never been trained on states with errors. These two aspects are critical within the HPLV industry and are valuable, not only to procedure understanding, but also to defect inspection of silicon wafers using SEM images.

Chapter 5 has presented significant advancements to error *localization* in objects during procedure executions by proposing StateDiffNet. This model is tasked with change segmentation between a rendered image of an assembly state (based on CAD models) and a real-world image from a procedure. StateDiffNet is trained exclusively on synthetic data, allowing for the explicit modeling of the expected variability between image pairs, and is not trained on assembly states with errors. Quantitative experiments on synthetic data and qualitative results on real-world data demonstrate that change detection can indeed be effectively applied for assembly error localization by pinpointing exact locations of potential errors. The framework can be applied beyond error localization in assembly images, as supported by a patent application for defect identification in SEM images.

Chapter 6 has proposed and tested STORM-PSR, a novel approach to PSR that is based on spatio-temporal features and is directly optimized for the PSR task. The STORM-PSR framework combines ASD, that is robust when objects are entirely visible and unoccluded, with a temporal model to predict step completions when objects are partially occluded. To effectively train the STORM-PSR, KFS and KCAS have been proposed. Experiments show that on the IndustReal dataset, STORM-PSR reduces the average delay between a step completion and the subsequent recognition of that step by a notable 26%, while maintaining state-of-the-art performance on POS and F_1 score. On the MECCANO dataset [22], that has been annotated in this chapter for PSR, the algorithm reduces the average delay by 11%.

Chapter 7 has focused on integrating the error localization algorithm proposed in Chapter 5 into an AR assembly guidance system. The performance of this system

on real-world data is quantified, achieving an ROC-AUC of 0.93 and an AP of 0.88 with a computation time of 1.83 ± 0.13 s. This performance is achieved with HPLV industry requirements in mind, namely (a) without training on any real-world data, (b) without explicit domain adaptation techniques, and (c) without pre-defining error states. Additionally, this chapter has addressed the user experience and acceptance of such an assistive system through an user study, comparing three user groups with varying expertise levels and domains. The user study demonstrates that over-warning users for potential errors leads to significantly more accurate assemblies compared to under-warning. While over-warning increases cognitive load, it does not significantly impact task-completion time. Importantly, these effects are mitigated by the type of background expertise of the user. Novices and domain experts perform better and experience increased flow when over-warned, whereas AR/VR experts exhibit similar patterns when being *under-warned*. Therefore, AR system design should consider tailoring its warning frequency to different types of users, underscoring the importance of selecting the appropriate testing group in industrial AR studies to ensure ecologically valid evaluation outcomes.

8.2 Discussion of the research questions

This section evaluates the proposed methods and findings with regards to the research questions presented in Section 1.5.2.

A Industrial benchmarks for procedure understanding

RQ2.1: Value proposition. *In what way can AR devices be used to provide support to engineers during the execution of procedures, such as maintenance and assembly?*

Chapter 2 has demonstrated the potential of AR devices equipped with advanced sensing capabilities in high-precision, low-volume (HPLV) settings. It has identified automated logging of procedural actions as a fundamental capability, providing direct benefits by alleviating engineers from repetitive administrative tasks, thereby ensuring comprehensive traceability and facilitating the identification and dissemination of best practices. Moreover, automated logging represents a foundational capability, upon which additional functionalities – such as safety-critical warnings and immediate detection of missed procedural steps – can be effectively integrated.

AR systems offering passive overlays of instructional content, such as those successfully implemented in aviation for the installation of wiring looms, also represent a viable path to value in the HPLV industry. However, passive systems generally provide value only to a specific subset of procedural tasks, not for the full range of tasks encountered in HPLV environments. Consequently, the introductory experiments have demonstrated a significant value proposition for *proactive* systems, capable

of actively interpreting the context and progress of procedures by focusing on on automated procedural logging as a foundational and enabling capability.

RQ2.2: Gap in literature. What is missing in current computer vision research to provide automated support to engineers during procedure executions?

Chapter 2 has described significant gaps within existing interdisciplinary research fields relevant to automated procedure logging, including procedural action recognition (ACR), assembly state detection (ASD), and AR-based assistive systems. Current ACR methods lack explicit evaluations of action completeness and correctness. Existing ASD algorithms are insufficiently scalable for HPLV industry requirements and inadequately investigate their performance on erroneous assembly states. Most AR-based assistive systems are primarily oriented towards training novices rather than supporting expert users during operational tasks, resulting in limited proactivity and flexibility outside of training tasks. Additionally, the broader literature on procedural understanding includes niche sub-fields such as key-step detection, which aims to identify critical moments within video executions. This aspect was not extensively covered in this chapter, but remains relevant particularly regarding the understanding of temporal dynamics within procedures. Consequently, several works on such research fields are detailed in the related work sections in later chapters.

RQ2.3: Requirements for automated support systems. What specific requirements can be induced from the high-precision, low-volume manufacturing industry for automated support systems?

To address this research question, Chapter 2 has developed a proof-of-concept framework for automated procedural logging, integrating insights from relevant literature. The resulting requirements, detailed in Section 2.6, are categorized into perception, procedural reasoning, and user experience. The most important perception requirements include the detection of procedural step completeness and correctness and algorithmic scalability. Procedural reasoning demands interpretable and searchable logs and handling unseen errors. User experience requirements emphasize granular, interpretable feedback tailored to user expertise.

Although the outlined requirements are highly relevant for the HPLV industry and primarily serve as a scope for this thesis. Several potentially relevant aspects, such as different AR interfaces (like a minimally invasive setup with e.g. Project Aria glasses [267]), cooperative multi-user procedures, and IT environment integration, have not been investigated due to this thesis's emphasis on perception challenges.

RQ3.1: Procedure understanding task definition. How can a task be defined, such that it combines visual perception and explicit procedural knowledge to reliably recognize (unique sequences of) correctly completed steps in procedural videos?

Chapter 3 has introduced the procedure step recognition (PSR) task, addressing a critical gap identified by the previous research questions, particularly concerning the evaluation of step completeness and correctness. PSR explicitly integrates visual

perception inputs with procedural knowledge and is defined such that it facilitates implementation into real-time systems. Additionally, the PSR task definition accommodates flexible execution strategies rather than enforcing strictly sequential steps. To evaluate PSR methods, three metrics are introduced: the procedure order similarity (POS), the F_1 score, and average delay (τ). The POS assesses the correctness of the recognized procedure execution order, F_1 evaluates recognition quality, and parameter τ quantifies the delay between actual completion and recognition of each step.

Despite its benefits, PSR faces inherent challenges in visually verifying certain procedural actions where visual information alone is insufficient, *e.g.* tightening a bolt to an exact torque specification. In such cases, integration with additional sensory inputs, such as automated torque tools, becomes essential. Moreover, some steps may inherently be invisible to a camera due to physical constraints, such as steps that rely on tactile feedback. Therefore, although PSR alone cannot autonomously validate all potential procedural steps, the task has been defined such that it can combine visual perception with complementary sensing tools and user interactions to create a comprehensive and robust understanding of procedure executions.

RQ3.2: Dataset. How can a dataset be created to effectively capture the variability in procedure executions, error types, and data modalities typical to industrial procedures?

Chapter 3 has introduced the IndustReal dataset, carefully designed to address significant shortcomings of existing datasets, *i.e.* the limited representation of procedural errors, absence of comprehensive 3D CAD models, a strict focus on sequential executions, and exclusion of maintenance tasks. IndustReal explicitly incorporates multiple execution errors, provides detailed CAD models, and supports flexible, subgoal-oriented procedural execution. Consequently, IndustReal more accurately reflects key requirements and challenges found in HPLV environments.

IndustReal does not fully represent all aspects of real-world HPLV procedures. While the chosen assembly task of a construction-toy set is inherently simpler than the typical HPLV industrial tasks, important parallels persist, such as the flexible execution orders and numerous execution errors. Therefore, the dataset remains challenging for state-of-the-art algorithms, enabling the meaningful study of PSR. IndustReal also lacks representation of the reflective or metallic parts common to the HPLV industry, which pose additional perception challenges. Since the benchmark experiments demonstrate that performance even on readily detectable components currently falls short of industry performance standards, these fundamental performance issues are prioritized. Therefore, despite these differences between IndustReal and practical HPLV settings, IndustReal effectively addresses the most crucial gaps that are not addressed by existing datasets, providing a valuable resource for developing and evaluating procedural understanding algorithms.

RQ3.3: Benchmark. To what extent can existing state-of-the-art methods from related tasks generalize to this newly proposed procedure understanding task?

Chapter 3 has also presented initial benchmark results for several baseline models that directly address the PSR task on the IndustReal dataset. Specifically, the state-of-the-art YOLOv8 model [268] is trained for assembly state detection and subsequently integrated into three distinct PSR baselines, each progressively leveraging more procedural knowledge. Results demonstrate a clear performance improvement corresponding to the increased use of procedural information. Importantly, the benchmarks highlight a substantial performance gap when models are trained exclusively on synthetic data versus models exposed to both synthetic and real-world data. However, the integration of explicit procedural knowledge significantly reduces this sim2real performance gap, underscoring the value of procedural context.

For example, Baseline 2 trained on real-world data achieves a POS of 0.731, F_1 of 0.860, and τ of 22.3 seconds, compared to significantly lower performance when trained only on synthetic data (POS of 0.240, F_1 of 0.573, delay of 44.4 seconds). The introduction of more procedural knowledge improves performance for the model trained on real and synthetic training by 9% for POS and 2.5% for F_1 with a similar τ , while synthetic-only training substantially improves POS (150%) and F_1 (28%) at a moderately increased delay (10%). These results provide the initial baseline for PSR performance and clearly show potential benefits from procedural context integration.

B Scalability of object recognition algorithms

RQ4.1: Scalability to procedure updates. *How can an ASR algorithm be designed, such that it does not require retraining when object states or procedures are modified?*

By employing representation learning rather than explicit memorization of specific assembly states, Chapter 4 has demonstrated that ASR models can generalize effectively to previously unseen assembly configurations. Consequently, these models do not require fine-tuning or retraining when minor procedural or component modifications occur. The chapter has also introduced improvements to contrastive loss functions by leveraging unlabeled transitional states, enhancing performance consistently across different backbone architectures and contrastive loss functions. Despite this promising scalability regarding procedural updates, the proposed algorithm still necessitates training with both synthetic and real-world data, a potential barrier to adoption within the HPLV industry. This dependency on real-world data is addressed in RQ5.2.

RQ4.2: Pre-defined error states. *How can erroneous assembly states be identified, without requiring the pre-definition of all potential procedure errors and what error recognition performance can be obtained by such an ASR algorithm, when it is not trained on any error assembly states?*

Assembly errors constitute a challenging subset of unseen assembly states due to their close visual similarity to correct configurations. To evaluate performance on such scenarios, the conducted experiments from Chapter 4 demonstrate the model's ability

to distinguish between correct and erroneous assembly states, given synthetic reference images of correct states. This capability is particularly notable for missing and incorrectly oriented components. The experiments demonstrate that the models do not uniformly generalize to all error types. While high performance is achieved for certain errors (average precision of 0.997 and 0.952 for missing and incorrectly oriented parts, respectively), results for incorrectly placed components and part-level errors are lower, with AP scores around 0.70. The lower performance on part-level errors is attributed to the fine-grained distinctions required, posing significant challenges even for human observers (*e.g.* camera resolution, partial occlusion). Given this level of visual detail, combined with the absence of any error examples during training, the achievement of approximately an AP of 0.70 on the most challenging errors and 0.95+ on other error types remains promising and indicates potential for further improvement.

RQ5.1: Error localization. How can an algorithm localize the exact regions containing a potential error, rather than merely recognizing that a mistake is present somewhere in the assembly?

Chapter 5 has demonstrated that global cross-attention can be used to accurately localize errors in assembly images. The approach involves detecting meaningful differences between rendered CAD images (guaranteed to be error-free) and corresponding real-world assembly images. As a result, detected changes directly indicate potential assembly errors. Furthermore, since the CAD renders are annotated with instance segmentation labels, the system can precisely pinpoint the affected components.

The quantitative results reported in Chapter 5 are only for synthetic data, whereas real-world results are only available in qualitative form due to the substantial annotation cost associated with pixel-level ground truths. While these qualitative results demonstrate promising localization capabilities across various error types, rigorous quantitative evaluation on real-world data remains necessary. That evaluation is conducted in Chapter 7 on a different assembly task, obtaining an AP of 0.88 and ROC-AUC of 0.93. Another limitation of the approach is that it requires the entire assembly object to be fully visible within the camera's field of view, and without occlusions. This condition is not frequently met in ego-centric procedural videos, unless operators are explicitly directed to capture such frames. Despite these limitations, the demonstrated error localization performance highlights the potential.

RQ5.2: Synthetic training data. To what extent can an algorithm be trained solely on synthetic data, while still accurately recognizing errors in real assembly states, and without being trained on potential errors that it may encounter in the real world?

The approach in Chapter 5 is trained exclusively on synthetic data without any assembly errors, removing the need to predefine error types. This setup enables full control over expected variability during training and has shown promising qualitative performance on real-world data. As previously discussed for RQ5.1, the quantitative evaluation of this algorithm on real-world data is presented in Chapter 7, obtaining

an AP of 0.88 and ROC-AUC of 0.93.

Under conditions of precise alignment between the renders from CAD models and procedural images, the approach demonstrates generalization to entirely unseen errors and parts. In practical scenarios where the camera and assembly objects move independently, this alignment cannot be reliably maintained, resulting in reduced performance on modified or new parts and necessitating retraining. Addressing this limitation through advanced attention mechanisms, or improved feature alignment techniques between the CAD renders and real-world images, represents a compelling direction for future research. Despite this limitation, achieving meaningful error localization on real data without error supervision is a notable result and provides a strong basis for further research.

C Temporal modeling of procedural steps

RQ6.1: Addition of temporal features. *How can a spatio-temporal neural network be trained directly for the PSR task?*

The approach proposed in Chapter 6 introduces a spatio-temporal neural network that is trained directly for the PSR task. It employs a spatial encoder that is pretrained using weakly supervised key-frame sampling (KFS), which maps identical step completions across different procedure executions to nearby points in feature space. These spatial features are then processed by a temporal model, and predictions are made using a multi-class classification head, allowing the use of standard cross-entropy loss to directly optimize for PSR labels. This formulation supports the recognition of multiple concurrently completed steps and enables temporal understanding.

A key limitation of such temporal models is their dependence on large-scale real-world training data. Procedural dynamics, especially realistic and intricate hand-object interactions, are difficult to simulate accurately, making synthetic generation of temporally consistent data an open challenge. Although the proposed KFS method helps to leverage synthetic data during pretraining, the model ultimately relies on real-world data for effective optimization. As a result, the approach demonstrates strong performance on IndustReal but does not achieve full independence from real-world training data.

RQ6.2: Partial visibility of objects. *As the algorithm is directly optimized for PSR, can it recognize step completions with a reduced delay, since it intrinsically does not have to wait for images with the assembly object unoccluded and entirely visible within the image?*

The spatio-temporal model that is presented in Chapter 6 is trained directly on step completions, enabling it to recognize actions even under partial occlusion. It complements temporal inference with robust predictions from the ASD model when the object is not occluded, combining both sources through equal weighting to improve overall robustness. This integrated approach achieves state-of-the-art performance on

IndustReal and reduces the average delay between step completion and recognition by 26%. While the model still inhibits the data dependency challenges outlined in RQ6.1, these results demonstrate that incorporating temporal reasoning substantially improves recognition timeliness in realistic ego-centric settings.

RQ6.3: Generalization. How does a spatio-temporal neural network for PSR perform on another dataset, and how does this compare to the approach outlined in Chapter 3?

Chapter 6 has extended the PSR task to the MECCANO dataset [22], [23] by generating new annotations, establishing the first benchmark performance for PSR on this dataset. Compared to IndustReal, the results are notably lower, with the best model achieving a POS of 0.377, an F_1 score of 0.497, and τ of 88.6 seconds.

This performance gap can be attributed to several factors. MECCANO lacks 3D CAD models, preventing the use of synthetic data during training. Its annotations are also significantly sparser, resulting in a smaller effective dataset. Moreover, frequent occlusions in MECCANO further challenge recognition. Despite these limitations, the results show that the proposed PSR framework maintains functionality under more challenging conditions, providing a useful baseline for future improvements on less rich datasets.

D User experience with industrial AI assistants

RQ7.1: System performance in a real setting. Can the algorithms presented in this thesis be integrated into a real-time AR system to provide advanced support during a procedure and what is the real-world system performance?

Chapter 7 has evaluated the real-world performance of the algorithm proposed in Chapter 5 by integrating it into a real-time AR assistive system for quadcopter assembly. The experimental system is tested on data collected from a user study involving 24 participants performing the assembly task. Without training on any real-world data, the system achieves an ROC-AUC of 0.93 and an average precision of 0.88, successfully identifying 81% of all errors made by participants.

Two limitations are observed during deployment. First, upon a prompt by the user, the system compares 4 video frames against 4 rendered CAD viewpoints each, which results in a delay of 1.8 seconds per evaluation (including streaming latency). While users have not reported any negative points about this delay, analysis shows that the computation time can be reduced by 57% without any loss in performance. Second, the model consistently misses a common error that requires extremely fine-grained visual discrimination, which is so subtle that even human annotators need to rewind the video to a moment prior to the evaluation to verify the error. Despite these limitations, the experiments demonstrate that the system can reliably detect the majority of real-world errors and performs comparably to human annotators for errors that are visually discernible within the provided frames.

RQ7.2: System feedback trade-off. *Should the system feedback be tuned to over-warn/under-warn users about potential mistakes, and how does this relate to the different user groups?*

The user study in Chapter 7 evaluates how users respond to two feedback strategies from an AI-based error detection system. Participants completed one assembly with over-warning (frequent alerts) and one with under-warning (minimal alerts). Over-warning has led to significantly more error-free assemblies without increasing task completion time, although it does raise cognitive load. Crucially, user expertise influences the outcome. Whereas novices and manufacturing experts benefited from over-warning, the AR/VR experts performed similarly or better with under-warning. This suggests that feedback strategies should be adapted to user expertise.

To ensure ecological validity, the study has been conducted under realistic conditions using a standard workbench and a workflow aligned with industrial practice. However, external validity remains limited by scope. The task is limited to a single assembly procedure and uses only one AR device (HL2). Moreover, while the three user groups are defined to isolate differences in domain and technological expertise, the grouping does not cover all combinations, such as AR/VR novices with manufacturing expertise, restricting the analysis of mixed-skill interactions. The classification also relies on self-reported thresholds from prior work, which may not fully reflect real-world expertise, and the sample size limits detection of smaller interaction effects. Despite these constraints, the study provides clear evidence that adaptive system feedback, tuned to user expertise, can enhance accuracy without compromising efficiency, and underscores the importance of selecting the appropriate testing group in industrial AR studies to ensure ecologically valid evaluation outcomes.

RQ7.3: User experience. *How do users, having different background expertise, experience such an AR assistive system and are they receptive to using it in the workplace?*

This research question is addressed in Chapter 7 through the thematic analysis of semi-structured interviews with the user-study participants, who reflect on usability, trust, and adoption after completing the study. Novices appreciate detailed guidance and report increased confidence, manufacturing experts favor error prevention but express skepticism toward automation, and AR/VR experts are tolerant of technical imperfections but critical of false positives. While critiques relating to field of view, warning frequency, and limited textual feedback remain, the overall acceptability remains high, with approximately 75% of participants across all groups expressing willingness to adopt the system in a workplace setting.

These findings suggest that successful deployment hinges less on raw system accuracy and more on tailoring information delivery to user expertise. For manufacturing experts in particular, adaptive alerts and optional detail layers may help to preserve trust. Addressing ergonomic constraints and providing contextual explanations for errors can further enhance long-term usability. Overall, the AR system is viewed as acceptable across heterogeneous user groups, provided that interface complexity and alert behavior dynamically align with the user domain knowledge.

8.3 Outlook on AI-assistance in industrial procedures

Advances in artificial intelligence are occurring at a rapid pace, with software companies placing large bets through investments into AI. As compute capacity keeps increasing, in part enabled by Moore's law and advances in lithography machines, neural networks are scaling rapidly in both size and capability. Generally applicable "foundation" models are being developed and released at an accelerated pace across modalities, *e.g.* vision models (semantic segmentation, depth estimation, object detection), large language models (LLMs) for natural language processing, and audio (transcription), as well as across domains (*e.g.* medical, automotive, robotics, surveillance, agriculture). There are significant benefits in applying these AI advances within the HPLV industry, where adoption remains limited. The cautious adoption of such technologies within the HPLV industry is understandable, considering the extremely tight tolerances and highly specific applications, which prevent direct reuse of off-the-shelf models (also) because of the lack of data due to the low volumes. However, through innovative methodologies, these advances can be modified and applied to fit the HPLV industry-specific requirements. As demonstrated in this thesis, automated procedure assistance is a high potential area for this because of the significant cost associated with downtime.

Current research on automated procedural assistance primarily concentrates on generalistic systems, such as ego-centric AI assistants that aim to handle a wide spectrum of tasks [269]. Although procedure action recognition is included in this vision, the importance of verifying step correctness and completeness is often overlooked, despite the important benefits that it can bring, as outlined in Chapters 2 and 3. Recent trends, such as video-question answering (VQA), are promising for generic systems. For example, the recently released HoloAssist dataset [270] features novice-expert interaction via question-answer pairs during procedural execution. However, this perspective tends to ignore the lack of annotated real-world data in industrial domains. This thesis has shown how other data modalities (*i.e.* CAD data) can help bridge this gap, yet research attention to these modalities remains limited. Future work should thus focus on leveraging such modalities and emphasize recognition of step correctness and completeness alongside VQA.

The solutions presented in this thesis extend beyond HPLV contexts. As evidenced by patent applications, the object-centric methods from Chapters 4 and 5 are applicable to a range of defect inspection problems. Beyond that, we anticipate that procedure step recognition may benefit consumer applications as well. Consider the example of a home coffee machine repair. Manuals and how-to instructions exist, but users may find the task challenging. In such scenarios, we argue that the recognition of correct and completed steps provides significant value. Instead of relying on manual user input to proceed, the system could automatically and proactively provide confirmation, enhancing the human-system interaction and reducing frustration. Systems benefit

from identifying not just what was done, but whether it was performed correctly. This may lower the barrier for people to performing such procedures, which in the case of technical repairs can additionally help reduce waste [271]–[273], and is applicable to a wide range of accessibility opportunities [274], [275].

With respect to the present thesis work, the promising future research directions are discussed hereafter. For error localization, significant potential lies in the selection of the best frames to analyze. In Chapter 7, it was shown that even humans need to scroll through video to specific viewpoints to identify mistakes. This capability of selecting informative viewpoints is not yet addressed by the localization methods presented in this thesis. In principle, temporal models can learn such cues, but as shown in Chapter 6 they lack the scalability of the proposed error localization algorithms. An essential open question for future work is whether systems can learn to identify such informative frames purely from synthetic data.

In addition to further perceptual capabilities, the implementation of a proof-of-concept system within a real industrial procedure is a next logical step. Integration with existing CAD data structures presents a major engineering challenge and is considered outside of the scope of this thesis, but is important for practical deployment. Results in this work suggest that such a system is technically feasible and indeed attractive. Evaluating user experience and system performance in such settings is valuable to determine the most impactful components for deployment.

System form factor is another relevant aspect not addressed in this thesis. Specifically, this thesis uses a single device (HoloLens 2), that is relatively heavy, heats up, and lacks high resolution and field of view. We anticipate a more practical solution that may not require AR projections at all. Instead, a lightweight camera (*e.g.* Project Aria [267]) may suffice when combined with a simple interaction device, such as a smartwatch. Close-up inspection could be handled through a pen-like camera for barcodes and fine-grained details, *i.e.* specific potential failures like damaged components. This line of research becomes increasingly relevant given the discontinuation of the HL2 AR glasses, without the announcement of a replacement device [276].

Another problem that has not been addressed in this thesis is coping with highly complex systems that require the handling of massive amounts of CAD data. A possible approach can be based on the fact that, when working on a particular part or with a specific procedure, only the parts relevant to that procedure need to be recognized. This demands a strategy to efficiently find the CAD models associated to the current task and, if required, to adapt this selection when steps are completed. Future work can investigate a hierarchical component representation that embeds visual and visibility information, thereby facilitating efficient and dynamic task-specific component selection.

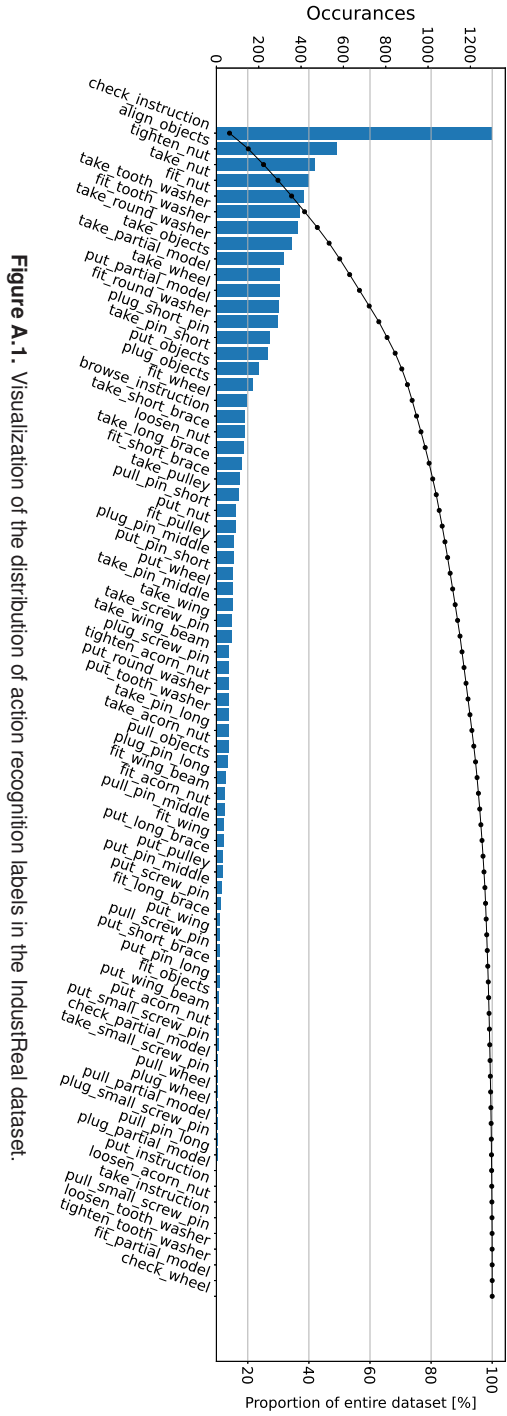
Finally, LLMs have matured significantly during the course of the research period of this thesis. They still fall short in understanding novel, complex procedures and systems, yet nonetheless pose various opportunities. For example, vision-language

models (VLMs) have been trained to establish a common ground between the two modalities. If such a model can be developed such that is also aligned with CAD data, *e.g.*, a vision CAD language model (VCLM), the applicability of such models within industrial settings becomes highly attractive.

In summary, this thesis has demonstrated that AI-assisted procedure understanding is feasible even within the HPLV industrial context. Future efforts should close the remaining data gaps to realize scalable systems, and re-think the form factor for the automated support of expert users in real-world deployments. When the aforementioned actions take place, the enabling of engineers during procedures through automated support based on the perception of the real physical world, is around the corner and will be certainly adopted by various companies both in the HPLV industry and beyond.

APPENDIX **A**

Appendix Chapter 3



APPENDIX **B**

Appendix Chapter 6

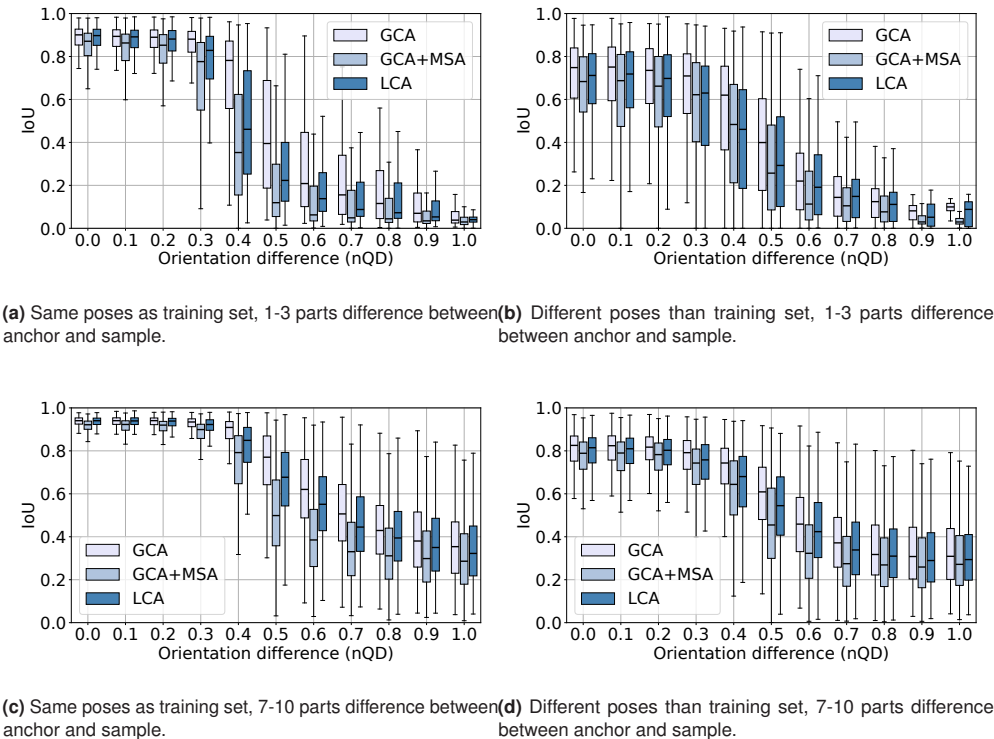


Figure B.1. Performance of three models with varying orientation difference, within the range of poses seen during training (a,c) as well as outside of this range (b,d), for various amounts of part difference between the anchor and sample image. When more parts differ between the images (c,d), the model scores a higher change IoU.

APPENDIX **C**

Appendix Chapter 7

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

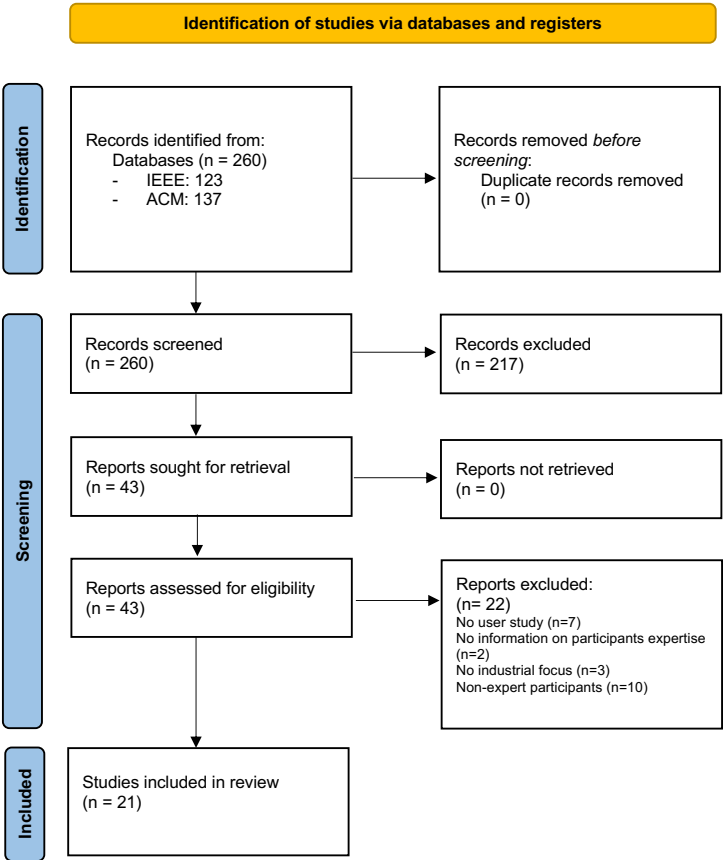


Figure C.1. The first Systematic Literature Review aimed to explore how experts have been defined in past AR-related literature and how this literature distinguishes between different types of experts. For this SLR, the following search string was applied:
"Publication Title": "augmented reality" OR "Publication Title": "AR" AND "All Metadata": expert*
This query was applied to the ACM Digital Library and IEEE Xplore databases, restricting results to publications from 2014 onward until the end of 2024 focusing on the industrial domain. Twenty-one papers were included in the final review, and an additional eleven papers were manually identified, resulting in a total of 32 papers.

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

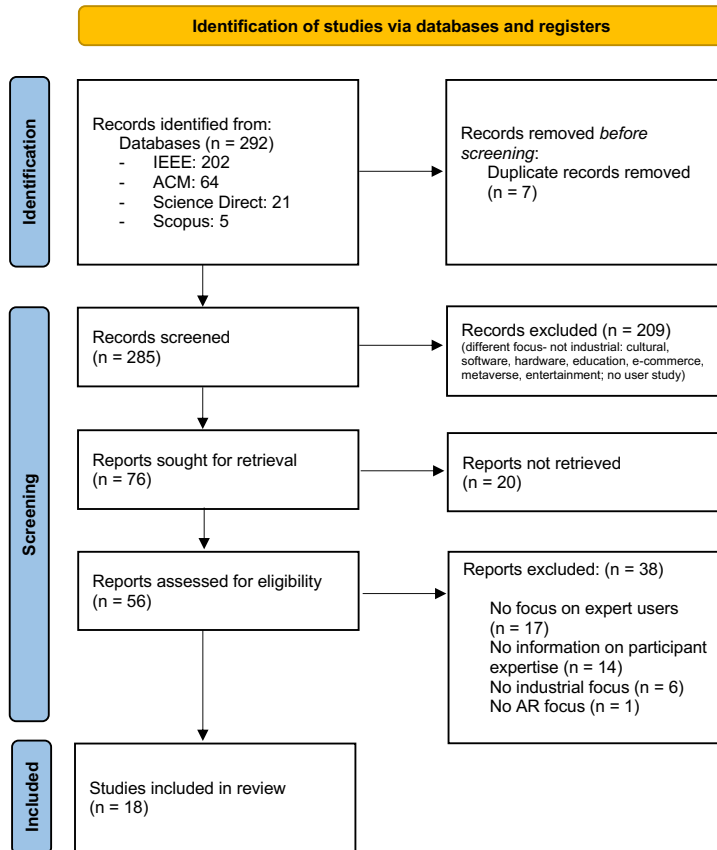


Figure C.2. The second Systematic Literature Review examined expert-related challenges that hinder the future adoption of AR in manufacturing and investigated how cognitive biases in experts influence AR system evaluation. For this SLR, the following search string was applied:
“Augmented Reality” AND “AR” AND (“Expert” OR “Specialist” OR “Professional”) AND (“Cognitive Bias” OR “Expert Bias”) AND “Industrial”.

This query was applied to the ACM Digital Library, IEEE Xplore, Scopus, and ScienceDirect, restricting results to publications from 2014 until the end of 2024. Eighteen papers were included in the final review.

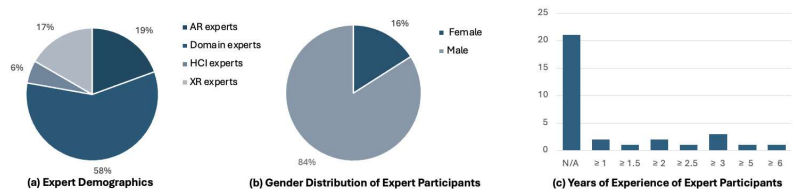


Figure C.3. Expert characteristics derived from the first Systematic Literature Review:
(a) Expertise profiles in industrial AR studies: The majority of the studies referred to domain experts (58%), followed by AR experts (19%), XR experts (17%) and HCI experts (6%).
(b) Gender distribution: In 53% of the studies that reported gender, 84% of the participants were male and 16% were female.
(c) Years of experience: The average experience level of expert participants was a minimum of 2.73 years (SD: 1.57). 66% of the reviewed studies did not report the experience levels of the expert participants.

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I was keen to do the PhD, but only if I could closely collaborate with industry. In my mind, this would prevent me from being alone with my project on a little island for four years. This turned out to be wishful thinking; I guess doing a PhD inherently means being isolated on a small island for quite some time. However, that was much more fun than I initially expected, and I look back on the past four years with great fondness. While I am the sole author of this dissertation, that does not reflect the incredible effort of all who have supported me throughout this journey. Without you, this thesis would not have been possible and I want to thank everyone who stood by my side and who, I believe, will continue to do so for many years to come.

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Curriculum vitae

Tim Jeroen Schoonbeek was born in Haren, Groningen, on May 2nd, 1996. He obtained his BSc degree in Electrical Engineering (Automotive) in 2019 from the Eindhoven University of Technology, where he explored depth estimation from disparity and segmented images for his research. Continuing his academic journey at the same university, Tim obtained a MSc (Cum Laude) degree in Automotive Engineering with a specialization in Mobile Perception Systems with dr. Gijs Dubbelman. His master thesis, titled “Learning to Predict Collision Risk from Simulated Optical Flow”, focused on sim-to-real domain generalization and was presented at the 2022 IEEE Intelligent Vehicles Symposium, where it was recognized among the top 5% of submissions. During his MSc program, Tim completed an internship at the Honda Research Institute Europe in Frankfurt, Germany, working on graph neural networks for trajectory prediction in highway driving scenarios.



Since September 2021, Tim has been working on his PhD at Eindhoven University of Technology in close collaboration with ASML Research. His research is focused on combining Computer Vision and AI algorithms with head-mounted augmented reality systems, to provide automated support and enhance human-robot collaboration during industrial procedure executions. This work has resulted into multiple peer-reviewed publications in the fields of AI, AR, and robotics, patent filings, and contributions to the ASML Technology Conference. Additionally, he participated in the International Summer School on Extended Reality and AI in Matera, Italy. During his PhD, Tim spent six months as a research collaborator at the

Cyber-Human Lab at the University of Cambridge, United Kingdom, researching human interaction with AI-based assistive systems in industrial environments. During this time, he also presented his PhD work as a start-up at the OxBridge AI Challenge, reaching the top 10 of over 200 teams.

In the summer of 2025, Tim joined the Interactive Multimodal AI Systems group for three months at Microsoft Research in Redmond, Washington, USA. Here, he contributed to the Situated Intelligence research effort, aiming to enable computers to reason about the everyday world in real time and collaborate fluidly with people in the physical space. Outside of his academic pursuits, Tim enjoys studying Portuguese, swimming, traveling, and riding and maintaining motorcycles.

